

Sharp Norms from Finite Structure: Graph Matrices and Structured Chaoses

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Abstract

Graph matrices encode dependencies in random matrices built from shared random variables. Their norm estimates are central to spectral algorithms, sum-of-squares (SoS), and high-dimensional statistics: precision can determine whether a signal dominates noise or a candidate moment matrix remains positive semidefinite. We develop a structural norm theory that connects a uniform sharp formula to extensions across models and to algebraic feasibility and tensor-network spectra. For every fixed simple graph shape in the dense Rademacher model, including overlapping or empty matrix boundaries, we prove

$$\mathbb{E}\|M_\alpha\| = \Theta_\alpha\left(n^{(v+h-s)/2}(\log n)^{a_*/2}\right).$$

Here v counts vertices, h counts isolated summation vertices, s is the minimum boundary-separator size, and a_* maximizes, over minimum separators, the number of components adjacent to the separator and meeting neither surviving boundary. Thus, despite high-order dependencies, two finite cut optimizations determine both growth exponents. The formula resolves the sharp-norm question in this model, replacing the polylogarithmic gap in separator estimates with an exact logarithmic exponent and matching lower bounds. It reveals that local fluctuations add at one cut and compete across cuts. An infinite family with identical coarse graph parameters but different sharp norm scales exhibits the structural information this rule detects. The proof converts label losses in high-moment expansions into layerwise separator excess, uniformly controlling every defect layer at growing moment order. Conditional flattening and synchronized fluctuations give the matching lower bound. Retaining the layers lets us track factor incidence, scalar tails, and label dimensions separately, yielding sharp estimates for specified independent-factor chaoses, local weights, and unequal dimensions. Separate distributional comparisons cover bounded asymmetric noise, Gaussian inputs, and fixed-degree Hermite inputs, retaining growing-moment control.

Two applications use this precision together with additional structure. For degree-four clique SoS in $G(n, 1/2)$, critical log-free estimates and an exact positive square give, with high probability, pseudoexpectations satisfying all constraints simultaneously for $9 \leq k \leq c\sqrt{n}$, for an absolute $c > 0$. This sharpens the Hopkins–Kothari–Potechin quartic-correction construction to the square-root scale without an asymptotic logarithmic loss. For fixed independent Gaussian tensor networks with equal edge dimension, we obtain necessary and sufficient deviation-convergence thresholds, sharp expected deviation scales, and entropy estimates with bounded additive error. For connected, loopless complex-Gaussian networks with nonempty input and output boundaries, an additional flow-quotient comparison and auxiliary-dimension transfer, combined with the tensor-GUE strong-convergence theorem of Chen, Garza-Vargas, and van Handel, show that the appropriately rescaled largest output-state eigenvalue converges in probability to the exact right edge of the known limiting law, determining the constant correction to min-entropy. The results connect finite structure to sharp growth scales, and additional algebraic and spectral structure to full feasibility and exact limiting constants.

1 Introduction

Many proofs about computation on random inputs turn on a spectral comparison: a random error must be smaller than a useful signal or a positive matrix that supports a relaxation. The matrices in these proofs often have polynomial entries built from the same random variables, so their entries are strongly dependent. Graph matrices encode these dependencies by finite graph templates. They are a central tool in sum-of-squares (SoS) lower bounds for planted clique, the Sherrington–Kirkpatrick model, tensor PCA, and the Wishart model of sparse PCA [3, 9, 17], and more recently for non-Gaussian component analysis, with consequences for robust statistics and mixture learning [7]. Their norm estimates help prove that candidate moment matrices are positive semidefinite, and hence that a relaxation remains consistent with an apparent solution.

Spectral algorithms for planted sparse vectors and random overcomplete tensor decomposition also require bounds on dependent polynomial matrices [13]; graph-matrix techniques give systematic proofs of key estimates in these analyses [1, Section 9]. Here the norm controls the random interference against which a signal must be detected. In both settings, the precision of the estimate can determine the parameter range in which the computational argument succeeds. This raises a basic structural question: *which features of a graph template determine the exact growth of its matrix norm?* We determine that growth for every fixed simple graph shape in the dense sign model, including the logarithmic factors that distinguish a removable loss from a real spectral fluctuation.

To see the object behind these applications, start with independent random signs: for every unordered pair $\{i, j\} \subseteq [n] = \{1, \dots, n\}$, choose $\xi_{\{i,j\}} = +1$ or -1 with equal probability. Equivalently, encode the edges and nonedges of $G(n, 1/2)$ by positive and negative signs. We call this the *dense Rademacher model*. The word “dense” refers to this ambient random input, not to the number of edges in a template. The signs are independent across unordered pairs; using a pair in the opposite order does not produce a new random variable.

A graph matrix uses a small graph as a recipe for multiplying and summing these signs. For example, the path $a - x - b$ defines the matrix

$$P_{ij} = \sum_{k \notin \{i,j\}} \xi_{\{i,k\}} \xi_{\{k,j\}} \quad (i \neq j), \quad P_{ii} = 0. \quad (1)$$

The endpoint labels i, j specify the row and column; the internal label k is summed out. Each summand follows the two edges of the path. Although the underlying signs are independent, different entries of P reuse them and are dependent. More generally, a finite graph, called a *shape*, specifies which signs to multiply. Its designated row and column vertices specify the matrix indices; the labels of all other vertices are summed over, with distinct vertices assigned distinct labels. The graph is a recipe for a dependent random matrix, not the large random graph on n vertices itself. For polynomial matrix-valued functions with the appropriate relabeling symmetry, grouping terms by shape gives a matrix analogue of a Fourier expansion. One can then analyze the graph-indexed components and the patterns produced by multiplying them [1, 17].

Norm estimates turn this representation into a quantitative proof. Consider, for illustration, a Hermitian block $B + E$, where B is positive definite and E is a random error. The implication

$$\left\| B^{-1/2} E B^{-1/2} \right\| < 1 \quad \implies \quad B + E \succ 0$$

expresses the basic role of spectral control: the error must be small relative to the available positive mass. Actual SoS arguments require substantially more structure, including suitable factorizations, preservation of positive terms, and treatment of kernels forced by the constraints. Graph-matrix norm bounds control the fluctuations and intersection errors that arise within these constructions.

Their precision can determine whether the resulting comparison closes at the desired parameter scale [9, 17].

Even a logarithmic loss can matter in such a comparison. In ellipsoid fitting, refined spectral estimates for particular correlated matrices enabled a construction to fit a constant multiple of d^2 independent Gaussian points in $\mathbb{R}^d$, improving a guarantee with a polylogarithmic loss [14]. Sharpening a matrix estimate thus changed the scale of a geometric feasibility result.

But not every logarithm is a loss in the proof. A single-edge shape already illustrates the distinction. If its two endpoints index the row and column, it gives the symmetric sign matrix Ξ , with $\Xi_{ij} = \xi_{\{i,j\}}$ for $i \neq j$ and $\Xi_{ii} = 0$. If instead one endpoint indexes *both* row and column and the other is summed out, the same edge gives a diagonal matrix D :

$$D_{ii} = \sum_{j \neq i} \xi_{\{i,j\}}, \quad D_{ij} = 0 \quad (i \neq j). \quad (2)$$

These two uses of the same one-edge graph have different norm scales:

$$\mathbb{E} \|\Xi\| \asymp \sqrt{n}, \quad \mathbb{E} \|D\| = \mathbb{E} \max_i \left| \sum_{j \neq i} \xi_{\{i,j\}} \right| \asymp \sqrt{n \log n}. \quad (3)$$

For the diagonal matrix, the operator norm selects the largest row sum: an extreme local fluctuation produces the extra logarithm. Thus even the number of vertices and edges does not determine the sharp scale. The placement of the matrix indices matters. For a general shape, which fluctuations can the norm select, and how much amplification do they cause?

A substantial theory already addresses these matrices. Separator-based bounds identify their polynomial norm scale up to polylogarithmic factors [1]. More general matrix-chaos inequalities express spectral bounds through coefficient flattenings, provide mechanically computable parameters for combinatorial models, and remove unnecessary dimensional factors in important examples [2]. Decoupling and linearization offer another systematic approach through lower-degree polynomial matrices [20]. These developments make a finer structural question natural: for each fixed graph shape, what is the exact logarithmic exponent, and which feature of the shape forces it? This is the sharp classification question recorded as Open Problem 31 [15].

We answer this question for every fixed finite simple graph shape in the sign model just described, allowing overlapping or empty row and column boundaries. A terminating finite combinatorial rule determines both exponents in

$$\mathbb{E} \|M_\alpha\| = \Theta_\alpha \left(n^{f(\alpha)} (\log n)^{g(\alpha)} \right).$$

The answer is governed by a competition between cuts. The smallest vertex cut between the two matrix boundaries determines the power of n , with a correction for isolated summation vertices. Among those smallest cuts, count the pieces that touch the cut but neither remaining boundary. The largest such count determines the power of $\log n$. Thus the classification identifies both the fluctuations the norm can select and the ones it cannot amplify simultaneously. Matching lower bounds show that the surviving logarithms are necessary. A finite graph calculation therefore determines when a log-free estimate is possible and how much amplification is unavoidable otherwise.

The proof develops a structural norm theory beyond graph edges. Its layerwise counting theorem separates the effects of factor incidence, scalar tails, and label-space dimensions. Together with matching lower-bound arguments and distributional comparisons, it yields sharp results for specified independent-factor, locally weighted, Gaussian, and fixed-degree Hermite models. These extensions explain how the cut rule changes when the random input or its geometry changes.

The applications carry this precision into two different conclusions. For degree-four SoS, critical log-free estimates and an exact positive square give full clique feasibility at $c\sqrt{n}$, removing the logarithmic loss in the cited construction's scale. For Gaussian tensor networks, uniform norm estimates give sharp deviation thresholds and entropy bounds. A further comparison, through a flow quotient and an auxiliary dimension, proves convergence of the largest eigenvalue to the exact edge of the known limiting law. The paper thus connects structural norm classification to algebraic feasibility and to spectral endpoints; reaching an exact endpoint requires the additional mechanism that preserves the leading constant.

1.1 A finite rule for the sharp norm

We now give the general definition behind Equation (1).

Definition 1.1 (Admissible shape and graph matrix). Let $\alpha = (W, E, U, V)$ be a fixed finite simple graph with ordered row and column boundaries $U, V \subseteq W$. Each boundary has distinct vertices, but the two boundaries may overlap or be empty. These are precisely the *admissible shapes* in the sign theorem; no additional connectivity condition is imposed. Write $v = |W|$, and use the independent uniform signs defined above. Rows are indexed by injective labels of U , and columns by injective labels of V . Thus $\mathbf{i}$ and $\mathbf{j}$ record the labels assigned to the designated row and column vertices. The graph matrix is

$$M_\alpha[\mathbf{i}, \mathbf{j}] = \sum_{\substack{\phi: W \hookrightarrow [n] \\ \phi|_U = \mathbf{i}, \phi|_V = \mathbf{j}}} \prod_{\{x, y\} \in E} \xi_{\{\phi(x), \phi(y)\}}. \quad (4)$$

Here $\phi: W \hookrightarrow [n]$ assigns distinct labels to all vertices; the boundary labels are fixed, and the others are summed over. We call this the *globally injective* model. There is no implicit normalization. If the boundary labels cannot be extended to such an injection, the entry is zero. In particular, assignments must agree on a common boundary vertex.

The graph is fixed throughout; constants may depend on it, but not on n . This convention is essential. A statement for every fixed graph is not a uniform statement for graphs whose sizes grow with n .

The classification asks for graph-dependent exponents satisfying

$$\mathbb{E} \|M_\alpha\| = \Theta_\alpha \left(n^{f(\alpha)} (\log n)^{g(\alpha)} \right). \quad (5)$$

The missing logarithm can reflect a genuine extremal phenomenon rather than slack in a concentration argument. The next statistic distinguishes these possibilities.

Definition 1.2 (Separators and active components). A set $S \subseteq W$ separates U from V if every path from U to V meets S ; vertices of the boundaries themselves may be deleted. In particular, every common boundary vertex belongs to every separator. Let s be the minimum size of such a separator. A component of $\alpha - S$ is *active* if it meets neither surviving boundary and is adjacent to S . Let $a(S)$ be the number of active components, and set

$$s = \min_{S: U|V} |S|, \quad a_* = \max_{\substack{S: U|V \\ |S|=s}} a(S). \quad (6)$$

A component already disconnected from both boundaries before the cut is not active. Such a component is a scalar random factor, with a different role in moment bounds.

Let h denote the number of isolated vertices outside $U \cup V$. Each such vertex contributes a summation without a random edge. The classification has the following form.

Theorem 1.3 (Sharp graph-matrix norm). *For every fixed admissible simple shape α , there are positive constants $c_\alpha, C_\alpha, n_\alpha$ such that, for $n \geq n_\alpha$,*

$$c_\alpha n^{(v+h-s)/2} (\log n)^{a_*/2} \leq \mathbb{E} \|M_\alpha\| \leq C_\alpha n^{(v+h-s)/2} (\log n)^{a_*/2}. \quad (7)$$

The formula turns a spectral extremum over a growing matrix into two optimizations on a fixed graph. First minimize the number of deleted vertices: with v, h fixed, each unit of separator size changes the polynomial exponent by one half. Only *among those minimum cuts* do we maximize the active-component count: each counted piece contributes a half-power of $\log n$. A larger cut cannot trade a loss in the power of n for a fixed logarithmic gain. The number of edges and the degrees of the pieces do not enter the exponents separately; they matter through these cut statistics and through the shape-dependent constants. This is the reason that shapes with the same polynomial scale can have different logarithmic behavior. Unlike an unspecified polylogarithmic upper bound, the formula decides which logarithms must remain and certifies their size by a matching lower bound.

The rule in Equation (6) is finite. Enumerate vertex subsets, test separation, retain those of minimum size, and count their active components. In particular, the logarithmic exponent is not defined through a limit of moment optimizations or through a list of small graphs. The same rule applies to a fixed graph of any size.

Detached components explain why adjacency to the cut is part of the definition. In the auxiliary *typed model*, each vertex role has a separate label set and each edge its own independent array. A detached component there multiplies every entry by the same random scalar: its size contributes a power of n , but the norm does not maximize it over n independent choices. Counting it as active would introduce a spurious logarithm. Section 2 compares this typed model with the original globally injective matrix; it does not factor the latter entrywise.

The formula recovers the contrast in Equation (3). For the sign matrix Ξ , the single-edge shape has $v = 2, h = 0, s = 1$, and $a_* = 0$. For the diagonal row-sum matrix D , the same edge has $v = 2, h = 0$, and $s = 1$, but deleting the common boundary vertex exposes one active component, so $a_* = 1$. Even the polynomial scale and minimum cut agree; the active count detects the square-root logarithm.

The distinction persists when the boundary placement and the usual coarse graph parameters are held fixed. Take the path $u - x - y - v$ with boundaries u, v and add two leaves. Attaching both leaves to x gives $a_* = 2$; attaching one to x and one to y gives $a_* = 1$. Both trees have six vertices, five edges, and minimum separator size one, but

$$\mathbb{E} \|M_{\text{concentrated}}\| \asymp n^{5/2} \log n, \quad \mathbb{E} \|M_{\text{distributed}}\| \asymp n^{5/2} \sqrt{\log n}. \quad (8)$$

In their equal-size typed coefficient representations, even the two scalar flattening summaries σ, ν of Bandeira et al. [2] agree. Section 2.1 defines these summaries and proves an infinite family of such separations. The maximum over minimum cuts detects information those scalars lose: the location at which several fluctuations can reinforce one another.

The same family shows what changes in the estimate. On a longer path, attach q_j leaves at internal site j , and put $Q = \sum_j q_j$. In the typed model, summing leaves gives independent diagonal gates between sign matrices. Multiplying their separate norm bounds pays $(\log n)^{Q/2}$. The sharp rule instead pays $(\log n)^{\max_j q_j/2}$: charges add at one cut and compete across cuts. The gain can therefore grow with the number of decorated sites in this family: separate maximization adds charges that the matrix cannot realize together. The elementary product bound is sharp when all leaves are at one site. Section C.2 also calculates specified direct literature bounds on the same

family. These comparisons concern those outputs and the two scalar summaries, rather than the full flattening profile or every possible refinement of earlier methods.

There is also an obstruction to reconstructing the answer from ordinary expected traces. An isolated diagonal root with a detached edge and a rooted triangle have identical expected trace moments at every order, yet their expected norms differ by $\sqrt{\log n}$. Their diagonal marginals agree; their joint dependence does not. The exact finite- n identities and the norm comparison appear in Section B.1. Thus the input to our rule is the shape and its boundaries, not merely its dimension, rank, or expected trace sequence.

1.2 A structural norm theory beyond graph edges

The next contribution is to separate the sources of spectral growth. Factor scopes specify which labels a random variable can see; scalar tails determine the size of fluctuations selected at a cut; unequal role sizes change the cost of that cut. We prove extensions that track these effects in explicit structural models. The common mechanism is the separator sequence in the high-moment expansion: it retains enough information to reweight the analysis when the input changes. Gaussian and fixed-degree Hermite comparisons also retain control over a growing moment window, which is needed to pass from moment estimates to the operator norm.

First replace each edge occurrence by a nonempty factor scope $e \subseteq W$ with its own independent random array. A matrix entry multiplies the selected array coordinates and sums over middle-role labels. The two-section graph joins roles that share a scope; its separators therefore record the factor incidence. Assume every role has a label set of size comparable to n . For each factor occurrence, let its coordinate law be fixed, symmetric, and of variance one, and suppose

$$\|\xi_e\|_{L^q} \asymp_e q^{\theta_e/2}, \quad q \geq 2, \quad 0 \leq \theta_e \leq 2.$$

These are two-sided, all-order assumptions on a law that does not change with n . Define

$$b_* = \max_{\substack{S: U|V \\ |S|=s}} \left(a(S) + \sum_{e \subseteq S} \theta_e \right). \quad (9)$$

The second term records factors supported wholly on the separator. Their extremes can contribute to the same label selection as the active components.

Theorem 1.4 (Sharp norm for the specified factor model). *After the explicit preprocessing of unused roles and detached scalar components, a fixed independent-factor shape with the laws above has*

$$\mathbb{E} \|M\| \asymp n^{(v+h-s)/2} (\log n)^{b_*/2}.$$

The constants may depend on the shape, the comparison constants for the role sizes, and the fixed factor laws.

The formula gives a precise interaction between structure and tails: component fluctuations and heavy factors add at the same cut, and the resulting charges compete across minimum cuts. Proving it requires a refinement of the total count in Equation (19). We retain the separator sequence (S_k) , assign each layer its factor moments, and only then sum over sequences. This reusable part of the argument distinguishes minimum separators that tie in size but carry different tail charges.

Symmetry can be dropped for bounded noise by a different argument. For independent centered variance-one coordinates bounded by K , Theorem 5.6 shows that replacing all Q occurrence arrays by

signs changes every L^p norm by at most $(2K)^Q$ in either direction. This is standard symmetrization and contraction applied to the original coefficient tensor, before simplifying its sign reference. It extends the sharp sign scale to dense centered Bernoulli and bounded-erasure factors, even with coordinate-dependent laws, but not to a shared unordered-edge array without a separate comparison argument.

A separate result, Proposition 5.4, treats certain n -dependent laws through a finite moment profile. The resulting moment-weighted cuts give an upper bound, with an expectation consequence under the stated cut-stability hypothesis. This is useful beyond a fixed distribution, but is not a matching sparse-law theorem. Detached random factors retain their actual moments in the trace, and an identity matrix factor retains its dimension; the profile does not absorb either into an unspecified constant.

Changing the dimensions affects the optimization itself, not just its constants: deleting a role with more labels incurs a larger polynomial cost, and a tie between cheapest cuts can determine the logarithm. Theorem 6.3 makes this precise for the typed independent-sign model with original factor scopes of arity at least two, nonempty role sets of size $m_x \asymp n^{\beta_x}$ for fixed $\beta_x \geq 0$, and deterministic real local amplitudes satisfying

$$c_e n^{\gamma_e} \leq |a_{e,n}(x_e)| \leq C_e n^{\gamma_e}, \quad 0 < c_e \leq C_e < \infty.$$

Each amplitude sees only the coordinates of its own occurrence. Write $\Gamma = \sum_e \gamma_e$. In the finite normal form of Section 6, bounded roles are eliminated, boundary factors are removed in the sign reference, and deterministic isolated roles and detached random components are recorded separately. Let B_+ be the sum of positive role exponents, h_β the sum of exponents of deterministic isolated roles, and d_0 the number of detached random components. In the remaining core, put

$$\kappa = \min_{S:U|V} \sum_{x \in S} \beta_x, \quad a_* = \max_{\sum_{x \in S} \beta_x = \kappa} a(S).$$

The theorem gives the explicit sharp scale

$$\mathbb{E} \|H\| \asymp n^{\Gamma + (B_+ + h_\beta - \kappa)/2} (\log(2n))^{a_*/2}. \quad (10)$$

Its logarithmic-window upper moments have the additional factor $q^{d_0/2}$. The optimization retains every minimum-cost separator, including ties. Bounded roles ($\beta_x = 0$) are included through a finite reduction. Section 6 explains both the reduction and a concrete transition where a change of minimum face changes the logarithmic exponent. For rational input exponents the statistics admit a terminating exact evaluation procedure; arbitrary real exponents require a representation permitting exact comparisons.

For Gaussian inputs, the next issue is the precision of the estimate as the moment order grows. Fixed-order limits alone do not control the operator norm; the extension retains the sharp scale throughout a logarithmic moment window. Consider a fixed coefficient-one independent-factor template with Q nonempty scopes, comparable role sizes, and standard real Gaussian coordinates; unary scopes are permitted. To express this model through sign statistics, for each occurrence adjoin a private middle role of size n and replace the occurrence by a sign factor on the enlarged scope. Let $(\hat{f}, \hat{g}, \hat{d}_0)$ be the sign norm and detached-component statistics of this lifted template. Theorem 7.1 gives

$$\begin{aligned} \mathbb{E} \|H_G\| &\asymp n^{\hat{f}-Q/2} (\log(2n))^{\hat{g}}, \\ |||H_G|||_{L^q} &\asymp n^{\hat{f}-Q/2} (\log(2n))^{\hat{g}} q^{\hat{d}_0/2}, \quad 2 \leq q \leq C_0 \log(2n). \end{aligned} \quad (11)$$

The second statement is uniform throughout this window for each fixed $C_0 > 0$ and all sufficiently large n , with constants allowed to depend on C_0 and the fixed template. The same result holds for standard circular complex coordinates. The private roles retain fluctuations from unary factors that cannot be removed as sign isometries; detached components supply the separate $q^{\widehat{d}_0/2}$ factor. This controls the sharp scale, while exact edge constants require the additional comparison described below.

Fixed-degree Hermite coordinates are covered by a separate all-moment comparison. Replace occurrence e by $\text{He}_{d_e}(g)/\sqrt{d_e!}$, using mutually independent standard real Gaussian coordinates and fixed positive integers d_e . Let H_{dec} replace that occurrence by d_e independent Gaussian occurrences on the same scope. With $A = \prod_e d_e^{d_e/2}/\sqrt{d_e!}$, Theorem 7.2 states

$$A^{-1} \lll H_{\text{dec}} \lll_{L^q} \leq \lll H_{\text{He}} \lll_{L^q} \leq A \lll H_{\text{dec}} \lll_{L^q}, \quad q \geq 1. \quad (12)$$

Consequently the norm and logarithmic-window moment exponents are those of an explicitly expanded Gaussian template. The comparison is uniform in dimension, not in the degrees. In particular, it accommodates fixed degrees above two and does not require the Hermite coordinate itself to have a symmetric law. This is a route beyond the separate tail theorem's $\theta_e \leq 2$ assumption, not a relaxation of that assumption inside its proof.

The same cut competition remains visible in concrete matrices. The Gaussian gated chain in Section 5 recovers the maximum-over-cuts rule of the sign leaf family through unary tail charges. The gated Khatri–Rao product instead adds degrees at a shared column role and has a short independent proof. These examples link the distributional extension to the same structural phenomenon.

Two obstructions explain why the hypotheses carry mathematical content. The coefficient example in Section 6 has identical entry magnitudes and random incidence but different operator scales: local amplitude control is not a substitute for arbitrary coefficient geometry. For a specified sparse iid family, Proposition 5.7 records the complete signed-Bernoulli phase diagram, including the $(\log n / \log \log n)^{1/2}$ correction at density $1/n$ and the rare-occupancy regime. This classical critical-density obstruction [19] delineates the fixed-law domain. The positive statements above are separate, proved extensions with their own hypotheses, not an assertion that every combination of their assumptions defines one covered model.

1.3 Application: degree-four SoS at the square-root scale

The first application shows how removing a spectral logarithm changes the scale of a computational lower bound. For the degree-four SoS relaxation of clique in a dense random graph, we establish feasibility through a constant multiple of $\sqrt{n}$. The candidate moment matrix is a structured sum of graph-matrix blocks. Turning their estimates into feasibility requires comparison with different positive scales on different subspaces, together with all Boolean, nonedge, normalization, and size constraints.

The quantitative conclusion of Theorem 8.2 is simultaneous feasibility through a constant multiple of the square-root scale: for some absolute $c > 0$, with probability at least $1 - n^{-10}$ over $G \sim G(n, 1/2)$,

$$\text{every real } 9 \leq k \leq c\sqrt{n} \text{ admits a feasible degree-four clique pseudoexpectation.}$$

The construction uses the quartic correction of Hopkins et al. [12]. Relative to the $\tilde{\Omega}(\sqrt{n})$ scale in the cited degree-four works [12, 18], the issue is the logarithmic loss, not a change of polynomial exponent. The statement concerns this relaxation and includes its full constraints.

The proof separates the incidence space from its kernel and retains an exact positive square before estimating errors. Only critical matching and path configurations need log-free estimates; the others have enough polynomial slack. A final algebraic extension enforces the full moment constraints on one graph event. The critical estimates also have short independent proofs, so this application illustrates the selective use of sharp bounds rather than a logical need for every case of the classification.

1.4 Application: tensor-network thresholds and exact spectral edges

The second application asks how accurately a random tensor network preserves lengths and how large an eigenvalue its output state can have. We obtain necessary and sufficient deviation thresholds, entropy bounds with bounded additive error, and, under the connected-network hypotheses below, an exact limiting largest eigenvalue. Consider fixed Gaussian random tensor networks, with independent standard circular complex coordinates at each tensor, independent tensors, and a common edge dimension N . Every connected component meets a terminal; the nontrivial deviation statement assumes at least one tensor vertex. Internal edges, including parallel edges, are distinct indices, and open edges end at distinct degree-one terminals. Here normalization is especially consequential. If H_N is the contraction map, mean-Gram normalization chooses M_N so that $\mathbb{E}M_N^*M_N = I$. Sample-state normalization instead considers $\rho_A = H_N H_N^* / \|H_N\|_F^2$. These answer different questions: approximate isometry concerns the former, while output-state entropies concern the latter. Random tensor network methods and their spectral consequences have been developed in several settings [6, 8, 10, 11].

For equal edge dimension N , write b for the number of input legs. For a nonempty set T of tensor vertices, let $c(T)$ be its internal edge boundary and let $a(T), b(T)$ count its output and input legs. The relevant signed cut gap is

$$\Delta = \min_{\emptyset \neq T} (c(T) + a(T) - b(T)). \quad (13)$$

Write $F_N = M_N^* M_N - I_{N^b}$ for the mean-Gram deviation. All Schatten norms here are unnormalized. Theorem 9.3 gives, for each fixed $1 \leq t < \infty$ and $\Delta \geq 0$,

$$\mathbb{E}\|F_N\|_{S_t} \asymp_{\mathcal{G},t} N^{b/t-\Delta/2}, \quad \mathbb{E}\|F_N\|_{\text{op}} \asymp_{\mathcal{G}} N^{-\Delta/2} \quad (\Delta > 0). \quad (14)$$

For every sign of Δ , the convergence criteria are both necessary and sufficient:

$$\|F_N\|_{S_t} \xrightarrow{\mathbb{P}} 0 \iff \Delta > 2b/t, \quad \|F_N\|_{\text{op}} \xrightarrow{\mathbb{P}} 0 \iff \Delta > 0. \quad (15)$$

For the sample-normalized state, the relevant statistic is instead the ordinary minimum input-output edge cut s_{cut} . It is distinct from the signed gap Δ . Theorem 9.4 gives a constant $C_{\mathcal{G}}$ such that, with probability $1 - O_{\mathcal{G}}(N^{-1})$,

$$N^{-s_{\text{cut}}} \leq \|\rho_A\|_{\text{op}} \leq C_{\mathcal{G}} N^{-s_{\text{cut}}}, \quad s_{\text{cut}} \log N - \log C_{\mathcal{G}} \leq H_{\alpha}(\rho_A) \leq s_{\text{cut}} \log N \quad (0 \leq \alpha \leq \infty). \quad (16)$$

All entropy orders hold on the same event. Here H_{α} is the Rényi entropy, with the von Neumann entropy at $\alpha = 1$, log rank at $\alpha = 0$, and min-entropy at $\alpha = \infty$. The error is bounded additively, not just negligible relative to $\log N$. The theorem thus controls the largest eigenvalue and the entropy endpoints, rather than only a fixed moment or a limiting empirical spectral distribution. It uses concentration of the realized Frobenius normalization as well as a cut-rank bound and the uniform Gaussian norm estimate.

The scale bound leaves a finer question: does the largest eigenvalue actually approach the edge predicted by the limiting bulk spectrum? Fixed normalized moments alone cannot exclude a bounded number of outliers. We resolve this question for the connected, loopless, independent complex-Gaussian network with $A, B \neq \emptyset$ and common edge dimension N . Let μ_G be the compact minimum-energy moment law of Fitter et al. [8, Theorems 4 and 7], and let $E_G = \sup \text{supp } \mu_G$. Then Theorem 10.3 gives

$$\begin{aligned} N^{s_{\text{cut}}} \lambda_{\max}(\rho_A) &\xrightarrow{\mathbb{P}} E_G, \\ H_{\infty}(\rho_A) &= s_{\text{cut}} \log N - \log E_G + o_{\mathbb{P}}(1). \end{aligned} \tag{17}$$

The new information is the exact right endpoint, not a new claim to the already-known fixed-moment law. Even at the same minimum cut, the constant can change: a single square Gaussian gate has $s_{\text{cut}} = 1$ and $E_G = 4$, whereas an N^2 by N gate has $s_{\text{cut}} = 1$ and $E_G = 1$. The cut dimension alone cannot distinguish their min-entropy corrections. The proof uses a flow-quotient comparison with an independently constructed Gaussian circuit and the tensor-GUE strong-convergence theorem of Chen et al. [5]; it is not a direct substitution into the graph-matrix norm formula. The cut-gap, entropy, and exact-edge statements retain their respective normalizations and hypotheses. In particular, the narrower last result does not replace the common-event entropy bound.

1.5 Proof ideas: sharp scales, structured extensions, and exact edges

Two obstacles govern the proofs. At the scale level, separately maximizing local fluctuations can overcount the logarithm, while counting only leading moment configurations misses the accumulated contribution of defects. Separator layers resolve both issues and support the structured extensions. At the exact-edge level, even a sharp estimate with an unspecified constant loses the endpoint; a comparison at an auxiliary dimension retains it. We describe these mechanisms below. The SoS application combines the required scale estimates with the positive square and subspace budget described above.

Counting every defect, not only the leading states. In $\mathbb{E} \text{tr}((MM^*)^p)$, the same random input occurs in all $2p$ copies. For signs, a label assignment survives precisely when each primitive variable has even multiplicity. Record its exact equality partitions, and call the loss in free labels relative to the maximum its defect δ . Each lost label costs a factor n^{-1} , but there can be many ways to lose it. Counting only defect zero cannot control a moment whose order grows like $\log n$. The key conversion associates intervals to role partitions and reads a separator S_k at each of the $p - 1$ integer layers. The label deficit is exactly the accumulated excess cut size:

$$\delta = \sum_{k=1}^{p-1} (|S_k| - s). \tag{18}$$

A backbone of disjoint boundary paths controls the path partitions; an encoding of off-path components charges repairs to the same deficit. The path count is calibrated by evaluating its exact positive expansion at a dimension of order p^2 , where independent-matrix moment bounds apply; it does not require an explicit listing of states. The encoding keeps the original backbone as well as its modifications, so that reconstruction remains injective. Active components account for the choices that remain free at minimum layers. For a core with r roles, the uniform estimate is

$$c_{\alpha,p,\delta} \leq C_r^{2p} p^{a_* p + (3r^2 + 10r + 2)\delta}. \tag{19}$$

Here $c_{\alpha,p,\delta}$ counts legal exact states, C_r depends only on r , and the bound holds for every integer $p \geq 2$ and $\delta \geq 0$. The important features are the coefficient a_* on p and a defect cost independent of p . Summing the label-weighted counts gives a geometric tail with ratio $p^{3r^2+10r+2}/n$. At $p \asymp \log n$, taking a $2p$ -th root turns p^{a_*p} into the required $(\log n)^{a_*/2}$. Section 3 and Appendix D give the expansion, intervals, and lossless encoding. The layerwise refinement, rather than the scalar count alone, is the input to the tail-charge extension.

Making all lower-bound fluctuations large at one label. Fix a minimum separator. Conditional flattening exposes products of active-component fluctuations indexed by its labels. The norm must find one label where *all* components are large: multiplying their separate maxima would not prove this. Moreover, candidate labels reuse internal randomness. We first condition on all of that randomness, retain a sufficiently large conditional good set, and only then expose fresh crossing variables. A simultaneous exponential tilt supplies the matching logarithmic gain. Detached components are handled as scalars before taking core trace moments; the trace collision above explains why this preprocessing cannot be omitted. Sections 2 and 4 establish the transfers to the globally injective model and the synchronized lower bound. Together with the count, they prove the classification.

Why the cut calculus survives changes of model. The extensions retain the separator sequence while changing its layer weights. For independent factors, replace every scope by a clique in its two-section graph. Factor legality then implies graph legality, but a scope e contributes its tail weight precisely on layers with $e \subseteq S_k$. We therefore count states with the full sequence (S_k) still visible. Minimum layers contribute at most b_* each, while nonminimum layers are charged to separator excess. For the matching lower bound, conditional flattening keeps the shared internal arrays fixed and exposes a crossing occurrence for each active component, allowing one cut to realize both the component and tail charges.

Unequal dimensions replace cardinality by the weighted cut cost $\sum_{x \in S} \beta_x$, and weighted coarea again converts label loss into accumulated excess. An integer weight vector preserving the entire minimum face lets us replace each role by a fiber of clones. Minimum clone separators use whole fibers and retain the same active-component maximum, so the original count controls the weighted defects; a separate flattening gives the lower bound at the actual dimensions. Bounded roles require a finite reduction.

For Gaussian factors, a private-role lift represents each occurrence by a normalized sign sum. Positive trace comparison is uniform for $p = O(\log n)$, and Banach-space contraction gives the reverse norm bound, retaining the sharp scale and detached-scalar contribution. Polarization similarly compares each fixed-degree Hermite coordinate with a product of independent Gaussian occurrences in every L^q . These dimension-free reductions retain their fixed-template and fixed-degree scope. Sections 5 to 7 state the results, and Appendices F, I and J give the complete reductions.

Preserving an exact endpoint by changing the comparison dimension. For the RTN route, a maximum terminal flow defines a directed graph; its full strongly connected components form a balanced acyclic quotient Q . This is a comparison of Wick counts, not a claim that contracting original tensors produces independent Gaussian gates. Let $F_G(p, N)$ be the normalized moment sum over all defects. The comparison bounds it by $F_Q(p, p^2)/(1 - p^{K_G}/N)$, with fixed $K_G = 4v + 2 + 8R$ and $N > p^{K_G}$; here v counts tensor vertices and R counts internal edges and open legs. The reference network uses the auxiliary dimension p^2 , not the original dimension N .

Theorem 10.4 states the underlying comparison for weighted positive expansions: a single map with controlled target defect and weighted fibers gives a geometric denominator. The global SCC map is the substantive RTN input to that elementary inequality. The sign-path calibration uses positivity directly, rather than the same fiber map.

The quotient is realized by independent Ginibre gates on potentially overlapping tensor factors. Tensor-GUE strong convergence identifies its limiting norm; concentration and a tail estimate turn this into $F_Q(p, p^2) \leq 2(E_G + \eta)^p$ for fixed $\eta > 0$ and large p . Thus the comparison keeps the exponential base E_G , which a generic shape-dependent constant would lose. Logarithmic moments exclude fixed right outliers; concentrated fixed moments supply the matching lower edge. Section 10 states the comparison and explains the mechanism; Appendix M proves both edge bounds and the passage through the random Frobenius normalization.

1.6 Dependencies and reading guide

The proofs follow the distinction between a sharp scale and an exact edge. Sections 2 to 4 develop the model, separator coarea mechanism, and matching lower bound, with the separating family in Section 2.1. Sections 5 to 7 then explain how the analysis changes with factor laws and label-space geometry. The gated chain and columnwise-product examples illustrate the tail rule. Sections 8 and 9 combine spectral estimates with the additional algebraic and normalization arguments; Section 10 develops the separate constant-preserving comparison for the exact edge.

The detailed proofs are organized by this same dependency order in the appendices, starting with the theorem-by-theorem guide in Appendix A. They include complete counting and encoding, conditional probability estimates, the factor and dimension reductions, the SoS algebra and positivity budget, and both RTN proof chains. The columnwise-product example has an independent proof, and the Wishart and sparse calibrations are also included. Thus a reader can follow the main argument first and then check any theorem's complete derivation without consulting a supplementary repository.

2 Models and reductions

A shape is a finite simple graph $G = (W, E)$ together with ordered, internally distinct tuples U, V of vertices. A vertex can occur in both tuples. We identify a tuple with its underlying set when discussing connectivity. All such shapes are admissible in the sign theorem. An empty tuple indexes a one-dimensional coordinate space. No symmetrization or normalization is implicit.

Definition 2.1 (Typed sign model). In the *typed model*, each role $x \in W$ has a finite label set I_x of size m_x . Each edge $e = \{x, y\}$ has its own independent sign array $\xi^e : I_x \times I_y \rightarrow \{-1, 1\}$, and all its coordinates are independent. The matrix H_G sums over the labels of $W \setminus (U \cup V)$. Agreement on common boundary roles is imposed exactly as in Equation (4). Different roles do not share label coordinates, even if their label sets have the same cardinality. Balanced sizes mean $c_x n \leq m_x \leq C_x n$ for fixed positive constants.

A separator meets every U – V path, including paths of length zero. For a separator S , let $a_G(S)$ count the components K of $G - S$ such that $K \cap (U \cup V) = \emptyset$ and $N_G(K) \cap S \neq \emptyset$. Set $s_G = \min_{S: U|V} |S|$ and $a_G^* = \max_{|S|=s_G} a_G(S)$. These definitions apply to disconnected graphs and empty boundaries.

Lemma 2.2 (Elementary reductions). *The following operations preserve the asserted separator statistics and give the stated matrix factors.*

1. *An edge contained in U , or contained in V , can be removed in the sign model by a diagonal sign isometry.*
2. *If h isolated middle roles are deleted from a globally injective shape on v roles, the exact multiplier is $(n - (v - h))_h$.*
3. *In the typed model, connected components give a Kronecker product, up to row and column permutations. A detached component is a scalar factor, and an isolated middle role contributes m_x .*
4. *Deleting isolated middle roles and detached components changes neither s_G nor a_G^* .*

Proof. An edge on one boundary contributes the same sign to every entry in its corresponding row or column. Its diagonal multiplier squares to the identity. Such an edge cannot create a new boundary-to-boundary path after a separator is deleted: the suffix after its last boundary endpoint would already be such a path. It can merge only components that meet that boundary, and therefore cannot change the active count.

For an injection of the remaining $v - h$ roles, exactly $n - (v - h)$ labels remain. Ordered injective choices for the deleted roles give the falling factorial in the second assertion. In the typed model the sums and products separate over connected components, and their primitive arrays are independent. Finally, a minimum separator never uses a vertex of a component that meets neither boundary: deleting that vertex from the separator would still separate the boundaries. Such a detached component is not adjacent to a minimum separator. Isolated middle vertices have the same property. $\square$

Definition 2.3 (Boundary core). We call the graph remaining after the deletions in the fourth assertion the *boundary core*. Every component of the core meets a boundary, and it has no isolated middle role. An empty core represents the scalar 1. Isolated boundary roles are retained.

Lemma 2.4 (Global upper transfer). *Suppose a typed norm upper bound holds uniformly for $0 \leq m_x \leq n$. It transfers to the globally injective sign model with a constant depending only on the number of roles. For $q \geq 1$, the same is true of an L^q bound, with no additional q -dependent constant.*

Proof. For $v \geq 1$, color every ambient label independently and uniformly by a role of W . Let H_c be the zero-padded matrix of injections respecting the role colors. Each fixed injection survives with probability v^{-v} , so, pointwise in the ambient signs,

$$M_G = v^v \mathbb{E}_c H_c. \quad (20)$$

For fixed colors the role classes are disjoint. Different edge roles therefore use disjoint unordered ambient pairs, giving the typed independence assumptions. Convexity proves the expected-norm claim; Minkowski proves the L^q claim. A zero class gives the zero matrix. For $v = 0$, the matrix is the scalar 1. $\square$

For the lower transfer, fix balanced disjoint role classes rather than averaging over them. Boundary compression alone does not isolate the desired middle-role assignments. The following Fourier extraction does so.

Lemma 2.5 (Global lower transfer). *After isolated middle roles are removed, there is a positive shape-dependent constant c_G such that $\|M_G\|_{L^q} \geq c_G \|H_G\|_{L^q}$ for every $q \geq 1$, where H_G uses fixed balanced disjoint role classes.*

Proof. Compress to rows and columns in the prescribed boundary classes. For every target edge-color pair introduce an auxiliary sign and multiply all ambient signs in that pair of classes by it. Extract the Fourier coefficient containing every target edge-color sign. Fourier extraction is an average of signed matrices, and each auxiliary sign flip preserves the law of the ambient signs. Compression is contractive.

There are exactly $|E|$ edges in each monomial. To survive this Fourier coefficient, each of the $|E|$ target pairs must occur oddly and hence exactly once. The induced map from roles to role colors is therefore an edge-bijective endomorphism. Every nonisolated target role is in its image. Isolated boundary roles are fixed by compression, and there are no isolated middle roles. Thus the role map is surjective and hence bijective. It is a boundary-fixing automorphism.

Reindexing the middle labels shows that every such automorphism gives the same typed matrix. The extracted coefficient is consequently $|\text{Aut}_{U,V}(G)|H_G$. Apply Minkowski to the contractive average. The automorphism count is positive and independent of q . $\square$

These transfers do not assert independence of different connected components in the globally injective model. Component independence is used only in the typed model, before the transfers.

Proposition 2.6 (Finite evaluation of the exponents). *For every fixed admissible shape, the pair $((v + h - s_G)/2, a_G^*/2)$ is computable by a terminating finite procedure.*

Proof. Record isolated middle roles and detached components, then enumerate all subsets of the boundary core. Test separation by reachability, retain every subset of minimum size, and count active components in each. There are at most $2^{|W|}$ subsets and finitely many graph operations per subset. The formulas use only these finite data. The proof makes no claim of polynomial running time in the shape size. $\square$

The norm classification cannot be recovered from the full trace sequence alone. The collision example in Section B.1 supplies the complete construction and calculation; the obstruction is structural, not a loss introduced by truncating the moment sequence.

2.1 Equal coarse data, different logarithmic scales

The active-component statistic detects information not contained in the minimum separator size, or even in two standard scalar summaries of coefficient flattenings. The following family isolates that information without changing the ambient sign law or the matrix boundaries.

Definition 2.7 (Leaf-decorated chain). Fix $L \geq 1$ and $\mathbf{q} = (q_1, \dots, q_L) \in \mathbb{N}_0^L$. The shape $\alpha_{L,\mathbf{q}}$ consists of the path $x_0 - x_1 - \dots - x_L - x_{L+1}$, with boundaries $U = (x_0)$ and $V = (x_{L+1})$, and q_j distinct leaves attached to x_j . Write $Q = \sum_j q_j$, $q_{\max} = \max_j q_j$, $v = L + 2 + Q$, and $d = L + 1 + Q$. These parameters are fixed before the label dimension grows.

Proposition 2.8 (Exact chain statistics and sharp scale). *The shape $\alpha_{L,\mathbf{q}}$ has v vertices, d edges, no isolated middle role and no detached component. Its minimum separator size is one, and its maximum active count over minimum separators is $q_{\max}$. Consequently, in the globally injective unordered-edge sign model,*

$$\mathbb{E} \|M_{\alpha_{L,\mathbf{q}}}\| \asymp_{L,\mathbf{q}} n^{(L+1+Q)/2} (\log n)^{q_{\max}/2}. \quad (21)$$

The same scale holds in the balanced typed model.

Proof. The unique boundary-to-boundary path is the displayed main path, so its singletons are exactly the minimum separators. Deleting an internal x_j leaves two components meeting the respective boundaries and q_j isolated leaves adjacent to the cut. Only these leaves are active. Deleting an endpoint exposes no active component. Hence $s = 1$, $h = 0$, and $a_* = q_{\max}$. Substitute in Theorem 1.3; the balanced typed assertion is established by its upper and lower proofs in Sections 3 and 4. $\square$

To compare coefficient data, use the equal-size typed model. Its deterministic coefficient tensor $\mathcal{A}$ has one noise axis of size n^2 for each edge and two matrix axes of size n . Its entries enforce equality of repeated role labels. In the conventions of Bandeira et al. [2, Section 2.2], let $\sigma(\mathcal{A})$ be the largest flattening norm with the matrix axes on opposite sides, and let $\nu(\mathcal{A})$ be the largest with both matrix axes on the column side and a nonempty set of noise axes on the row side. We use ν for that reference's $v(\mathcal{A})$ to avoid a collision with the vertex count. Lemma C.1 computes both quantities.

Corollary 2.9 (Same coarse parameters, different sharp logarithms). *Fix $L \geq 2$ and $r \geq 1$. The concentrated decoration $(Lr, 0, \dots, 0)$ and the balanced decoration $(r, \dots, r)$ have the same $v, d, |U|, |V|, s, h$, and the same functions σ, ν for their equal-size typed coefficient tensors. Nevertheless, their globally injective expected norms have ratio*

$$\Theta_{L,r}((\log n)^{(L-1)r/2}). \quad (22)$$

For $L = 2, r = 1$, both trees have six vertices and five edges; their scales are $n^{5/2} \log n$ and $n^{5/2} \sqrt{\log n}$, respectively.

Proof. Both have $Q = Lr \geq 1$, and Lemma C.1 gives $\sigma = \nu = n^{(v-1)/2}$ in each case. All remaining equalities follow from the definition. The maximum decorations are Lr and r ; divide the two-sided estimates in Equation (21). $\square$

There is a concrete matrix interpretation of the difference. In the typed model, summing the leaf labels first gives the exact identity

$$H_{\alpha_L, \mathbf{q}} = A_0 \tilde{D}_1 A_1 \cdots \tilde{D}_L A_L, \quad \tilde{D}_j(i, i) = \prod_{h=1}^{q_j} \left(\sum_{a=1}^n \xi_{i,a}^{j,h} \right), \quad (23)$$

where the A_j and leaf arrays are mutually independent sign arrays. An empty product is one. The globally injective matrix does not have this independent product identity; Lemmas 2.4 and 2.5 are the transfers between the models.

The direct literature bounds and an adapted elementary bound make different uses of this structure. After removing the common polynomial factor $n^{(v-1)/2}$, Theorem 20 of Tulsiani and Wu [20], evaluated at logarithmic Schatten order, gives $(\log n)^v$. The smaller direct output of Theorems 2.4 and 2.5 of Bandeira et al. [2], as well as their graph-specific Theorem 4.8, gives $(\log n)^{d/2}$ when $Q \geq 1$. The complete substitutions are in Section C.2.

Even without those general bounds, multiplying the individual matrix and gate norms in Equation (23) gives

$$\mathbb{E} \|M_{\alpha_L, \mathbf{q}}\| \leq C_{L, \mathbf{q}} n^{(L+1+Q)/2} (\log n)^{Q/2}. \quad (24)$$

This bound is already sharp for concentrated decorations. For distributed decorations it pays $(Q - q_{\max})/2$ extra logarithmic powers: it charges the maximum at every gate separately. The separator formula charges leaves together at one cut, then maximizes over the competing cuts. Thus Q is replaced by $q_{\max}$, not by zero.

Appendix C proves the coefficient formula, all literature substitutions, and the elementary bound. The separation concerns the two specified scalar flattening summaries of the typed representation, not the complete flattening profile or the capabilities of every adapted use of earlier methods. The Gaussian chain in Example 5.3 will exhibit the same maximum-over-cuts rule through separator-supported tail charges rather than leaf sums.

3 Replica partitions and the upper bound

The trace method must retain two kinds of information at once: the number of free labels and the number of states realizing that freedom. Expand $\text{tr}((MM^*)^p)$ using exact equality partitions of the $2p$ replicas at each role. A surviving sign assignment contributes once, with a falling-factorial label count. It is not weighted by the number of Gaussian pairings that could refine its equality blocks. Appendix D includes a complete fourth-moment example illustrating this distinction.

The mechanism linking these states to graph structure is an interval. With τ the cyclic row matching, an even partition π receives

$$I(\pi) = [|\pi| - |\tau \vee \pi|, p - |\tau \vee \pi|]. \quad (25)$$

Neighboring roles have compatible intervals, so each integer layer defines a boundary separator. The coarea identity below equates the total label defect with the sum of the excess separator sizes. We prove this structural step here, explain how it pays for the encoding, and retain the final moment-to-norm conversion. The complete path count, off-path reconstruction, and seed-entropy calculation are in Appendix D.

3.1 Exact expansion and the counting theorem

Throughout this section G is a boundary core on r roles. Let $p \geq 2$, and index replicas by $\{0, \dots, 2p-1\}$. Define the matchings

$$\tau = \{\{1, 2\}, \{3, 4\}, \dots, \{2p-1, 0\}\}, \quad \eta = \{\{0, 1\}, \{2, 3\}, \dots, \{2p-2, 2p-1\}\}.$$

Their join is the one-block partition. A partition is *even* if all its blocks have even size. We use $\pi \wedge \psi$ for common refinement and $\pi \vee \psi$ for the equivalence relation generated by their union.

Definition 3.1 (Legal replica state). A legal state assigns an even partition π_x to each role, with τ refining π_x on U , η refining π_x on V , and $\pi_x \wedge \pi_y$ even for each edge xy . Let $|\pi|$ denote its number of blocks.

Lemma 3.2 (Exact positive expansion). *For the typed sign model,*

$$\mathbb{E} \text{tr}((H_G H_G^*)^p) = \sum_{\pi \text{ legal}} \prod_{x \in W} (m_x)_{|\pi_x|}. \quad (26)$$

Proof. Expand the trace into label assignments on the $2p$ replicas. Matrix multiplication imposes τ on row roles and η on column roles. At each role record the exact equality partition of its labels. For an edge, the primitive coordinate classes are exactly the cells of the meet of its endpoint partitions. Independence and symmetry make its expected product zero unless every such cell is even; if all are even the expectation is one. A nonisolated role then has an even partition as a union of even cells. An isolated core role is a boundary role and its trace constraint also forces evenness. Assigning distinct labels to the blocks gives the falling factorials. Each label assignment occurs once. $\square$

Definition 3.3 (Replica defect). Write $s = s_G$, $a_* = a_G^*$ and define the defect by

$$\sum_x |\pi_x| = p(r - s) + s - \delta. \quad (27)$$

Let $c_{G,p,\delta}$ count the states of this defect.

Theorem 3.4 (Uniform all-defect count). *For all integers $p \geq 2$ and $\delta \geq 0$,*

$$c_{G,p,\delta} \leq C_r^{2p} p^{a_* p + K_r \delta}, \quad K_r = 3r^2 + 10r + 2, \quad (28)$$

where one may take $C_r = 200^r (2r)^{2r^2} 4^{r^2}$ for $r \geq 1$. For $r = 0$, take $C_0 = 1$ and $K_0 = 0$.

3.2 Partition intervals

Definition 3.5 (Partition interval). For an even partition put

$$c(\pi) = |\tau \vee \pi|, \quad \ell(\pi) = |\pi| - c(\pi), \quad u(\pi) = p - c(\pi), \quad d(\pi) = p - |\pi|.$$

Thus $I(\pi) = [\ell(\pi), u(\pi)]$ is an interval in $[0, p - 1]$ of integer length $d(\pi)$.

Lemma 3.6 (Refinement and compatibility). *If θ refines π and both are even, then $I(\theta) \subseteq I(\pi)$. If $\pi \wedge \psi$ is even, their intervals intersect.*

Proof. Coarsening θ to π takes $|\theta| - |\pi|$ binary merges. Each merge decreases the number of blocks after joining with τ by zero or one. Consequently

$$0 \leq |\tau \vee \theta| - |\tau \vee \pi| \leq |\theta| - |\pi|.$$

Substitution gives both endpoint inequalities. If the meet is even, pair inside each of its cells to obtain a perfect matching ρ refining both partitions. Its interval is the singleton $\{p - |\tau \vee \rho|\}$, which is contained in both intervals. $\square$

Lemma 3.7 (Separator coarea identity). *For each legal state, the sets*

$$S_k = \{x : \ell(\pi_x) < k \leq u(\pi_x)\}, \quad 1 \leq k \leq p - 1,$$

are separators, and

$$\delta = \sum_{k=1}^{p-1} (|S_k| - s) \geq 0. \quad (29)$$

Proof. On U we have $\ell = 0$, and on V we have $u = p - 1$. A vertex outside S_k is either below the layer, with $u < k$, or above it, with $\ell \geq k$. Compatible intervals intersect, so no edge joins the below class to the above class. A surviving row boundary vertex is below, and a surviving column boundary vertex is above. Common boundary vertices have the full partition and belong to every layer. This proves separation, including length-zero paths. Each vertex belongs to exactly $d(\pi_x)$ layers. Summing and using Equation (27) gives the identity. $\square$

3.3 How coarea controls the encoding

The identity above converts a loss of labels into a budget for larger separator layers. To use that budget, choose vertex-disjoint paths between the boundaries and first count the partitions on this backbone. A path has no active component. Its count has only a constant raised to the moment order, together with a polynomial cost for each lost label; Lemma D.2 proves this by comparison with a product of independent sign matrices.

The reusable step is positive evaluation at a chosen dimension: each exact state in one defect layer contributes a known minimum weight, so a bound on the total moment bounds the number of states. Lemma D.1 isolates that step. Its positivity concerns exact state weights, not coefficients after expanding falling factorials into powers of the dimension.

The remaining components are encoded by a seed matching and local changes from that seed. Boundary-touching components have prescribed seeds. A boundary-free component can have a free seed only across layers in which it is active. Coarea therefore charges its free pairing choices to the active-component statistic, while every repair is charged to the defect. The crucial counting direction is to bound the range of possible repaired backbones and then the fibers over each one. One must not assume that a repair is independent of the seed, or reconstruct the original backbone from its repair alone.

Appendix D gives the path estimate, the deterministic record and decoder, the seed-entropy bound, and the complete exponent budget. Together these prove Theorem 3.4; the conversion of that count into an operator bound is short enough to retain here.

3.4 From counting to a norm bound

Proposition 3.8 (Core upper bound). *For fixed $C_0 > 0$, uniformly for $m_x \leq n$ and $2 \leq q \leq C_0 \log(2n)$,*

$$\| \| H_G \| \|_{L^q} \leq C_{G,C_0} n^{(r-s)/2} (\log(2n))^{a_*/2}$$

for sufficiently large n .

Proof. The exact expansion and Theorem 3.4 give

$$\mathbb{E} \operatorname{tr}((H_G H_G^*)^p) \leq C_r^{2p} n^{p(r-s)+s} p^{a_* p} \sum_{\delta \geq 0} (p^{K_r}/n)^\delta. \quad (30)$$

The exact sum is finite; extending it is an upper bound. For any fixed logarithmic window the ratio is at most $1/2$ eventually. Choose $p = \max(2, \lceil \log(2n) \rceil, \lceil q/2 \rceil)$. The $2p$ -th root bounds the L^q norm and leaves a dimension factor $n^{s/(2p)} \leq e^{s/2}$. This proves the assertion. $\square$

For a nonisolated detached connected typed component K on k roles, orthogonality of distinct edge monomials gives $\mathbb{E} Z_K^2 = \prod_{x \in K} m_x$, so $\mathbb{E} |Z_K| \leq n^{k/2}$. These expectation factors multiply independently of the core. Isolated middle roles contribute at most n^h . Consequently the full typed expected norm is at most $C_G n^{(v+h-s)/2} (\log(2n))^{a_*/2}$. Lemma 2.4 proves the upper half of Theorem 1.3. Detached factors are restored here in expectation, not inside the large core trace moment.

4 The matching lower bound

4.1 Proof overview: synchronized separator fluctuations

The polynomial exponent has a dimension-versus-rank interpretation. The graph supplies summation variables, while a separator limits the number of labels that must be shared by the row and column

sides. The resulting scale is $n^{(v+h-s)/2}$. This interpretation alone does not explain the logarithm: two graphs with the same vertex count and separator size can have different norm scales.

Fix a minimum separator S . After conditioning on labels in S , an active component K produces a scalar fluctuation $T_K(x_S)$. Several active components produce a product

$$w(x_S) = \prod_{K \text{ active}} T_K(x_S).$$

The operator norm can select a favorable separator label. A single component has a square-root logarithmic extreme-value gain; $a(S)$ components can produce $a(S)$ such gains at the *same* separator label. This synchronization is the lower-bound mechanism. The proof cannot replace a maximum of a product by a product of separate maxima: those separate maxima might occur at incompatible labels.

There is another dependence issue. Candidate separator labels reuse the internal randomness of each active component. The lower bound therefore conditions on *all* internal random variables before selecting a family of fresh trials. On a sufficiently large conditional good set, the remaining crossing variables yield independent trials with useful success probabilities. An exponential tilt then makes all active components simultaneously large. The order of these operations is part of the proof, not a matter of presentation.

For the upper bound, the same logarithm appears as entropy in a trace expansion. Near-leading replica configurations possess a limited number of pairing choices that do not cost a free label. The active-component statistic counts precisely this residual freedom. Configurations with fewer labels incur a factor $n^{-\delta}$, where δ is their defect. The crucial estimate makes their additional entropy only polynomial in the moment order, with an exponent linear in δ . At a moment order of order $\log n$, the label loss then dominates the entropy increase.

We first work with balanced typed classes and remove isolated middle roles. Fix a minimum separator S . Let D be the union of its active components, and write $C = W \setminus D$. The proof first exposes a maximum over separator labels, then produces a simultaneous large value of all active components.

4.2 Conditional flattening

Lemma 4.1 (A lower stacking inequality). *Let A_i be deterministic rectangular matrices and ε_i independent uniform signs. Then*

$$\mathbb{E} \left\| \sum_i \varepsilon_i A_i \right\| \geq 3^{-1/2} \max \left\{ \left\| \sum_i A_i A_i^* \right\|^{1/2}, \left\| \sum_i A_i^* A_i \right\|^{1/2} \right\}.$$

For a decoupled degree- q matrix chaos with coefficient tensor A , every flattening that keeps its original row and column indices on opposite sides satisfies

$$\mathbb{E} \|X_A\| \geq 3^{-q/2} \|A_{[R|C]}\|. \quad (31)$$

The assertion also holds conditionally if A is measurable with respect to a sigma-field independent of the integrated sign groups.

Proof. For Hilbert-space vectors v_i , let $Z = \|\sum_i \varepsilon_i v_i\|$ and $a^2 = \sum_i \|v_i\|^2$. Expansion gives $\mathbb{E} Z^2 = a^2$ and $\mathbb{E} Z^4 \leq 3a^4$. Interpolation yields $\mathbb{E} Z \geq (\mathbb{E} Z^2)^{3/2} / (\mathbb{E} Z^4)^{1/2} \geq a/\sqrt{3}$. Apply this to the transpose of the random matrix acting on a top eigenvector of $\sum_i A_i A_i^*$, and then to the transposed problem. For the second claim, integrate one independent noise group at a time. Choose vertical or horizontal stacking according to the side of the prescribed flattening receiving that group. Each step costs at most $\sqrt{3}$. Conditioning freezes the other coefficient data and leaves exactly the same argument. $\square$

Condition on all arrays with an endpoint in D . Summing each active component produces a weight $T_K(x_S)$, and their product is $w(x_S)$. All remaining edge arrays are independent of these weights.

Lemma 4.2 (Exact separator flattening). *For q_0 remaining edge types,*

$$\mathbb{E}_{\text{fresh}} \|H_G\| \geq 3^{-q_0/2} \left(\prod_{x \in C \setminus S} m_x \right)^{1/2} \max_{x_S} |w(x_S)|. \quad (32)$$

Proof. Assign the components of $G - S$ meeting U to the row side and those meeting V to the column side. A component cannot meet both. Assign any detached component to the row side. The active components have already been removed into w . Put fresh edges within or incident to a row-side component on the row side, and do the analogous operation on the column side. Put edges contained in S on the row side. Keep the original matrix indices on their respective sides.

Let R_0, C_0 be the sets of roles observed by the resulting primitive coordinates. Then $R_0 \cup C_0 = C$ and $R_0 \cap C_0 = S$. For the union, no isolated middle role remains, and deleting D cannot isolate a role outside S . For the reverse inclusion in the intersection, use inclusion-minimality of S : for every $z \in S$, there is a boundary-to-boundary path meeting S only at z . This path observes z on both sides, through an edge or through an original boundary coordinate. It cannot enter an active component and then reach a boundary without returning to z or meeting another separator vertex. A common boundary role is observed on both sides directly.

The coefficient tensor has value $w(x_S)$ when repeated role coordinates agree, and zero otherwise. The assignment is recoverable from these coordinates, so there is no additional summation multiplicity. Delete consistency-zero rows and columns and permute the rest. Its flattening is exactly

$$\bigoplus_{x_S} w(x_S) \mathbf{1}_{M_L} \mathbf{1}_{M_R}^*, \quad M_L = \prod_{x \in R_0 \setminus S} m_x, \quad M_R = \prod_{x \in C_0 \setminus S} m_x.$$

Its norm is $\sqrt{M_L M_R} \max |w|$. Apply Lemma 4.1 conditionally to the fresh arrays. $\square$

4.3 From flattening to simultaneous witnesses

The flattening reduces the lower bound to making all active-component scalars large at one separator tuple. For an active component, condition on its entire internal array. Second and fourth moments give a positive-measure set of internal realizations on which the remaining crossing sum has the correct variance. Higher fixed moments rule out a single dominant coefficient. A weighted exponential tilt then gives a polynomially small, rather than exponentially small, chance of reaching the square-root logarithmic scale.

Choose separator tuples that are distinct in every separator coordinate. Conditionally, their crossing arrays are disjoint; the component events are also independent across components. Taking the tilt constant small enough makes the combined success exponent smaller than the number-of-trials exponent. Thus at least one tuple makes all factors large together. Appendix E proves the conditional good-set estimates, the tilt inequality, and this synchronization, with their quantifiers and failure probabilities. No product of separate maxima is used.

Completion of Theorem 1.3. Insert Equation (69) into Equation (32). If r roles remain after deleting isolated middle roles, the power of n is $(r - |D| - s)/2 + |D|/2 = (r - s)/2$. Choose a minimum separator maximizing $a(S)$. This proves the balanced typed lower bound, including detached components among the fresh arrays. Apply Lemma 2.5, then restore $(n - (v - h))_h \asymp n^h$. The

exponent becomes $h + (v - h - s)/2 = (v + h - s)/2$. If there are no residual roles, the matrix is the scalar $(n)_h$. Edgeless residual shapes consist of all-ones factors and common-coordinate identity factors, and the same formula is immediate. Together with the upper bound in Section 3, this proves the theorem. $\square$

5 Sharp norms for independent factor chaoses

Replacing graph edges by nonempty factor scopes puts tensors, repeated independent interactions, and random diagonal gates in one incidence model. The new logarithmic contribution comes from factors supported entirely on a separator: their extremes add to the fluctuations of active components outside the cut.

Three routes govern the distributions. Fixed symmetric tail-regular laws admit sharp bounds with explicit charges. Bounded centered laws, without symmetry or identical distributions, inherit the sign scale by Section 5.5. A finite moment profile gives dimension-dependent laws an upper bound with a weighted cut cost. Each route requires independent occurrences, even for equal scopes. The chain and columnwise-product examples explain the charges; the sparse example delineates the fixed-law conclusion.

5.1 The factor model and its exact preprocessing

Definition 5.1 (Independent-factor chaos). An independent-factor shape consists of a finite role set W , boundaries U, V , and a finite list $\mathcal{F}$ of nonempty scopes. Equal scopes can occur repeatedly; each occurrence has its own independent array, with independent identically distributed coordinates. Define

$$H = \sum_{x \in \prod_{v \in W} I_v} \prod_{e \in \mathcal{F}} \xi_{x_e}^{(e)} e_{x_U} e_{x_V}^*. \quad (33)$$

This is a typed Cartesian sum. No global injectivity restriction or shared-array identification is implicit. The two-section graph joins two distinct roles when they occur in one factor. Unary factors mark a random singleton without adding an edge.

Remove unused middle roles and then every factor-connected component disjoint from both boundaries, including unary-only singletons. If their scalar sums are $Z_1, \dots, Z_J$, then

$$H = d(n) \prod_{j=1}^J Z_j H_c, \quad d(n) = \prod_{\text{unused middle } v} m_v. \quad (34)$$

All random factors in this identity are independent. The core keeps every boundary coordinate, including unused common-boundary identity factors.

Write r for the number of roles in H_c , and let s be the minimum separator size in its two-section graph. For such a separator, $a(S)$ counts attached components meeting neither surviving boundary, as in the graph model. These are the core statistics used below.

5.2 Tail-regular factors

Assume $m_v \asymp n$ and the fixed symmetric variance-one laws satisfy $\|\xi_e\|_q \asymp_e q^{\theta_e/2}$ for every $q \geq 2$, with $0 \leq \theta_e \leq 2$. For a core separator define $\Theta(S) = \sum_{e \subseteq S} \theta_e$ and $b_* = \max_{|S|=s} (a(S) + \Theta(S))$.

Theorem 5.2 (Fixed-law factor theorem). *With v the original role count and h the unused middle count,*

$$\mathbb{E} \|H\| \asymp n^{(v+h-s)/2} (\log n)^{b_*/2}.$$

For every fixed $A > 0$, the core satisfies

$$\mathbb{P}\{\|H_c\| > C_A n^{(r-s)/2} (\log n)^{b_*/2}\} \leq n^{-A}$$

for all sufficiently large n .

The proof is in Appendix F. Its counting input is a layerwise strengthening of Theorem 3.4, not an assumption that a factor meet is equivalent to all its pairwise meets. Only the forward implication is used for the upper bound. The lower bound chooses one crossing occurrence for each active component and conditions on the complete shared internal arrays.

Boundary factors with nonconstant absolute values are retained. For example, an independent Gaussian factor supported on a separator has $\theta = 1$ and supplies a square-root logarithmic charge. A sign factor there has $\theta = 0$. Removing both as diagonal isometries would therefore give the wrong theorem.

Example 5.3 (A chain of independent Gaussian gates). Consider $n \times n$ matrices $G_0, \dots, G_L$ with iid standard real Gaussian entries and diagonal gates $D_j = \text{diag}(\prod_{h=1}^{q_j} z_i^{(j,h)})$, where all matrix entries and all $z_i^{(j,h)}$ are mutually independent standard real Gaussians. Here $L \geq 1$ and $q_j \geq 0$ are fixed integers. The path representation of $G_0 D_1 G_1 \cdots D_L G_L$ has singleton minimum separators, no active components after deleting them, and unary charge q_j at the j th internal role. Thus

$$\mathbb{E} \|G_0 D_1 G_1 \cdots D_L G_L\| \asymp n^{(L+1)/2} (\log n)^{\max_j q_j/2}. \quad (35)$$

This example illustrates how separator-local tails, rather than the sum of all gate degrees, determine the logarithm.

The contrasting columnwise product in Section 5.7 places all gates at one role, where their charges do add. The sign leaf family in Section 2.1 gives a complementary realization: there leaf sums create the gates, whereas here unary Gaussian occurrences carry the charges directly. Both optimize over the same competing cuts. No equality in distribution or growing-order central-limit replacement between the two models is used.

5.3 Dimension-dependent laws and moment-weighted cuts

A distinct result accommodates a finite moment profile for laws that may depend on n . Retain comparable role sizes $m_v \asymp n$ and real symmetric variance-one coordinates, independent by coordinate and occurrence as in Equation (33). Suppose, for fixed $C_0 > 0$, fixed nonnegative β_e, γ_e , and fixed constants C_e independent of n ,

$$\mathbb{E} |\xi_e|^{2t} \leq C_e^{2t} n^{\beta_e(t-1)} t^{\gamma_e(t-1)}, \quad 1 \leq t \leq C_0 \log n,$$

for every integer t in the displayed window and all sufficiently large n . Put $\kappa(S) = |S| - \sum_{e \subseteq S} \beta_e$ and $b(S) = a(S) + \sum_{e \subseteq S} \gamma_e$.

Proposition 5.4 (Core profile bound). *For every integer $2 \leq p \leq C_0 \log n$,*

$$\mathbb{E} \operatorname{tr}((H_c H_c^*)^p) \leq C^{2p} \sum_{(S_k)} n^{pr - \sum_k \kappa(S_k)} p^{\sum_k b(S_k) + K'_r \sum_k (|S_k| - s)}, \quad K'_r = 3r^2 + 11r + 2, \quad (36)$$

where the sum ranges over separator sequences of length $p-1$. If every minimizer of κ has cardinality s , then, with $\kappa_* = \min \kappa$ and $b_*^{\text{mw}} = \max_{\kappa(S)=\kappa_*} b(S)$,

$$\mathbb{E} \|H_c\| \leq C n^{(r-\kappa_*)/2} (\log n)^{b_*^{\text{mw}}/2}.$$

A corresponding n^{-A} upper tail holds for every fixed $A > 0$ by changing the threshold constant within the same moment window.

For the full matrix one must multiply the trace bound by $d(n)^{2p} \prod_j \mathbb{E}|Z_j|^{2p}$. The profile hypothesis alone does not supply sharp L^1 estimates for arbitrary dimension-dependent detached factors. No matching sparse-law lower bound is asserted.

5.4 Why the structural rule survives the change of law

Two inputs replace sign parity. First, occurrence-wise independence keeps the trace expansion positive for symmetric laws. Each surviving factor cell has a moment cost, and the layerwise count charges that cost only to separators containing its whole scope. This gives the tail-weighted cut statistic instead of a bound involving the largest scalar moment everywhere. Second, the lower bound separates the internal component fluctuations from the factors supported entirely on the separator. The former use a conditional moderate deviation; the latter use the assumed lower moment growth.

The distinction between expectation and high moments is also essential. Detached components contribute their first absolute moments to the expected norm, but each can add a square-root moment factor to a tail bound. They must be restored after the core estimate, not at the trace order chosen to detect its largest singular value. Appendix F supplies the complete detached-component count, fixed-law moment inequalities, layerwise refinement, weighted trace sum, and conditional lower argument. It also proves the profile bound without assuming that a dimension-dependent law is a fixed law.

5.5 Sharp factor norms for bounded noise without symmetry

Symmetry is needed for the positive replica expansion, but not for every distributional extension of its conclusion. For bounded noise, a coefficient-level comparison transfers the sign estimates before any shape reduction. This also permits different laws at different coordinates. The structural conclusion is stated first: the sign-reference exponents continue to govern the expected norm and logarithmic-window moments. We then prove the dimension-free comparison that transfers them. Its method is standard symmetrization and contraction; the sharp reference scale is the input supplied by the preceding theory.

Corollary 5.5 (Sharp bounded-noise factor norms). *Fix a typed independent-factor shape with scopes of arity at least two and comparable role sizes. Replace its sign coordinates by mutually independent centered variance-one coordinates bounded by a common K , allowing their laws to vary with the coordinate and with n . Compute f, g, d_0 in the sign reference: remove boundary-contained sign factors, remove unused middle roles, split detached random components, and use*

the minimum-separator statistics of the resulting core. Here d_0 counts those detached random components. Then

$$\mathbb{E} \|H_X\| \asymp n^f (\log(2n))^g.$$

For every fixed $C_0 > 0$ and $1 \leq p \leq C_0 \log(2n)$,

$$\| \|H_X\| \|_{L^p} \asymp n^f (\log(2n))^g p^{d_0/2}. \quad (37)$$

The implicit constants depend only on the fixed shape, K , the role-size comparison constants, and C_0 . For each fixed $A > 0$,

$$\mathbb{P}\{\|H_X\| > C_A n^f (\log(2n))^{g+d_0/2}\} \leq n^{-A}$$

for all sufficiently large n .

Proof. Use rectangular matrices with operator norm as B , and apply Theorem 5.6 to the coefficient tensor of the original factor network. The number Q counts occurrences *before* any boundary reduction. The sign norm and moment estimates proved above then give the expectation and Equation (37). Markov's inequality at $p = \log(2n)$ gives the tail after enlarging C_A . The bounded-noise matrix itself is never simplified by deleting boundary factors: these factors need not be isometries and may vanish. $\square$

The comparison used in this proof applies to the same coefficient tensor under two laws. Stating it in a Banach space makes clear that its constants do not depend on the number of matrix rows, columns, or primitive variables.

Theorem 5.6 (Dimension-free bounded-noise comparison). *Let B be a real Banach space, let $Q \geq 0$ be an integer, and let $J_1, \dots, J_Q$ be finite sets. For deterministic B -valued coefficients set*

$$F(X) = \sum_{i_t \in J_t} a_{i_1, \dots, i_Q} \prod_{t=1}^Q X_{i_t}^{(t)}.$$

Assume all primitive coordinates are mutually independent and satisfy $\mathbb{E} X_i^{(t)} = 0$, $\mathbb{E} (X_i^{(t)})^2 = 1$, and $|X_i^{(t)}| \leq K$ almost surely, where $K \geq 1$ is common to all coordinates. No symmetry or identical-law assumption is imposed. If $F(\varepsilon)$ uses independent uniform signs and the same coefficient tensor, then, for every $1 \leq p < \infty$,

$$(2K)^{-Q} \|F(\varepsilon)\|_{L^p} \leq \|F(X)\|_{L^p} \leq (2K)^Q \|F(\varepsilon)\|_{L^p}. \quad (38)$$

The constants are independent of p , the index-set sizes, and B . For $Q = 0$ the comparison is equality.

The complete proof is given in Appendix G (page 53).

For example, if $Z \sim \text{Bernoulli}(\vartheta)$ and $\eta \leq \vartheta \leq 1 - \eta$ for fixed $0 < \eta \leq 1/2$, then $(Z - \vartheta)/\sqrt{\vartheta(1 - \vartheta)}$ satisfies the hypothesis with $K = \sqrt{(1 - \eta)/\eta}$. The parameters may vary by coordinate and dimension within this interval. Independent erasures $\varepsilon Z/\sqrt{\vartheta}$ with $\vartheta \geq \eta > 0$ also qualify.

This comparison is a transfer between laws for the *same* coefficient tensor, not a classification of arbitrary tensors by incidence. It requires separate linearity and independent groups; it neither replaces X_i^2 by an independent product nor transfers the theorem automatically to a globally shared unordered-edge array. If the density tends to zero, K can diverge and the constants in Equation (38) can change polynomial exponents.

5.6 A sparse boundary case: the complete iid phase diagram

The last restriction has a concrete consequence even for a single matrix. The following phase diagram concerns one specified iid family, not all sparse factor networks. It is an elementary consequence of Seginer's row/column norm comparison and binomial occupancy estimates. The critical-density obstruction already appears in the introduction of Seginer [19]; we include it to delineate the fixed-law theory, not as a new sparse-matrix phenomenon.

Proposition 5.7 (Sparse signed Bernoulli phase diagram). *Let X be $n \times n$ with independent entries $X_{ij} = \rho^{-1/2} \varepsilon_{ij} B_{ij}$, where the signs and $B_{ij} \sim \text{Bernoulli}(\rho)$ are mutually independent. For fixed $\beta \geq 0$ put $\rho = n^{-\beta}$. As $n \rightarrow \infty$,*

$$\mathbb{E} \|X\| \asymp_{\beta} \begin{cases} n^{1/2}, & 0 \leq \beta < 1, \\ n^{1/2} (\log n / \log \log n)^{1/2}, & \beta = 1, \\ n^{\beta/2}, & 1 < \beta \leq 2, \\ n^{2-\beta/2}, & \beta > 2. \end{cases} \quad (39)$$

The full proof, including expectation tails in every regime, is in Appendix O. At $\beta = 1$, no fixed pair f, g represents this scale as $\Theta(n^f (\log n)^g)$: successive logarithmic comparisons force $f = g = 1/2$, leaving a ratio $(\log \log n)^{-1/2}$ tending to zero. For $\beta > 2$, the matrix is zero with probability tending to one; the expectation instead comes from a rare occupied entry. Neither behavior contradicts Theorem 5.2, whose laws are fixed independently of n . Nor does this example establish that its positive class is maximal. Here $\mathbb{E} X_{ij}^4 = \rho^{-1}$ already determines the density, so occupancy is not additional information missing from the full permitted moment data.

5.7 Worked example: a gated columnwise tensor product

The chain in Equation (35) and the columnwise product below isolate two different geometries of the same tail rule. Along the chain, different gates sit at different singleton cuts, so the largest gate degree determines the logarithm. In the columnwise product, all gates share one column role and their charges add at that cut. The following independent proof makes this distinction visible directly in the matrix, through a selected column and a positive-semidefinite Schur product.

Theorem 5.8 (Gated Khatri–Rao product). *Fix integers $r, q \geq 1$. Let $A_1, \dots, A_r$ be independent $n \times n$ real standard Gaussian matrices. Their columnwise tensor product $K \in \mathbb{R}^{n^r \times n}$ is defined by*

$$K[:, j] = A_1[:, j] \otimes \cdots \otimes A_r[:, j].$$

Let D be diagonal with $D_{jj} = \prod_{h=1}^q g_{jh}$, where all g_{jh} are independent standard real Gaussians independent of the matrices. Then

$$\mathbb{E} \|KD\| \asymp_{r,q} n^{r/2} (\log(2n))^{q/2}. \quad (40)$$

The complete proof is given in Appendix H (page 53).

In the factor representation, the column role is the center of a star, the r row roles are leaves, and the center carries q unary Gaussian gates. For $r \geq 2$ the center is the unique minimum separator. For $r = 1$ both endpoints are minimum separators, with charges q at the gated endpoint and zero at the other. Maximizing over the full minimum-separator family gives the logarithmic exponent $q/2$ in both cases. Thus the maximum over cuts in the chain and the sum of charges at one cut in the star are two manifestations of the same separator-supported tail statistic. The proof in Appendix H establishes this example independently of the general counting argument.

6 Local weights and polynomial aspect ratios

The next extension changes the geometry of the label spaces and the deterministic amplitudes. It is useful to separate these two changes. A role with n^{β_v} labels contributes a cost β_v to a cut; a local amplitude of size n^{γ_e} contributes its exponent to the entire matrix. Thus the polynomial scale is governed by a weighted separator, but its cost alone still does not determine the logarithm. All minimizing separators, including ties, must be retained.

The weight restriction is local: each amplitude sees only the labels of its own factor, and its magnitude has fixed two-sided bounds. This permits an all-moment contraction argument. It does not permit an arbitrary coefficient tensor on the full set of roles. We prove the positive-aspect result first and then eliminate bounded roles by a finite Fourier argument. This last step is needed for a closed statement at nonnegative, rather than strictly positive, aspects.

This is a sign-model extension, separate from the tail-law theorem in Section 5. We do not silently combine every hypothesis of the two theorems into a single larger model.

6.1 Weighted minimum cuts and the sharp formula

Definition 6.1 (Locally weighted sign model). A separate sign-model extension, with all original factor scopes of arity at least two and nonempty role sets, allows $m_v \asymp n^{\beta_v}$ with $\beta_v \geq 0$ and deterministic real local amplitudes, with fixed $\gamma_e \in \mathbb{R}$ and fixed constants $0 < c_e \leq C_e < \infty$, satisfying $c_e n^{\gamma_e} \leq |a_{e,n}(x_e)| \leq C_e n^{\gamma_e}$. The amplitudes depend only on their own factor coordinates. Write $\Gamma = \sum_e \gamma_e$.

Definition 6.2 (Weighted core and minimum-face statistics). Restrict roles and boundaries to $P = \{v : \beta_v > 0\}$ and restrict each scope to P . Discard empty scopes and scopes contained in one boundary, retaining their exponents in Γ . Unary factors on a middle role are absorbed into an incident higher-arity sign array when one exists. A graph-isolated middle role with unary factors is one detached random scalar; without any factor it is deterministic. Let $B_+ = \sum_{v \in P} \beta_v$, let h_β be the sum of exponents of deterministic isolated roles, and let d_0 be the number of detached random components. In the remaining core set

$$\kappa = \min_{S: U|V} \sum_{v \in S} \beta_v, \quad a_* = \max_{\beta(S)=\kappa} a(S).$$

Theorem 6.3 (Locally weighted aspect theorem). *For the fixed typed independent-sign model just specified,*

$$\mathbb{E} \|H\| \asymp n^f (\log(2n))^{a_*/2}, \quad f = \Gamma + \frac{B_+ + h_\beta - \kappa}{2}.$$

For fixed $C_0 > 0$ and $2 \leq q \leq C_0 \log(2n)$,

$$\| \|H\| \|_{L^q} \leq C n^f (\log(2n))^{a_*/2} q^{d_0/2}.$$

Consequently the upper-tail logarithmic exponent is $(a_* + d_0)/2$. For rational input exponents the displayed statistics have a terminating exact evaluation procedure.

The proof in Appendix I establishes each reduction. In particular, bounded roles are eliminated by a finite Fourier projection, not by selecting a summand and assuming that its norm is a lower bound. For positive aspects, an integer surrogate preserves all minimum-cost separators, and a clique expansion transfers the counting theorem. Constants can depend on this expansion; no uniformity near changing minimum faces is claimed. For arbitrary real input exponents, effective evaluation requires an exact comparison representation.

6.2 Why the whole minimum face matters

Two features distinguish this result from substituting dimensions into the equal-size formula. The active count must be maximized over the *entire* minimum-cost face, including ties. A rational surrogate and true-twin expansion preserve that face in the counting proof. Also, $\beta_x = 0$ is covered by a finite Fourier reduction, not by a limiting argument that still needs polynomially many fresh labels. Local amplitudes are removed by all-moment contraction; the two-sided local hypothesis is what permits this step.

Example 6.4 (A change of minimum-separator face). Consider the path $u - x - v$ with a leaf attached at x , with row boundary u and column boundary v . Give u, v , and the leaf exponent one, and give x exponent $1 + \varepsilon$, where $\varepsilon > -1$ is fixed. The only singleton cuts are the two endpoints and x . If $\varepsilon > 0$, the endpoint cuts minimize cost and expose no active component. If $\varepsilon = 0$, all three cuts minimize cost, and the middle cut exposes the leaf. If $\varepsilon < 0$, the middle cut is the unique minimum. The logarithmic exponent is consequently zero in the first regime and one half in the other two.

This change is not a contradiction with continuity of a matrix at a fixed dimension. The aspect vector is fixed before n tends to infinity. For fixed positive ε , the extra cost $n^{-\varepsilon/2}$ eventually outweighs the logarithmic gain. The theorem is therefore pointwise in the aspect vector, not uniform across a changing minimum face. The minimum-face reduction preserves all ties instead of selecting one convenient minimum cut. The proof has three logically distinct reductions. Local contraction removes the amplitudes in every moment norm with constants independent of the moment order. For positive aspects, an integer surrogate preserves the entire minimum-separator face, and cloning roles converts its weighted defect into the ordinary counting problem. Finally, bounded roles are eliminated by Fourier projection and unary signs are absorbed by an equality in distribution. These last operations cannot be replaced by keeping an arbitrary summand of the matrix. The complete argument, including the unequal-dimension conditional estimate and the finite evaluation procedure, is in Appendix I.

6.3 Scope of the coefficient restriction

The local-weight hypothesis is substantive. Let $Z_n = \sum_{k,l} \varepsilon_{kl}$ and $H_n = Z_n F_n$, where F_n is a deterministic n by n sign matrix. Then $\mathbb{E}|Z_n| \asymp n$. For $F_n = J_n$, $\mathbb{E}\|H_n\| \asymp n^2$. For an n by n compression of a Sylvester Hadamard matrix of least order $N \geq n$, one has $\sqrt{n} \leq \|F_n\| \leq \sqrt{2n}$, and hence $\mathbb{E}\|H_n\| \asymp n^{3/2}$. The entry magnitudes and random incidence are identical in these two constructions.

Choosing the first coefficient matrix for even n and the second for odd n gives no fixed power-log representation at all. This does not rule out richer coefficient-sensitive invariants. It does show why a structural factor theorem cannot be restated as a theorem for arbitrary global weights.

7 Gaussian and Hermite factor chaoses

This section supplies the uniform Gaussian input used later for the operator-norm and sample-state endpoints in Section 9. It concerns coefficient-one incidence sums with independent occurrence arrays and comparable role sizes. Zero-arity Gaussian factors are excluded; they can instead be multiplied separately with their own scalar moments. Unary factors are allowed.

The Gaussian case also belongs to the fixed-law factor theory, but a private-role lift gives a useful additional conclusion: it transfers the full logarithmic-window moment statement, including the distinction between core and detached fluctuations. It provides the precise uniform input needed

for sharp operator-norm scales; fixed-order Gaussian moments alone do not provide that input. Exact RTN edge constants require the separate comparison in Section 10. Hermite coordinates are then handled by a dimension-free polarization comparison.

7.1 The uniform Gaussian theorem

For each of the Q occurrences on scope e , add one private middle role of size n and replace the occurrence by an independent sign factor on the enlarged scope. Denote this lifted template by $\hat{\mathcal{F}}$. Let $(\hat{f}, \hat{g}, \hat{d}_0)$ be its sign norm and moment statistics: its expectation scale is $n^{\hat{f}}(\log(2n))^{\hat{g}}$ and each detached random component contributes one factor $q^{1/2}$ in the logarithmic moment window.

Theorem 7.1 (Uniform Gaussian lift). *For every fixed coefficient-one independent-factor template with nonempty scopes, comparable role sizes, and standard real Gaussian coordinates,*

$$\mathbb{E} \|H_G\| \asymp n^{\hat{f}-Q/2}(\log(2n))^{\hat{g}}.$$

For every fixed $C_0 > 0$, uniformly for $2 \leq q \leq C_0 \log(2n)$ and all sufficiently large n ,

$$\| \|H_G\| \|_{L^q} \asymp n^{\hat{f}-Q/2}(\log(2n))^{\hat{g}} q^{\hat{d}_0/2}.$$

The same assertions hold for standard circular complex coordinates. Constants depend on the fixed template, role-size comparisons, and, for the moment assertion, C_0 .

The lift is finite and exact at the level of the comparisons used below. It is not an appeal to a central limit theorem at growing moment order. In particular, its private roles record fluctuations that would be lost if unary Gaussian factors were removed as sign isometries.

The proof uses an exact finite comparison of even moments between a Gaussian coordinate and a normalized sum of independent signs. Representing the latter by a private summation role permits a direct application of the sign calculus. The comparison is uniform at the logarithmic moment orders needed for operator norms; a central limit theorem would not provide that uniformity. Sign-magnitude conditioning gives the reverse norm inequality. All normalization factors and the choice of the private dimension are proved in Section J.1.

7.2 Fixed Hermite coordinates

For fixed positive integers d_e , replace the coordinates of occurrence e by $\text{He}_{d_e}(g)/\sqrt{d_e!}$, with independent underlying standard real Gaussian arrays. Here $e^{tx-t^2/2} = \sum_d \text{He}_d(x)t^d/d!$. Write H_{He} for the resulting matrix and H_{dec} for the template obtained by replacing each occurrence e by d_e independent standard real Gaussian occurrences on the same scope.

Theorem 7.2 (Fixed-degree Hermite comparison). *For the fixed independent-occurrence model just specified, put $A = \prod_e d_e^{d_e/2}/\sqrt{d_e!}$. Then, in every dimension and for every $q \geq 1$,*

$$A^{-1} \| \|H_{\text{dec}}\| \|_{L^q} \leq \| \|H_{\text{He}}\| \|_{L^q} \leq A \| \|H_{\text{dec}}\| \|_{L^q}.$$

Consequently the expected-norm and logarithmic-window moment exponents are those of the expanded Gaussian template.

This comparison is uniform in the number of labels, not in the degrees d_e . It lets the same structural calculus treat a fixed Hermite coordinate without mistaking an uncentered Gaussian power for a homogeneous chaos. It does not require the Hermite coordinate itself to have a symmetric

law, and it permits fixed degrees above two without changing the $\theta_e \leq 2$ hypothesis of the separate tail-law theorem.

The complete proof is given in Section J.2 (page 58).

For original arities at least two, the logarithmic charge becomes $\frac{1}{2} \max_{|S|=s} (a(S) + \sum_{e \subseteq S} d_e)$. This does not cover ordinary powers g^d without Hermite subtraction, arbitrary sums of chaoses, shared arrays across occurrences, or growing degrees.

8 A square-root-scale degree-four SoS certificate

The first principal application assembles correlated graph-matrix blocks into a feasible semidefinite moment system. The norm estimates provide blockwise error scales. The additional structure is algebraic: an exact positive square protects the large correction, critical log-free bounds control the remaining errors, and an extension from the pair block enforces positivity and all constraints on the full moment matrix.

The application concerns the degree-four sum-of-squares relaxation for clique in $G(n, 1/2)$. A feasible pseudoexpectation at a prescribed size shows that this particular relaxation cannot exclude that size. It is a statement about a specified proof system, not a computational lower bound for all algorithms. The quantitative target here is a fixed positive constant times $\sqrt{n}$, without a denominator growing polylogarithmically with n .

8.1 Problem and quantitative statement

Definition 8.1 (Degree-four clique feasibility). For a graph G , degree-four clique feasibility at size k means a linear functional $\mathcal{L}_k$ on polynomials of degree at most four satisfying normalization, positivity on squares of degree-two polynomials, and the Boolean, nonedge, and size constraints through their allowed degrees.

Theorem 8.2 (Square-root-scale degree-four feasibility). *There exist absolute constants $c > 0, n_0$ such that, for $G \sim G(n, 1/2)$ and $n \geq n_0$, with probability at least $1 - n^{-10}$, simultaneously for every real $9 \leq k \leq c\sqrt{n}$, there is a functional $\mathcal{L}_k$ satisfying*

$$\begin{aligned} \mathcal{L}_k(1) &= 1, & \mathcal{L}_k(p^2) &\geq 0 & (\deg p \leq 2), \\ \mathcal{L}_k((x_i^2 - x_i)q) &= 0 & & & (\deg q \leq 2), \\ \mathcal{L}_k(x_i x_j q) &= 0 & & & (\{i, j\} \text{ a nonedge}, \deg q \leq 2), \\ \mathcal{L}_k((\sum_i x_i - k)q) &= 0 & & & (\deg q \leq 3). \end{aligned}$$

The construction follows the corrected quartic ansatz of Hopkins et al. [12]. The relevant comparison with Raghavendra and Schramm [18] is at the logarithmic scale: those cited versions give a $\tilde{\Omega}(\sqrt{n})$ certificate. The statement above has a fixed positive constant multiplying $\sqrt{n}$; it does not change the polynomial exponent, give an optimized constant, or establish a lower bound for every algorithm.

8.2 The obstruction and the mechanism

A bound on each random summand is not by itself a PSD proof. The deterministic mean has different positive scales on the incidence space and its kernel. Moreover, the quartic correction is correlated with the same graph that generates the uncorrected moments. Applying the triangle

inequality to the correction and the dangerous cross term separately discards precisely the positive square that makes the budget work.

Let D be the signless vertex-pair incidence matrix, and write P for the projection onto $\text{range}(D^*)$ and $P_0 = I - P$. The comparison metric is

$$B = \omega^3 n^2 P + \omega^2 n^2 P_0.$$

The random correction contains $\rho Q Q^*$; the local fluctuation contains $\alpha_0(D^* Q^* + Q D)$. Their sum is an exact square minus $\alpha_0^2 D^* D / \rho$. This negative term acts only on P , so it is charged against the larger $\omega^3 n^2$ scale. Charging it against the smallest eigenvalue on the whole pair space would lose this advantage.

The remaining obstruction is logarithmic, but only in particular blocks. Seven matching-containing patterns and fifteen edge-plus-path patterns are at the critical polynomial scale and need log-free bounds. The proof gives elementary same-scale references for both families. Other fluctuations may retain a fixed logarithmic factor, because a polynomial margin remains after normalization. The norm input is therefore selective: log-free control is needed at the critical scale, while a coarser bound suffices away from it. The elementary references establish the critical estimates independently of the full classification; the PSD assembly uses both types of estimates.

The final normalized budget has the schematic form

$$K(\gamma^{-1} + x + x^2 + \gamma x^3) + o(1), \quad x = \omega / \sqrt{n}.$$

Choose the fixed correction strength γ first, then a sufficiently small fixed upper bound on x , and finally n large. The term $o(1)$ is uniform over the allowed internal parameter interval. This order of choices is the reason the conclusion has a constant multiple of $\sqrt{n}$, rather than a dimension-dependent loss.

A PSD pair block still does not finish the application. The lower-degree moments must be corrected using the same quartic data, the target size must hold exactly, and constants and linear polynomials must be included in positivity. The last two subsections of Appendix K perform these steps on one common graph event. No independence of the tuned parameter from the graph, and no union bound over uncountably many target sizes, is used.

The complete construction and verification are in Appendix K (page 59). In particular, Equation (91) displays the exact positive square, and Equation (93) identifies the actual clique-pair compression. The proof then bounds every residual block in its own metric, tunes the size constraint on one simultaneous event, and recovers the constant and linear rows by an explicit congruence. Thus positivity is not asserted only for the pair block or for a relaxed subset of the constraints.

For the SoS interface, the first step is exact algebra. Multiplication of graph matrices identifies inner boundary labels and partitions the remaining terms by their cross-collisions; repeated quotient edges cancel in pairs in the sign model. The output is a finite list of shapes with specified coefficients. Applying logarithmic moments and a finite union bound gives one event on which every required block estimate holds. The PSD argument then works pointwise on that event. In particular, a parameter chosen from the observed graph may enforce a size constraint without needing independence from the random blocks it multiplies.

This interface also explains why a matching norm theorem is not a black-box PSD theorem. Triangle inequality combines upper bounds for a finite sum, but separate lower bounds need not survive cancellation. The correction and its cross term must first be assembled into a positive square. Only its residual negative part is charged to the appropriate subspace. The distinction is algebraic, not a missing logarithm in the norm estimate. The argument in Section 8 keeps both stages explicit. Both reductions fix template sizes and degrees before dimension grows. The finite norm budgets

therefore do not supply uniform constants for growing-degree relaxations. The application also chooses its own normalization, whose effect remains explicit in the argument.

9 Spectral thresholds for random tensor networks

A tensor-network contraction gives a different use of the norm theory. Its primitive random objects are tensors, so its natural incidence description is already an independent-factor chaos. To study approximate isometry, however, one must first center its Gram matrix. This creates a sum indexed by active tensor subsets. Decoupling each term produces a new factor template whose separator is computed in a doubled network. The norm theorem becomes an input only after these operations.

We distinguish two questions throughout. Mean-Gram normalization asks whether a linear map is close to an isometry on its input space. Sample-state normalization asks about the spectrum and entropy of a density matrix obtained from one realization. These normalizations are not interchangeable. Even for the mean-Gram deviation, operator and unnormalized Schatten norms can have different convergence thresholds because a finite Schatten norm retains a dimension factor.

9.1 The model, its normalizations, and the cut gap

Definition 9.1 (Gaussian tensor network and normalizations). Fix a finite network with Q tensor vertices, internal edges joining distinct vertices, and open edges ending at distinct degree-one terminals. Parallel internal edges are distinct indices. Split terminals into a outputs and b inputs. Every component meets a terminal. Each edge has dimension N , and each tensor has independent standard circular complex Gaussian coordinates, independently of the other tensors. Let e be the internal edge count and $r = e + a + b$. Contracting internal indices gives $H_N : \mathbb{C}^{N^b} \rightarrow \mathbb{C}^{N^a}$.

Set

$$M_N = N^{-(r-b)/2} H_N, \quad F_N = M_N^* M_N - I_{N^b}, \quad \rho_A = \frac{H_N H_N^*}{\|H_N\|_F^2}.$$

Covariance identifies the two copies of every tensor coordinate: each input label must agree, and each internal or output edge has one free label. Thus $\mathbb{E} H_N^* H_N = N^{r-b} I$. This proves the mean-Gram normalization. The state normalization is random and is treated separately.

If $Q = 0$, this incidence model has no edges or terminals. The empty contraction is the scalar one, the deviation is zero, and all entropies vanish. Below assume $Q \geq 1$.

Definition 9.2 (Signed cut gap). For nonempty tensor sets T , write $c(T)$ for the internal edge boundary and $a(T), b(T)$ for incident output and input legs. Put

$$\delta(T) = c(T) + a(T) - b(T), \quad \Delta = \min_{\emptyset \neq T} \delta(T), \quad d = N^b.$$

Placing precisely T on the input side gives a cut of size $b + \delta(T)$. Thus $\Delta > 0$ means that the input-terminal cut is uniquely minimum.

9.2 Sharp deviation scales and their interpretation

All Schatten norms below are unnormalized. In particular, $\|I_d\|_{S_t} = d^{1/t}$ for finite t , whereas $\|I_d\|_{\text{op}} = 1$.

Theorem 9.3 (Mean-Gram deviations). *Assume at least one tensor vertex. If $\Delta \geq 0$, then for every fixed $1 \leq t < \infty$,*

$$\mathbb{E} \|F_N\|_{S_t} \asymp_{\mathcal{G}, t} N^{b/t - \Delta/2}.$$

There are positive constants c_t, p_t such that the norm exceeds $c_t N^{b/t - \Delta/2}$ with probability at least p_t , uniformly in N . For all signs of Δ ,

$$\|F_N\|_{S_t} \xrightarrow{\mathbb{P}} 0 \iff \Delta > 2b/t, \quad \|F_N\|_{\text{op}} \xrightarrow{\mathbb{P}} 0 \iff \Delta > 0.$$

If $\Delta > 0$, the uniform Gaussian norm bound additionally gives $\mathbb{E}\|F_N\|_{\text{op}} \asymp N^{-\Delta/2}$.

The signed gap has a direct meaning. The input-terminal cut has size b . Moving a nonempty set T of tensors to the input side changes its size by $\delta(T)$. A strictly positive minimum change ensures that the original cut is uniquely minimum, which is the operator-isometry criterion. For a finite Schatten norm, that same gap must also pay for the N^b input directions; the required extra margin is $2b/t$.

For example, consider a single tensor, viewed as an N^2 by N Gaussian matrix. Here $a = 2$, $b = 1$, and $\Delta = 1$. The expected operator deviation is of order $N^{-1/2}$ and converges to zero. The expected Frobenius deviation is of constant order and does not converge to zero in probability. The expected trace-norm deviation is of order $N^{1/2}$. Thus one family of maps can be approximately isometric in operator norm and fail the corresponding unnormalized Frobenius or trace criterion. The independent finite-dimensional Wishart calculation in Appendix N verifies this dimension dependence directly.

The proof separates three inputs. Fixed trace moments give the finite-Schatten upper bounds and qualitative operator convergence. An exact second moment and a fourth-moment bound yield nonrare spectral mass, proving the lower scales and excluding convergence at equality. The uniform Gaussian theorem in Section 7 is needed for the sharp operator expectation rate; a fixed-order moment estimate alone would leave an arbitrarily small polynomial loss.

9.3 The sample-state spectrum and entropy

For $\alpha > 0$, $\alpha \neq 1$, write $H_\alpha(\rho) = (1 - \alpha)^{-1} \log \text{tr}(\rho^\alpha)$. Use the continuous extension at $\alpha = 1$, and set $H_0(\rho) = \log \text{rank } \rho$ and $H_\infty(\rho) = -\log \|\rho\|_{\text{op}}$. Let s be the ordinary minimum input-output edge cut, not the signed gap Δ .

Theorem 9.4 (Sample-normalized spectral endpoint). *For the fixed network model above, ρ_A is defined almost surely. There is a constant C_G such that, with probability $1 - O_G(N^{-1})$,*

$$N^{-s} \leq \|\rho_A\|_{\text{op}} \leq C_G N^{-s}.$$

On the same event,

$$s \log N - \log C_G \leq H_\alpha(\rho_A) \leq s \log N \quad (0 \leq \alpha \leq \infty)$$

simultaneously for all the indicated orders.

This theorem controls the entropy endpoint at the scale level; it does not identify a leading spectral constant. For connected networks with both boundary sets nonempty, Theorem 10.3 supplies the sharper conclusion by a separate proof. We retain the present theorem for its broader model and its simultaneous guarantee for all entropy orders. Its lower bound on the largest eigenvalue comes from cut rank. Its upper bound requires both the uniform operator estimate for H_N and concentration of the random normalization $\|H_N\|_F^2$. Replacing the latter by its expectation without a probability estimate would not prove the statement.

Random tensor networks and their spectral consequences have been studied in several settings [6, 8, 10, 11]. Our statements specify independent Gaussian coordinates, fixed network size, and equal edge dimensions. The mean-Gram result is not asserted for a network obtained by independently normalizing every local tensor. Such scalar normalizations cancel in ρ_A , but they change the mean-Gram question. The proof in Appendix L keeps this distinction explicit.

9.4 Why the thresholds use different moment inputs

For $\Delta \geq 0$, the lower argument gives a nonvanishing probability of the matching scale, excluding convergence at equality and below the finite-Schatten threshold. For $\Delta < 0$, a rank obstruction excludes convergence in all the stated norms. Thus the dimension factor changes the criterion for a finite Schatten norm, rather than merely changing a constant in an operator estimate.

A single Gaussian tensor gives a useful calibration. If its output and input dimensions are N^a and N^b , then $\Delta = a - b$. Operator-norm convergence requires $a > b$, while unnormalized trace-norm convergence requires $a > 3b$. Confusing these two statements changes the threshold even in a Wishart matrix. We include the second and fourth Wishart moments as an independent check.

Fixed trace moments establish the finite Schatten thresholds and the qualitative operator-norm criterion. They do not alone give the sharp operator-norm expectation rate: obtaining that rate requires a moment estimate uniform up to logarithmic order. We keep these dependencies separate. For the RTN interface, centering the Gram matrix produces a sum indexed by active tensor subsets. Quadratic Gaussian decoupling turns each term into an independent-factor template in a doubled network. Its separator cost gives the cut-gap penalty, while its matrix dimension supplies the factor retained by a finite Schatten norm. Fixed moments suffice for the finite-order conclusions. Uniform moments become essential when obtaining sharp operator scales without a residual polynomial loss; this is the additional output of Theorem 7.1 used here.

The entropy consequence has one more step. The output state is normalized by its realized Frobenius norm, not by its expected Gram matrix. A deterministic cut bounds rank, while a norm bound controls the largest eigenvalue after this normalization. These two quantities bound the Rényi entropies from opposite sides. Keeping this step separate from approximate isometry explains why the state theorem is a consequence within the RTN application, rather than a third independent application. The full proof is in Appendix L (page 64). It includes the exact doubled-network role count, the second-moment identity, a positive-probability lower bound excluding convergence at the threshold, and the separate normalization step needed for the entropy statement.

10 Exact spectral edges of Gaussian tensor networks

The minimum cut determines the dimension scale of the largest eigenvalue, but it does not determine its leading constant. That constant records finer network structure. We now identify it for every fixed connected network in the independent complex-Gaussian, equal-dimension model. The result upgrades the sample-state scale in Theorem 9.4 to convergence of the right spectral edge.

This requires a separate argument. The all-defect estimates for graph matrices determine powers and logarithms up to shape-dependent constants; they do not identify the exact exponential base of an RTN moment sequence. Here a global flow quotient and a comparison at a different auxiliary dimension preserve that base. The quotient is realized as a product of overlapping Ginibre gates, to which the tensor-GUE strong convergence theorem of Chen et al. [5, Section 9.4, Theorem 9.8] applies. The reduction and dimension comparison below connect that external theorem to the original network; they do not identify contracted original tensor clusters with independent Gaussian gates.

10.1 The model and the exact right edge

Definition 10.1 (Connected equal-dimension RTN model). In this section the tensor-vertex graph is fixed, finite, connected, and loopless. Parallel internal edges are allowed and retain distinct index coordinates. The sets A, B of output and input legs are both nonempty. All internal edges and

open legs have dimension N . Each actual tensor vertex carries an independent array of independent standard circular complex Gaussians with $\mathbb{E}|g|^2 = 1$. Contractions use the unnormalized pairing $\sum_{i=1}^N e_i \otimes e_i$.

Attach all input legs to a terminal $\mathbf{b}$ and all output legs to a terminal $\mathbf{a}$, obtaining an augmented multigraph G . Write V_G for its actual tensor vertices, $v = |V_G|$, and $R = |E(G)|$ for the number of internal edges plus open legs. Let $s \geq 1$ be the minimum $\mathbf{b}$ – $\mathbf{a}$ edge-cut size, allowing terminal legs in the cut. Define

$$\begin{aligned} Y_N &= N^{s-R} H_N H_N^*, & Z_N &= N^{-R} \text{tr}(H_N H_N^*), \\ X_N &= N^s \rho_A = Y_N / Z_N. \end{aligned} \tag{41}$$

Cut factorization gives $\text{rank } H_N \leq N^s$. Whenever we use $N^{-s} \text{tr}$, we normalize the nonzero spectrum padded to N^s eigenvalues; the original boundary matrix need not have that dimension.

Definition 10.2 (Permutation energy and limiting right edge). For $p \geq 1$, set $\gamma_p = (1\ 2\ \cdots\ p)$ and $|\sigma| = p - \# \text{cyc}(\sigma)$ on S_p . The transposition metric is $d(\sigma, \tau) = |\sigma^{-1}\tau|$. For labels $\pi_u \in S_p$ at actual tensor vertices, fix $\pi_{\mathbf{b}} = \text{id}$, $\pi_{\mathbf{a}} = \gamma_p$, and put

$$\begin{aligned} \mathcal{H}_G^{(p)}(\pi) &= \sum_{\{u,w\} \in E(G)} d(\pi_u, \pi_w), & \Delta_G(\pi) &= \mathcal{H}_G^{(p)}(\pi) - s(p-1), \\ A_G(p, \delta) &= \#\{\pi : \Delta_G(\pi) = \delta\}. \end{aligned} \tag{42}$$

Let μ_G be the compactly supported probability measure with positive moments $m_p = A_G(p, 0)$, and set $E_G = \sup \text{supp } \mu_G$.

This is the minimum-energy moment law of Fitter et al. [8, Theorems 4 and 7]. Its existence and the identification of its right endpoint also follow from the construction below.

Theorem 10.3 (Exact RTN right edge). *Under the fixed connected model of this section,*

$$\|Y_N\| \xrightarrow{\mathbb{P}} E_G, \quad N^s \lambda_{\max}(\rho_A) \xrightarrow{\mathbb{P}} E_G. \tag{43}$$

In particular $1 \leq E_G < \infty$, and

$$H_\infty(\rho_A) = s \log N - \log E_G + o_{\mathbb{P}}(1). \tag{44}$$

The theorem excludes right outliers at every fixed positive distance from E_G , a conclusion not implied by convergence of fixed normalized moments. It identifies a constant through a uniquely specified compact law, not a finite closed-form algorithm for every graph. No local law, edge fluctuations, or smallest-nonzero-eigenvalue assertion is included.

For a single N by N Gaussian gate, $s = 1$ and the zero-defect permutations are the noncrossing interval $[\text{id}, \gamma_p]$. Their Catalan moments give the Marchenko–Pastur edge $E_G = 4$. For a single N^2 by N gate, the energy is $|\pi| + 2d(\pi, \gamma_p)$; its unique minimizer is $\pi = \gamma_p$. Thus $\mu_G = \delta_1$ and $E_G = 1$. Both networks have the same minimum cut, but their min-entropy corrections differ. These are classical single-gate calibrations; the theorem treats arbitrary fixed connected topology.

If each internal link instead uses the normalized Bell pairing, the raw Gram matrix gains a factor $N^{-e_{\text{int}}}$. Multiplying it by $N^{s-|A|-|B|}$ then gives exactly Y_N . The sample-normalized state itself is unchanged. Likewise, normalizing each local Gaussian tensor by its own norm multiplies the full contraction by one nonzero scalar, which cancels from ρ_A . This observation transfers the state conclusion to those local Haar-vector tensors, but not the deterministic normalization of Y_N or the mean-Gram conclusions of Section 9.

10.2 Overview of the dimension-transfer mechanism

Preserving an exact endpoint requires greater precision than preserving norm exponents. Let $A_G(p, \delta)$ count the permutation labelings of the RTN Wick expansion with energy defect δ , and let $F_G(p, N) = \sum_{\delta \geq 0} A_G(p, \delta) N^{-\delta}$. A maximum terminal flow defines a directed graph whose full strongly connected components form a balanced acyclic quotient Q . One anchor per component gives a graph-wide label comparison, including every positive-defect layer. If v is the number of tensor vertices and R counts internal edges and open legs, it yields

$$F_G(p, N) \leq \frac{F_Q(p, p^2)}{1 - p^{K_G}/N}, \quad K_G = 4v + 2 + 8R, \quad N > p^{K_G}. \quad (45)$$

The quotient is realized by independent square Ginibre gates on possibly overlapping tensor factors. Strong convergence of tensor GUE, after expressing each gate through two GUE matrices, identifies its limiting norm [5]. Gaussian concentration and a high-moment bound on the exceptional event then show $F_Q(p, p^2) \leq 2(E_G + \eta)^p$ for each fixed $\eta > 0$ and all sufficiently large p . The auxiliary dimension p^2 suppresses that exceptional contribution without changing the limiting exponential base. Taking p proportional to $\log N$ proves the upper edge; fixed-moment concentration proves the matching lower edge.

Lemma M.8 and Corollary M.9 state the analytic bridge separately from the circuit construction. Small exceptional probability alone does not control a growing moment: the rare-spike example in Section M.5 explains why the high-moment estimate on that contribution is also verified.

This argument never assumes that an original contracted tensor cluster remains Gaussian. Its independent Gaussian circuit is a comparison ensemble introduced only after a deterministic count. It is independent of the graph-matrix classification, although both proofs control nonleading replica states through their dimension loss. The exact-edge statement retains its narrower model; it does not replace the broader common-event entropy bounds above.

The comparison has an elementary form that can be used beyond this network model. Its inputs separate the combinatorial task from the subsequent choice of dimension: construct one global map, bound its effect on defect, and control the total weight of each fiber.

Theorem 10.4 (Positive defect-fiber transfer). *For each integer $p \geq 2$, let Ω_p, Λ_p be finite state sets with nonnegative, dimension-independent weights w_ω, v_λ and defects $\delta(\omega), j(\lambda) \in \mathbb{N}_0$. Define*

$$F_\Omega(p, N) = \sum_{\omega} w_\omega N^{-\delta(\omega)}, \quad F_\Lambda(p, M) = \sum_{\lambda} v_\lambda M^{-j(\lambda)}.$$

Suppose one map $\phi_p : \Omega_p \rightarrow \Lambda_p$ satisfies, for a fixed $\kappa \geq 0$ and numbers $r_p \geq 1$,

$$j(\phi_p(\omega)) \leq \kappa \delta(\omega), \quad (46)$$

$$\sum_{\substack{\omega: \phi_p(\omega) = \lambda \\ \delta(\omega) = \delta}} w_\omega \leq r_p^\delta v_\lambda \quad \text{for every } \lambda, \delta. \quad (47)$$

Then, for $M \geq 1$ and $N > r_p M^\kappa$,

$$F_\Omega(p, N) \leq \frac{F_\Lambda(p, M)}{1 - r_p M^\kappa / N}. \quad (48)$$

Also, $a_{p, \delta} := \sum_{\delta(\omega) = \delta} w_\omega$ satisfies $a_{p, \delta} \leq r_p^\delta M^{\kappa \delta} F_\Lambda(p, M)$.

Proof. Group source states by defect and image. The hypotheses give

$$a_{p,\delta} \leq r_p^\delta \sum_{\lambda: j(\lambda) \leq \kappa\delta} v_\lambda \leq r_p^\delta M^{\kappa\delta} \sum_{\lambda} v_\lambda M^{-j(\lambda)}.$$

Multiply by $N^{-\delta}$ and sum the geometric series. $\square$

The theorem does not construct the map. For the RTN, that task is performed by the full-SCC anchor construction in Lemmas M.3 and M.4. Wick weights are one, $r_p = p^{4v}$, and $\kappa = 1 + 4R$. For general weights, a cardinality bound alone cannot replace Equation (47). Nor can one replace the full positive target sum by its zero-defect coefficient. The path count uses a related but different input: direct evaluation of positive falling-factorial weights, formalized in Lemma D.1. These are two uses of positivity, not an identification of their state spaces or encodings.

The quantitative comparison can be stated independently of the limiting law. Orient a fixed maximum family of terminal flow paths forward and each unused edge in both directions. Let Q be the quotient by the full strongly connected components, retaining edge multiplicities. For this quotient, $F_Q(p, M)$ denotes the positive Wick moment polynomial defined just as for G . Put $C_0 = 1 + 4R$.

Theorem 10.5 (Auxiliary-dimension comparison). *For integers $p \geq 2$, $M, N \geq 1$ with $p^{4v} M^{C_0} < N$,*

$$F_G(p, N) \leq \frac{F_Q(p, M)}{1 - p^{4v} M^{C_0}/N}. \quad (49)$$

In particular, with $K_G = 4v + 2 + 8R$,

$$F_G(p, N) \leq \frac{F_Q(p, p^2)}{1 - p^{K_G}/N} \quad (N > p^{K_G}). \quad (50)$$

The complete proof is given in Section M.2 (page 69).

The entire proof of the edge theorem is in Appendix M. In particular it constructs the quotient, bounds the fibers of one global anchor map, identifies the independently sampled comparison circuit, checks the hypotheses of the external strong-limit theorem, and derives the growing-moment and normalization estimates. The strong-limit input itself is attributed, not claimed as a new theorem proved here.

10.3 Comparison and scope

The max-flow approach of Fitter et al. [8] already identifies the fixed-moment law, establishes weak convergence, and derives entropy corrections. In its series-parallel setting the law has an explicit convolution description. Our additional conclusion is right-edge convergence for the fixed connected Gaussian network, rather than another identification of that law. The strong-convergence input for overlapping tensor GUE is due to Chen et al. [5], building on the weak model of Charlesworth and Collins [4].

The bridge specific to this proof is the all-defect global comparison with a balanced acyclic quotient, followed by auxiliary-dimension transfer. Its proof is independent of the graph-matrix classification in Sections 3 and 4. Both arguments exploit dimension loss to control nonleading configurations, but their combinatorial states and precision targets differ.

All topology and support sets are fixed before N grows. The argument gives no practically uniform threshold over growing graphs or shrinking tolerances. Nonflat link weights, shared tensors across vertices, and non-Gaussian coordinates are outside this exact-edge statement. Nor does the

min-entropy correction $-\log E_G$ serve as the correction for every finite Rényi order. The broader coarse bound in Theorem 9.4 and the mean-Gram criteria in Theorem 9.3 retain their separate models and conclusions.

11 Concluding remarks

The sharp classification separates two sources of spectral growth that are not visible in the minimum-cut size alone. A separator determines the polynomial cost of sharing labels, while the components exposed by a minimizing separator determine which local fluctuations can be selected by the operator norm. The matching upper and lower bounds identify the resulting logarithm as a feature of the matrix, not an artifact of concentration. The equal-coarse-data chains in Corollary 2.9 isolate the information detected by this rule. The coarea argument explains why the description survives beyond the leading trace configurations: excess separator cost pays for the additional choices in every defect layer.

The extensions separate incidence, label-space geometry, and scalar tails as inputs to the norm problem. The sign leaf family and the Gaussian gated chain have different primitive variables but share the same competition among minimizing cuts. Local weights and aspect ratios alter the scale and the minimum-separator family, while bounded noise and Hermite comparisons transfer conclusions between specified laws. These results do not furnish a joint theorem for arbitrary coefficients or dimension-dependent laws. Critical sparse occupancy introduces a $\log \log n$ correction, and the global-coefficient examples show that local magnitudes do not determine operator geometry.

The distinction between spectral scale and spectral precision is also important for applications. In the degree-four construction, removing the logarithmic loss from the critical blocks is enough to reach a constant multiple of the square-root scale, but only after the correlated terms are assembled into a positive square. For an exact RTN edge, even a matching norm scale leaves the essential constant undetermined. The separate quotient comparison preserves the exponential base of the moment sequence and connects the original network to an ensemble with a known strong limit. These arguments illustrate two different ways in which sharper spectral information becomes useful; neither reduces an application to bounding each summand separately.

The fixed-shape classification leaves a concrete algorithmic question. Finite enumeration evaluates the exponents, but does not efficiently maximize the active-component count over all minimum separators. A second question is whether the all-defect argument can be made sufficiently uniform in the shape to treat growing-degree semidefinite relaxations. The present encoding and the weighted clone reduction allow constants to depend on the template; controlling that dependence would require more than the termination proved here.

The Lean formalisation is ongoing and can be found here.¹

AI Disclosure. We used OpenAI’s ChatGPT and Codex (GPT-family models) to assist in developing and completing portions of the proofs, including proposing intermediate lemmas and alternative arguments, checking technical steps and boundary cases, performing auxiliary computations, and improving the exposition. The tools materially affected selected proof developments in Sections 3–9 and the corresponding appendices. The research questions, theorem statements, overarching proof architecture, and final selection and validation of all arguments were determined by the authors. The authors independently reviewed and verified all AI-assisted material incorporated into the manuscript, checked the computations and references, and take full responsibility for the correctness and originality of the work.

¹The current development is available at <https://github.com/xxxhbb/sharp-norms-from-finite-structure>.

A Proof guide and external inputs

The main text states the results, explains their scope, and retains the separator coarea argument, coefficient flattening, and conversion from moments to norms. The appendices below contain the technical arguments needed to complete those proofs. All arguments needed for the stated results are contained in the paper. The following guide locates every numbered theorem, lemma, proposition, and corollary. References to established external theorems are distinguished from results proved here; their use does not constitute a new proof of those external theorems.

A.1 The graph-matrix classification

Theorem 1.3 is assembled at the end of Section 4. Its upper bound uses Proposition 3.8, proved immediately after its statement in Section 3; its conditional lower bound uses Appendix E. The full logarithmic-window moment claim is completed by the detached-component argument in Appendix F. Lemmas 2.2, 2.4 and 2.5 and Proposition 2.6 have their proofs adjacent to their statements in Section 2. Thus the typed/global comparison and finite evaluation are part of the proof, rather than assumptions imported from experimental data.

Proposition 2.8 and Corollary 2.9 have their deductions in Section 2.1. Their coefficient input, Lemma C.1, is proved in Appendix C, which also contains all literature substitutions and the adapted product bound. The sharp family scales depend on the classification; they are not an independent proof of it.

Lemmas 3.2, 3.6 and 3.7 are proved in Section 3. Theorem 3.4 is completed in Appendix D, including Lemma D.2, its proof, the off-path record and decoder, and the seed-entropy budget. Lemma D.1 is proved immediately before the path count, which instantiates it with exact falling-factorial weights. Lemmas 4.1 and 4.2 are proved in Section 4; Lemma E.1 and the full conditional estimates and synchronization are proved in Appendix E. The trace obstruction has its complete construction and calculation in Appendix B.

A.2 The structured extensions

Theorems 1.4 and 5.2 and Proposition 5.4 are proved in Appendix F. In particular, the layerwise refinement Equation (73) is proved there before it is used in the weighted trace expansion; it is not inferred from the weaker aggregate count. Theorem 5.6 has its dimension-free proof in Appendix G; Corollary 5.5 then has its short deduction next to its statement in Section 5.5. Theorem 5.8 has an independent proof in Appendix H. The sparse statement Proposition 5.7 is proved, regime by regime, in Appendix O.

Theorem 6.3 is proved in Appendix I, in the order local contraction, positive-aspect minimum-face reduction, conditional lower bound, bounded-role elimination, and unary absorption. The Gaussian result Theorem 7.1 is proved in Section J.1, and Theorem 7.2 in Section J.2. Each comparison retains its own hypotheses; these proofs do not claim their unrestricted simultaneous combination.

A.3 The applications and the exact edge

Theorem 8.2 is proved in Appendix K. This includes the exact collision algebra, the full finite pattern families, the two-scale PSD budget, lower-degree cleanup, tuning on one common event, and the congruence recovering the entire degree-four moment matrix. Theorems 9.3 and 9.4 are proved in Appendix L, with separate arguments for fixed Schatten order, the uniform operator estimate, the probability lower bound, and sample normalization.

Theorem 10.3 is proved in Appendix M. Lemmas M.1 to M.4 are proved there before Theorem 10.5, whose proof is at Section M.2. They establish a deterministic comparison of positive moment polynomials. Next, Lemmas M.5 to M.7 and M.10 have adjacent proofs establishing the independent comparison circuit and its growing moments. Finally Lemma M.11 has its proof before the fixed-moment lower edge and the final normalization. These are not consequences of the graph-matrix classification.

The weighted abstraction Theorem 10.4 has its complete proof next to its statement in Section 10; the SCC lemmas verify its hypotheses for the network. The general analytic bridge Lemma M.8 and Corollary M.9 are proved in Section M.5, followed by the rare-spike example and the quantitative circuit verification. The two positive-evaluation uses are distinguished rather than presented as the same map.

A.4 What is used from the literature

The Boolean hypercontractive inequality used in Section E.1 is the fixed-degree estimate $\|P\|_q \leq (q-1)^{d/2} \|P\|_2$ for a degree-at-most- d polynomial in independent signs and $q \geq 2$ [16]. Its applications here have fixed degree and fixed moment order. Paley–Zygmund, conditional Jensen, finite-dimensional net bounds, and integral max-flow/min-cut are used in their standard forms; the specific estimates and graph reductions are derived in the proofs.

For the sparse iid matrix, the external Segner comparison is stated in Equation (116) and attributed to Segner [19, Theorem 1.1]. The complete occupancy estimates following that comparison are included here. This is a calibration using an established norm theorem, not a new theorem about all sparse chaoses.

For the exact RTN edge, the substantial external input is strong convergence for fixed polynomials in independent GUE matrices embedded on fixed nonempty, possibly overlapping tensor supports [5, Section 9.4, Theorem 9.8]. The application following Lemma M.5 specifies the number of variables, their supports, the coefficient dimension, and the polynomial. The companion fixed-moment input is Charlesworth and Collins [4, Theorem 4]. The limiting moment law agrees with that of Fitter et al. [8]; the argument also constructs it from the comparison circuit. We do not reprove these external strong- and weak-limit theorems. We do prove the quotient comparison, the passage from qualitative norm convergence to growing moments, both bounds on the original edge, and the sample-normalization step that use them.

Accordingly, “complete proof” here means a complete derivation of the stated contributions from explicitly identified standard and external inputs, with all manuscript-specific counting, conditioning, and application verifications included in this PDF.

A.5 Application dependencies at a glance

The conclusions require different inputs in addition to norm control. Table 1 separates those inputs; in particular, the exact RTN edge uses the independent comparison of Section 10.

Table 1: Application interfaces. The exact-edge row uses a separate proof route for the connected RTN model with nonempty boundaries.

Conclusion	Norm or moment input	Additional argument
Degree-four SoS feasibility	Log-free critical shapes; tails for the remaining blocks	Subspace metric, positive square, full moment constraints
Finite Schatten thresholds	Fixed moments; second/fourth-moment lower bound	Centering, decoupling, doubled-network cuts
Sharp operator expectation	Logarithmic-window Gaussian moments	Centered expansion; nonrare lower scale
State spectrum and entropy	Uniform Gaussian norm bound	Sample normalization; deterministic cut-rank bound
Exact RTN right edge	Tensor-GUE strong convergence; auxiliary growing moments	Flow-SCC comparison, dimension transfer, fixed-moment lower edge

B The full-trace obstruction

B.1 Why the full trace sequence is not enough

There is a useful obstruction to any approach that treats an ordinary trace sequence as a complete summary of the matrix. Consider two three-role shapes with one common row and column role. In the first, that root is isolated and the two middle roles form an edge. In the second, all three roles form a triangle. Both matrices are diagonal. For root label i , their diagonal entries are respectively

$$X_i = 2 \sum_{\substack{a < b \\ a, b \neq i}} \xi_{ab}, \quad Y_i = 2 \sum_{\substack{a < b \\ a, b \neq i}} \xi_{ia} \xi_{ib} \xi_{ab}. \quad (51)$$

The factor two comes from the two orders of the middle roles in the globally injective sum.

For each fixed i , these two random variables have exactly the same distribution. Indeed, condition on every root-incident sign ξ_{ia} . The remaining signs ξ_{ab} are independent, and multiplying them by the fixed signs $\xi_{ia}\xi_{ib}$ preserves their joint law. Each diagonal entry therefore has the distribution of twice a sum of $\binom{n-1}{2}$ independent signs. Since the trace of a power of a diagonal matrix sums coordinate powers,

$$\mathbb{E} \operatorname{tr}(M_0^{2p}) = \mathbb{E} \operatorname{tr}(M_1^{2p}) \quad \text{for every integer } p \geq 1. \quad (52)$$

This is an exact finite- n identity, not agreement of finitely many leading moments. Odd expected trace moments vanish as well. When $n \equiv 0$ or $3 \pmod{4}$, the number $\binom{n-1}{2}$ is odd, so none of these sign sums can be zero. On this infinite subsequence both matrices have deterministic rank n for every realization.

Nevertheless their expected norms differ: $\mathbb{E}\|M_0\| \asymp n$, whereas $\mathbb{E}\|M_1\| \asymp n\sqrt{\log n}$. The first scale is particularly transparent. Write $Z = 2 \sum_{a < b} \xi_{ab}$, so $X_i = Z - 2 \sum_{a \neq i} \xi_{ia}$. The shared scalar has L^1

size comparable to n , while the maximum of the row-sum corrections has expected size $O(\sqrt{n \log n})$. The upper bound follows by triangle inequality and a scalar subgaussian union bound; the lower bound already follows from the marginal L^1 norm of one X_i . In the rooted triangle, removing the mandatory root exposes a connected active component, and the lower-bound mechanism of this paper yields the additional square-root logarithm.

The difference is in the joint dependence of the coordinates, which an ordinary expected trace does not record. Direct parity counting gives, for $i \neq j$, $\text{Corr}(X_i, X_j) = (n-3)/(n-1)$ but $\text{Corr}(Y_i, Y_j) = 2/(n-1)$. In the first case almost all fluctuation is common to the coordinates. In the second it can be amplified by selecting the root label. We do not use small pairwise correlations as a substitute for a lower-bound proof; the fully conditioned trial argument supplies that proof.

This example explains an important feature of the upper-bound strategy. We do not feed the undifferentiated trace sequence of the full shape into an optimizer and expect it to reconstruct the expected norm. We first identify genuinely detached components from the shape itself and take their L^1 contribution separately. Only then do we use high trace moments on the boundary core. The preprocessing retains information lost by Equation (52). Adding more ordinary trace moments to the unprocessed matrix cannot repair this loss.

The example also clarifies what the finite rule computes. It is a structural classification of a supplied shape, not a spectral reconstruction theorem. Matrix dimension, exact rank, and even all ordinary expected trace moments can agree while the logarithmic exponent differs. A complete classification can therefore be simpler to evaluate on a graph than to infer from a seemingly richer list of spectral statistics.

C Coefficient calculations for the separating family

The graph family and its sharp scales are stated in Section 2.1. This appendix supplies the coefficient calculations and comparisons used there. The sharp norm statement is a corollary of the classification, not an independent proof of it.

C.1 Exact coefficient flattenings

Lemma C.1 (Role-frontier formula). *For any partition of the axes of the equal-size typed coefficient tensor $\mathcal{A}$ into rows and columns, let T consist of the roles having appearances on both sides. Then*

$$\|\mathcal{A}_{[R|C]}\| = n^{(v-|T|)/2}. \quad (53)$$

In particular,

$$\sigma(\mathcal{A}) = n^{(v-1)/2}, \quad \nu(\mathcal{A}) = \begin{cases} n^{(v-1)/2}, & Q \geq 1, \\ n^{(v-2)/2}, & Q = 0. \end{cases} \quad (54)$$

Proof. Split each edge axis into its two endpoint-coordinate axes without moving either coordinate across the row-column partition. After row and column permutations, the coefficient matrix is a Kronecker product of rolewise equality tensors. A role appearing on only one side gives an equal-label vector with n nonzero entries, hence norm $\sqrt{n}$. A role appearing on both sides gives $\sum_{i=1}^n e_i^{\otimes a} (e_i^{\otimes b})^*$, where $a, b \geq 1$. Both families of vectors are orthonormal, so this factor has norm one. Every role appears, and multiplying these norms proves Equation (53).

For σ , the main path connects matrix axes on opposite sides. If no role crossed the partition, consecutive appearances along the path would all be on the same side, a contradiction. Thus $|T| \geq 1$. Putting all noise axes on the row side makes only x_{L+1} cross, so the minimum is attained. For ν ,

the role-axis incidence graph is connected and meets both sides, hence again $|T| \geq 1$. When $Q \geq 1$, putting a single leaf edge on the row side makes its parent the only crossing role. When $Q = 0$, every nonempty segment of row edges has two frontier roles, counting an endpoint role when the segment reaches a matrix boundary. Thus $|T| \geq 2$, attained by a single row edge. These observations give Equation (54). $\square$

For unequal positive role sizes the same factorization gives $\prod_{x \notin T} \sqrt{m_x}$. In particular, the resulting upper bounds are uniform for $m_x \leq n$. If a role class is empty, the matrix is zero. This is the uniformity required by Lemma 2.4; it is not an assertion that the globally injective coefficient tensor has the same flattenings.

C.2 Direct outputs of the specified literature bounds

All shape parameters are fixed in these comparisons. In Tulsiani and Wu [20, Theorem 20], specialize the Bernoulli density to $1/2$, so the normalized edge variables are signs. With $t \geq 2$, the stated Schatten-moment bound specializes to

$$\mathbb{E} \|M_\alpha\|_{S_{4t}}^{4t} \leq (48tv)^{4tv} (Cd)^d n^v n^{2t(v-1)}.$$

The density-dependent factor is one. Taking the $4t$ -th root and using $\|M\| \leq \|M\|_{S_{4t}}$ and Jensen gives $\mathbb{E} \|M_\alpha\| \leq C_\alpha t^v n^{(v-1)/2 + v/(4t)}$. Set $t = \max(2, \lceil \log n \rceil)$ to obtain

$$\mathbb{E} \|M_\alpha\| \leq C_\alpha n^{(v-1)/2} (\log n)^v. \quad (55)$$

This deduction uses an expectation moment bound, not a conversion from a fixed-failure-probability statement.

For the typed degree- d chaos, every noise vector has coordinate dimension $m = n^2$ and the matrix dimensions are n, n . Theorem 2.4 of Bandeira et al. [2] gives $C_d (\log n)^{d/2} \sigma(\mathcal{A})$ for signs. Their Theorem 2.5 gives $C_d [\sigma(\mathcal{A}) + (\log n)^{(d+2)/2} \nu(\mathcal{A})]$. When $Q \geq 1$, Equation (54) makes the first bound smaller. The unequal-size calculation above and Lemma 2.4 transfer it to the globally injective model.

The graph-specific Theorem 4.8 in that reference gives the same output directly. Its logarithmic-count parameter is

$$s - |U \cap V| + |W \setminus (U \cup V)| - h = 1 + L + Q = d.$$

For its prescribed intermediate-flattening construction, the unique main path uses $L + 1$ edges and covering the Q off-path leaves requires all Q leaf edges. The iteration count is therefore also d . These are comparisons with the displayed bounds in the cited versions, not impossibility results for the underlying methods.

When $Q = 0$, the situation is different: Equation (54) gives $\nu/\sigma = n^{-1/2}$, so the strong-NCK bound is already asymptotically log-free. No logarithmic improvement over that output is asserted for an undecorated path.

C.3 The adapted product bound

Identity Equation (23) follows by interchanging finite sums, first summing each leaf coordinate. For a rectangular sign matrix with both dimensions at most n , zero padding and the net estimate in Lemma D.2 give expected norm $O(\sqrt{n})$. For one leaf array, put $S_i = \sum_{a=1}^m \xi_{i,a}$, $m \leq n$. The sign exponential-moment bound yields

$$\mathbb{P}\{\max_{i \leq n} |S_i| > u\} \leq \min\{1, 2ne^{-u^2/(2n)}\}.$$

For $u_0 = \sqrt{2n \log(2n)}$, tail integration gives

$$\mathbb{E} \max_i |S_i| \leq u_0 + 2n \int_{u_0}^{\infty} e^{-u^2/(2n)} du \leq u_0 + \frac{n}{u_0} \leq C\sqrt{n \log(2n)}.$$

Each gate norm is at most the product of its leaf-array maxima. All these arrays and all mixing matrices are independent. Consequently submultiplicativity and independence give

$$\mathbb{E} \|H_{\alpha_L, \mathbf{q}}\| \leq (C\sqrt{n})^{L+1} (C\sqrt{n \log(2n)})^Q.$$

This estimate is uniform over all role sizes at most n ; Lemma 2.4 proves Equation (24). It is sharp for a concentrated decoration by Equation (21). For a distributed decoration its logarithmic excess is precisely $(Q - q_{\max})/2$. Thus the comparison includes a direct structural argument as well as the general literature bounds.

D Complete all-defect counting and encoding

This appendix proves Theorem 3.4. The definitions of legal states, defect, and partition intervals are in Section 3. We keep the original backbone in the encoding record; the repaired backbone is used only to bound the number of possible seeds.

D.1 A worked exact-state expansion

A fourth-moment calculation on a single typed edge illustrates what the exact expansion retains. Let X be an m by m sign matrix and take $p = 2$. The row and column trace matchings are the two different matchings on four positions. They cannot both remain unchanged in a legal state, because their meet has singleton cells. There are exactly three possibilities: make the row partition full, make the column partition full, or make both full. Their respective label weights are $m^2(m-1)$, $m^2(m-1)$, and m^2 . Thus

$$\mathbb{E} \operatorname{tr}((XX^*)^2) = 2m^3 - m^2.$$

The negative coefficient in the expanded polynomial does not come from a negative replica contribution. All three exact-state weights are nonnegative; it comes from expanding the falling factorials. This is one reason to keep the exact positive representation during comparisons.

The same example shows why a pairing in the code is not an extra Wick pairing in the moment. An even equality block may admit several matching refinements. We choose a canonical refinement when encoding that block; we do not multiply its Rademacher expectation by the number of possible refinements. For Gaussian coordinates, by contrast, repeated-coordinate moments do carry pairing multiplicities. The private-role comparison used later transfers those multiplicities through an explicit moment inequality, rather than identifying the two expansions.

The defect parameter is similarly concrete. In the single-edge fourth moment, the two leading states have three free label blocks and defect zero. The fully identified state has two free blocks and defect one. At higher orders, the path estimate counts every allowed intermediate partition with its actual defect. On a general graph, the off-path encoding adds only a polynomial price per lost block. The counting theorem is therefore a stability statement about the entire neighborhood of the leading configurations, not merely a description of those leading configurations.

This perspective makes the role of a logarithmic moment order transparent. With a fixed defect penalty n^{-1} and an entropy increase p^{K_r} , all positive defects form a controlled geometric tail when $p = O(\log n)$. The leading factor p^{a*p} becomes $p^{a*/2}$ after taking the $2p$ -th root. Both scales in the final norm formula are already visible before optimizing any analytic concentration inequality.

D.2 Counting a path backbone

Before specializing to paths, we record exactly what positive evaluation provides. This also explains why the negative power-basis coefficient in the preceding fourth-moment example is harmless.

Lemma D.1 (Positive evaluation and coefficient extraction). *For each integer $p \geq 2$, let Ω_p be finite, with integer defects $\delta(\omega) \geq 0$ and weights $w_\omega \geq 0$. For a real number D_p , suppose at an auxiliary integer $M \geq 1$ that*

$$P_p(M) = \sum_{\omega \in \Omega_p} W_\omega(M), \quad W_\omega(M) \geq c_p w_\omega M^{D_p - \delta(\omega)}, \quad c_p > 0.$$

If $a_{p,\delta} = \sum_{\delta(\omega)=\delta} w_\omega$, then

$$a_{p,\delta} \leq c_p^{-1} M^{\delta - D_p} P_p(M). \quad (56)$$

No expansion of the W_ω into signed monomials is required.

Proof. Retain only defect- δ states in the nonnegative sum and apply their weight lower bound. Divide by $c_p M^{D_p - \delta}$. $\square$

For a path of $l \geq 1$ edges, the next proof applies this lemma with $w_\omega = 1$, $D_p = pl + 1$, $M = 2p^2$, and $c_p = e^{-(l+1)}$. The exact weight is $\prod_x (M)_{|\pi_x|}$, not the corresponding monomial. Thus a moment estimate for independent sign matrices can calibrate the count without enumerating all path states. The general graph still requires the off-path encoding and seed bound below.

Fix s vertex-disjoint boundary-to-boundary paths, with unit capacity on every vertex, including endpoints. The vertex form of max-flow min-cut provides these paths. Common boundary vertices give singleton paths. Trim other paths so that their internal vertices are not boundary vertices. Let P be their union and $Q = W \setminus P$. A component of $G[Q]$ meets at most one boundary side, since otherwise it would supply another vertex-disjoint path.

For the i th path put $\delta_i = \sum_{x \in P_i} d(\pi_x) - (p - 1)$. By Lemma 3.7 on the path, $\delta_i \geq 0$. Set $\delta_{\text{on}} = \sum_i \delta_i$ and $D = \sum_{x \in Q} d(\pi_x)$. Then

$$\delta = \delta_{\text{on}} + D. \quad (57)$$

The vector consisting of the path defects and individual off-path defects has at most r coordinates. Its number of possibilities is at most $2^{\delta+r} \leq 2^{rp}$, since $\delta \leq (r - s)(p - 1)$.

Lemma D.2 (Path count). *A path with l edges has at most $100^{2p(l+1)} p^{2t}$ legal states with $\sum_x |\pi_x| = pl + 1 - t$. A singleton path has one state, at $t = 0$.*

Proof. For $l \geq 1$, evaluate its positive moment expansion at auxiliary dimension $m = 2p^2$. For $1 \leq b \leq p$, $(m)_b \geq e^{-1} m^b$, by $\log(1 - x) \geq -2x$ for $0 \leq x \leq 1/2$. The typed path matrix is a product $X_1 \cdots X_l$ of independent m by m sign matrices.

To bound a single factor, take two $1/4$ -nets of the unit sphere, each of size at most $9m$. Its operator norm is at most twice the largest absolute bilinear form on the nets. Each such form is a sign sum with squared coefficients summing to one. The exponential moment inequality for signs and a union bound give

$$\mathbb{P}\{\|X\| > 2(t_0 + z)\} \leq e^{-z^2/2}, \quad t_0^2 = 4m \log 9 + 2 \log 2 \leq 11m.$$

Tail integration and Minkowski imply $\|X\|_{L^{2p}} \leq 12\sqrt{m}$ for $p \leq m$. Independence and submultiplicativity, using order $2p$ for every factor, now give

$$\mathbb{E} \text{tr}((HH^*)^p) \leq m \mathbb{E} \prod_{j=1}^l \|X_j\|^{2p} \leq 12^{2pl} m^{pl+1}.$$

Every state under consideration contributes at least $e^{-(l+1)}m^{pl+1-t}$. Positivity therefore bounds its count by $e^{l+1}12^{2pl}2^tp^{2t}$. Since $t \leq l(p-1)$, the asserted constant suffices. $\square$

Ignoring non-backbone edges enlarges the set of allowed backbone states. For fixed defects, Lemma D.2 bounds their number by

$$100^{2pr}p^{2\delta_{\text{on}}}. \quad (58)$$

D.3 A lossless off-path encoding

For perfect matchings define $d_M(\rho, \sigma) = p - |\rho \vee \sigma|$. Their union decomposes into alternating cycles, and a cycle of t pairs takes $t-1$ two-pair switches to resolve. Thus d_M is the switch distance. A radius- t ball has at most $(4p^2)^t$ elements for $t \geq 1$, and one for $t = 0$.

If ρ, σ both refine an even π , then $d_M(\rho, \sigma) \leq d(\pi)$. Across a legal edge xy , choose a matching refining $\pi_x \wedge \pi_y$. For any chosen refinements σ_x, σ_y , the triangle inequality gives

$$d_M(\sigma_x, \sigma_y) \leq d(\pi_x) + d(\pi_y). \quad (59)$$

A partition with at most p blocks has at most p^{2b} coarsening records of b binary merges. Across r roles, records of at most B merges number at most $(2rp^2)^B$. Finally, if ξ refines π , then

$$|\pi| - |\pi \vee \rho| \leq d_M(\xi, \rho). \quad (60)$$

Indeed, joining with π contracts the sequence of merges generating $\xi \vee \rho$ from ξ .

Fix a component C of $G[Q]$ and write $D_C = \sum_{x \in C} d(\pi_x)$. Choose a deterministic matching refinement σ_x of each π_x . If C meets U , anchor it there and set $\rho_C = \tau$; if it meets V , set $\rho_C = \eta$. Otherwise use $\rho_C = \sigma_w$ for a fixed anchor $w \in C$. Only the latter seeds are free. Along a spanning tree, Equation (59) gives

$$d_M(\rho_C, \sigma_x) \leq 2D_C \quad (x \in C). \quad (61)$$

At a boundary anchor the initial distance from the prescribed matching is at most its own defect, which is included in this bound.

For an attachment edge xy with $x \in P, y \in C$, choose ξ refining $\pi_x \wedge \pi_y$. Then $d_M(\rho_C, \xi) \leq 2D_C + d(\pi_y) \leq 3D_C$. By Equation (60), making ρ_C refine the partition at x costs at most $3D_C$ merges. Normalize the state by

$$\hat{\pi}_y = \rho_C \quad (y \in C), \quad \hat{\pi}_x = \pi_x \vee \bigvee_{C: x \in N_P(C)} \rho_C \quad (x \in P).$$

If b is the total number of added backbone merges, then

$$b \leq 3rD, \quad \hat{\delta} = \delta_{\text{on}} + b. \quad (62)$$

The normalized state is legal. Coarsening endpoint partitions joins even intersection cells; within Q the endpoint matchings agree; across an attachment the seed refines the new backbone partition. Boundary constraints are unchanged.

The code retains the *original* backbone from Equation (58), followed by its forward coarsening record, at cost at most $(2rp^2)^{3rD}$. Once seeds are known, reconstruct each original off-path partition by a matching within distance $2D_C$ of its seed, then by $d(\pi_y)$ merges. The number of these records is at most

$$\prod_C \prod_{y \in C} (4p^2)^{2D_C} p^{2d(\pi_y)} \leq (4p^2)^{2rD} p^{2D}. \quad (63)$$

All original partitions are recoverable from these records. No inverse splitting of a repaired backbone is assumed.

More explicitly, fix orders on roles, replicas, tree edges, and components once and for all. Within each even cell choose the matching that pairs successive replicas in that order. Choose the first shortest switch word whenever several words have the same length; use the first merge word realizing a required coarsening. These conventions make the encoder a function of the original state, rather than a relation with an unspecified multiplicity. The defect vector fixes the word length budgets. Shorter words can be padded by a null operation; this is absorbed by the factors $4p^2$ and $2rp^2$ already used above.

The decoder first reads the defect vector and the original backbone partitions. It applies the recorded forward merges to those partitions, obtaining the repaired backbone. It then reads one seed per component (or uses the prescribed boundary seed). For each off-path role, its switch word recovers the chosen matching refinement, and its merge word recovers the original partition. Block identifiers can always be the smallest replica in the current block, so every merge is unambiguous. The resulting partitions at all roles are exactly the original state. In particular, two states with the same record are equal. Invalid records may be included when counting possibilities; they only enlarge the upper bound. No step of this decoder needs to infer an original partition by splitting a join.

D.4 Charging the seed entropy

Fix the original backbone and a forward coarsening that permits a normalized state. For a boundary-free component C of Q , put

$$\theta_C = \bigwedge_{x \in N_P(C)} \hat{\pi}_x, \quad e_C = p - |\theta_C|.$$

Its neighbor set is nonempty, because the core has no detached component. A compatible seed refines θ_C , so its cells must be even. If their sizes are $2t_1, \dots, 2t_j$, the seed count is

$$\prod_{i=1}^j (2t_i - 1)!! \leq (2p)^{\sum_i (t_i - 1)} = (2p)^{e_C}. \quad (64)$$

Apply the separator layers to the normalized state. By interval containment, every layer crossing $I(\theta_C)$ contains all backbone neighbors of C . No role of C lies in any layer, because its partition is a matching. Consequently C is an active component at each of these e_C layers. At a minimum layer there are at most a_* such components. There are at most $\hat{\delta}$ nonminimum layers, each with at most r components. Thus

$$\sum_C e_C \leq a_*(p - 1) + r\hat{\delta}. \quad (65)$$

The crude bound $\sum_C e_C \leq r(p - 1)$ absorbs the powers of two in Equation (64). All free seeds together have at most

$$2^r p^{a_* p + r\delta_{\text{on}} + 3r^2 D}$$

choices. Boundary-touching components have prescribed seeds.

The coarsening can depend on the seed in the construction. This does not invalidate the count: first count its range, then count the fiber above each fixed coarsening. No probabilistic independence between these combinatorial data is used.

Formally, let $\mathcal{B}$ denote a recorded original backbone and forward merge word. For each fixed $\mathcal{B}$ in the range of the encoder, its repaired partitions, and hence the partitions θ_C , are determined.

Sum the seed bound over this range, and for each seed sum the local reconstruction bound. The number of full records is at most the number of possible $\mathcal{B}$ times the largest such fiber bound. This elementary range–fiber inequality is valid even though the map producing $\mathcal{B}$ used the seed. It is the reason the normalization does not reverse the counting direction.

Completion of Theorem 3.4. Multiply the counts for the defect vector, the backbone, its forward merges, the off-path reconstruction, and the seeds. The exponent of p is

$$a_*p + (r + 2)\delta_{\text{on}} + (3r^2 + 10r + 2)D \leq a_*p + K_r\delta.$$

The remaining factors are bounded by

$$2^{2rp} 100^{2pr} (2r)^{3rD} 4^{2rD} \leq [200^r (2r)^{2r^2} 4^{r^2}]^{2p},$$

using $D \leq rp$. If $r = 0$, only the empty state at defect zero exists. This proves all cases. $\square$

E Conditional deviations and synchronized lower witnesses

We complete the conditional estimate used after Lemma 4.2. All internal randomness is exposed before the separator trials are selected. This order is essential: unconditional independence of those trials would be false.

E.1 A uniform conditional moderate deviation

Fix an active connected component K with k roles, and let $B \neq \emptyset$ be its roles adjacent to S . Condition on all internal arrays of K . Summing over $K \setminus B$ gives coefficients γ_{x_B} . At a fixed separator label,

$$T_K = \sum_{x_B} \gamma_{x_B} \prod_{z \in B} \eta_z(x_z),$$

where the η_z are independent sign vectors: each is the product of the crossing-edge signs incident to that role.

We use the fixed-degree Boolean hypercontractive inequality $\|P\|_q \leq (q - 1)^{d/2} \|P\|_2$ for a sign polynomial of degree at most d and $q \geq 2$ [16]. Here all applications have fixed degree and fixed q ; their constants do not grow with n .

Choose $z_0 \in B$ and put

$$Q = \sum_{x_B} \gamma_{x_B}^2, \quad R_j = \sum_{x_{B \setminus \{z_0\}}} \gamma_{j, x_{B \setminus \{z_0\}}}^2.$$

Distinct summation assignments give distinct Walsh monomials, because K is connected. For a singleton component use $\gamma = 1$. Thus $\mathbb{E}Q = \prod_{x \in K} m_x \asymp n^k$ and $\mathbb{E}R_j \asymp n^{k-1}$. Hypercontractivity and Minkowski give

$$\mathbb{E}Q^2 \leq C(\mathbb{E}Q)^2, \quad \mathbb{E}R_j^{16} \leq Cn^{16(k-1)}.$$

Paley–Zygmund, followed by Markov and a union bound, yields an internal event $\mathcal{G}_K$ of probability at least $c > 0$ on which

$$cn^k \leq Q \leq Cn^k, \quad \max_j R_j \leq n^{k-3/4}. \quad (66)$$

The last failure probability is $O(n^{-3})$: its one-coordinate bound is $O(n^{-4})$ and there are $O(n)$ coordinates. Increasing the constant in the upper bound for Q leaves a positive probability for the intersection.

Here is the good-set calculation with the constants separated from the dimension. If $\mathbb{E}Q \geq c_1 n^k$ and $\mathbb{E}Q^2 \leq C_1(\mathbb{E}Q)^2$, Paley–Zygmund gives $\mathbb{P}\{Q \geq (\mathbb{E}Q)/2\} \geq (4C_1)^{-1} =: q_1$. Choose a fixed A so that $\mathbb{P}\{Q > An^k\} \leq q_1/4$ by Markov. For each j , the sixteenth-moment bound gives $\mathbb{P}\{R_j > n^{k-3/4}\} \leq Cn^{-4}$; summing over at most $C'n$ indices leaves $O(n^{-3}) \leq q_1/4$ for large n . The intersection therefore has probability at least $q_1/2$. All choices depend on the fixed component and label-size comparison constants, not on a separator tuple.

For every full internal realization in $\mathcal{G}_K$, expose the vectors η_z for $z \neq z_0$. The remaining sum is $Y = \sum_j z_j \eta_{z_0}(j)$, with $\sigma^2 = \sum_j z_j^2$. Uniformly over those internal realizations,

$$\mathbb{E}\sigma^2 = Q, \quad \mathbb{E}\sigma^4 \leq CQ^2, \quad \mathbb{E}|z_j|^{32} \leq CR_j^{16} \leq Cn^{16k-12}.$$

A second Paley–Zygmund argument, an upper Markov bound, and the union bound at $|z_j| = n^{k/2-1/4}$ show that, with conditional probability at least $c' > 0$,

$$c'n^k \leq \sigma^2 \leq C'n^k, \quad \max_j |z_j| \leq n^{k/2-1/4}. \quad (67)$$

The coefficient failure probability is again $O(n^{-3})$. If $|B| = 1$, there is no exposure at this step; the assertions follow directly from Equation (66).

For clarity, after a full internal realization is fixed, the same Paley–Zygmund argument applies to the nonnegative random variable σ^2 , whose conditional mean is Q . Its second moment is $\mathbb{E}\sigma^4 \leq CQ^2$. A sufficiently large fixed upper cutoff removes less than a quarter of the resulting positive probability. For the coefficients the displayed thirty-second moment yields $\mathbb{P}\{|z_j| > n^{k/2-1/4} \mid \mathcal{I}\} \leq Cn^{-4}$, because the denominator is n^{16k-8} and the numerator is at most Cn^{16k-12} . The union bound again costs only $O(n)$. These estimates hold for every internal realization in $\mathcal{G}_K$, not merely on average over that event.

Lemma E.1 (Weighted sign tilt). *For $Y = \sum_j z_j \varepsilon_j$ and $\sigma^2 = \sum_j z_j^2 > 0$, suppose $t \geq \sigma$ and $8t \max_j |z_j|/\sigma^2 \leq 1$. Then*

$$\mathbb{P}\{Y \geq t\} \geq \frac{1}{2} \exp(-96t^2/\sigma^2).$$

Proof. Tilt by $\lambda = 8t/\sigma^2$, so the density is $e^{\lambda Y}/\mathbb{E}e^{\lambda Y}$. Since $x/2 \leq \tanh x \leq x$ on $[0, 1]$, the tilted mean lies between $4t$ and $8t$, and the tilted variance is at most σ^2 . Chebyshev gives probability at least $1 - 1/9 - 1/16 > 1/2$ to the band $t \leq Y \leq 12t$. On this band the inverse likelihood ratio is at least $e^{-12\lambda t}$, since $\sum_j \log \cosh(\lambda z_j) \geq 0$. Integrating on the band proves the claim. $\square$

Take $t = \varepsilon n^{k/2} \sqrt{\log n}$. For each fixed $\varepsilon > 0$, Equation (67) satisfies the hypotheses for all sufficiently large n , uniformly in the internal realization. Integrating only over the event in Equation (67) gives

$$\mathbb{P}\{|T_K(x_S)| \geq \varepsilon n^{k/2} \sqrt{\log n} \mid \mathcal{I}\} \geq cn^{-C\varepsilon^2} \quad \text{on } \mathcal{G}_K, \quad (68)$$

where $\mathcal{I}$ contains all internal arrays. The constants and the threshold for n are uniform over the full realizations in $\mathcal{G}_K$.

E.2 Synchronizing the witnesses

Let $K_1, \dots, K_a$ be the active components. Choose $N = \Theta_G(n)$ separator tuples whose labels are distinct in every separator role across the tuples. Condition on all component-internal arrays. Crossing coordinates for distinct tuples and distinct components are disjoint, so these trials are independent under this full conditional law.

The internal good events use disjoint arrays and have joint probability at least a positive constant. On this joint event, Equation (68) gives success probability at least $cn^{-C\varepsilon^2}$ for simultaneous success in all a components at any one tuple. Choose ε so that $C\varepsilon^2 < 1/2$. The conditional probability that no tuple succeeds is then at most $\exp(-c\sqrt{n})$. Therefore

$$\mathbb{E} \max_{x_S} \prod_{j=1}^a |T_{K_j}(x_S)| \geq c_G n^{|D|/2} (\log n)^{a/2}. \quad (69)$$

This integration uses independence given $\mathcal{I}$, not independence given only the good event. Explicitly, if $\mathbb{P}(\mathcal{G}) \geq c_0$ and each of N conditionally independent trials has conditional success probability at least ρ_n on every full realization in $\mathcal{G}$, then

$$\mathbb{P}(\mathcal{G} \cap \{\text{some success}\}) \geq c_0[1 - (1 - \rho_n)^N].$$

If $a = 0$, the product is one. If S is empty, then $a = 0$ and no packing is needed.

F Factor moments, layerwise counting, and lower tails

This appendix proves Theorem 5.2 and Proposition 5.4 and the factor statement Theorem 1.4. The law and preprocessing are those of Section 5; the separate weighted sign theorem is not an additional hypothesis here.

The proof retains the original separator layers before summing over replica states. This is the point at which a graph-only count would lose the factor-specific information. We first account for scalar components, then prove the layerwise count and the conditional lower bound.

F.1 Detached sign components

We first prove the scalar count needed for the moment statement Equation (75). For a connected detached graph K with k roles and at least one edge, let $\delta = \sum_x (p - |\pi_x|)$. There are constants B, L depending only on K such that the number of legal scalar states is at most

$$B^{2p} p^{p+L\delta}. \quad (70)$$

To see this, fix a rooted spanning tree and a canonical matching refinement at every role. The root matching has at most $(2p)^p$ choices. Across a tree edge, Equation (59) bounds the matching distance by the sum of endpoint defects. Counting switch records gives at most $p^{3(d_x+d_y)}$ choices for a child refinement. Recover its original partition with at most p^{2d_y} merge records, and recover the root partition using at most $p^{2d_{\text{root}}}$ additional merge records. Multiplying over the tree gives $(2p)^p p^{(3k+2)\delta}$. Defect allocations contribute a fixed exponential base in p , proving Equation (70).

The positive scalar expansion consequently gives

$$\mathbb{E}|Z_K|^{2p} \leq B^{2p} n^{pk} p^p \sum_{\delta \geq 0} (p^L/n)^\delta.$$

For a matching lower bound, impose the same perfect matching at every role. These $(2p-1)!!$ distinct exact states are all legal, each with weight $\prod_x (m_x)_p$. For $p \leq \min_x m_x/2$ this weight is at least $\prod_x (m_x/2)^p$, and $(2p-1)!! \geq p! \geq (p/e)^p$. Taking roots and using monotonicity between even moments proves

$$\|Z_K\|_q \asymp n^{k/2} \sqrt{q}, \quad 2 \leq q \leq C_0 \log(2n). \quad (71)$$

At $q = 1$, second and fourth moments give $\mathbb{E}|Z_K| \asymp n^{k/2}$. For a higher-arity sign component, factor legality implies two-section legality for the upper count, and the common-matching states remain legal for the lower count. A unary-only singleton is an ordinary sign sum and satisfies the same assertion.

In the typed model, Equation (34) makes these L^q norms multiply independently of the core. Proposition 3.8 and the core L^1 lower bound supply its sharp logarithmic factor. The global upper and lower transfers have constants independent of q , proving Equation (75) with d_0 equal to the number of detached random components. For $1 \leq q < 2$, sandwich between L^1 and L^2 . Markov at $q = C_A \log(2n)$ gives the upper-tail exponent $(a_* + d_0)/2$.

F.2 Fixed moments for independent factor laws

We need a dimension-free fixed-order replacement for Boolean hypercontractivity. If independent symmetric variance-one variables have bounded $2M$ moments, then for Hilbert-space vectors v_i ,

$$\left\| \sum_i \xi_i v_i \right\|_{L^{2M}} \leq A_M \left(\sum_i \|v_i\|^2 \right)^{1/2}. \quad (72)$$

Expand the M inner products in the norm power. Odd multiplicities vanish by symmetry. Bound inner products by products of lengths. Every surviving index word admits a pairing, and summing over all $(2M - 1)!!$ pairings only overcounts. The bounded variable moments contribute a constant depending on M , not on the number of indices. This proves the inequality.

Iterate it through the fixed number of independent occurrence groups, using a Hilbert direct sum for coefficient lists and Minkowski for their squared norms. Orthogonality identifies the final coefficient norm with the L^2 norm. Thus every such Hilbert-valued multilinear chaos satisfies $\|T\|_{2M} \leq A_{M,\mathcal{F}} \|T\|_2$, also for deterministic coefficients produced by conditioning. Interpolation at orders one, two, and four gives the lower stacking inequality of Lemma 4.1 with a law-dependent positive constant per occurrence.

F.3 The layerwise refinement

For a fixed original separator sequence $\mathbf{S} = (S_k)$, let $N(\mathbf{S})$ be the number of graph-legal states inducing it. Then

$$N(\mathbf{S}) \leq C_r^{2p} p^{\sum_k a(S_k) + K'_r \sum_k (|S_k| - s)}, \quad K'_r = 3r^2 + 11r + 2. \quad (73)$$

Here the statistics refer to the original, not normalized, slices.

Use the encoding of Appendix D. An effective backbone merge increases interval length by one and preserves containment. Normalizing off-path roles deletes all their D memberships. Thus, with $b \leq 3rD$,

$$\sum_k |S_k \triangle \hat{S}_k| = b + D \leq (3r + 1)D.$$

Since an active count is between zero and r , it follows that

$$\sum_k a(\hat{S}_k) \leq \sum_k a(S_k) + (3r^2 + r)D.$$

The seed interval argument now gives $\sum_C e_C \leq \sum_k a(\hat{S}_k)$ instead of Equation (65). The normalized slices depend only on the repaired backbone: all off-path intervals have width zero, regardless of the

position of their singleton interval. Therefore this inequality is valid in a fiber with fixed original slices. Count only repaired-backbone fibers admitting a completion with those original slices.

Adding the path, coarsening, reconstruction, and seed exponents gives

$$\sum_k a(S_k) + 2\delta_{\text{on}} + (3r^2 + 11r + 2)D.$$

The same fixed-base and defect-vector estimates as before prove Equation (73). This proves the required refinement without assuming it follows from the final unweighted bound.

F.4 Weighted trace expansion

For occurrence e , put $\lambda_e = \bigwedge_{v \in e} \pi_v$. Its cells of size j contribute $\mathbb{E}\xi_e^j$ in the exact trace. Symmetry removes all states with an odd factor cell; every remaining term is nonnegative. A covered role is even as a union of even factor cells, and an uncovered core role is even by its boundary matching. Factor legality implies pairwise legality in the two-section.

Write the cell sizes of λ_e as $2t_1, \dots, 2t_j$ and $d_e = \sum_i (t_i - 1) = p - |\lambda_e|$. Tail regularity implies

$$\prod_i \mathbb{E}|\xi_e|^{2t_i} \leq \tilde{C}_e^{2p} p^{\theta_e d_e}. \quad (74)$$

Indeed $(2t)^{\theta t} \leq 2^{2\theta t} p^{\theta(t-1)}$ for $1 \leq t \leq p$. Under the finite profile, the corresponding bound is $C_e^{2p} n^{\beta_e d_e} p^{\gamma_e d_e}$.

By interval containment, every layer in $I(\lambda_e)$ contains all roles of e . Hence, for nonnegative weights w_e ,

$$\sum_e w_e d_e \leq \sum_k \sum_{e \subseteq S_k} w_e.$$

The label contribution is at most $C^{2p} n^{pr - \sum_k |S_k|}$. Combining these statements with Equation (73) proves Equation (36).

For fixed laws, minimum layers have charge at most b_* . Nonminimum layers are at most δ in number, and their charges are bounded by a fixed constant B . There are at most $2^{r(p-1)}$ subset sequences. Thus

$$\mathbb{E} \operatorname{tr}((H_c H_c^*)^p) \leq C^{2p} n^{p(r-s)+s} p^{b_* p} \sum_{\delta \geq 0} (p^{K'_r + B} / n)^\delta.$$

Taking $p = \lceil \log n \rceil$ proves the core upper bound of Theorem 5.2. Increasing the threshold constant in Markov's inequality gives each prescribed polynomial upper tail.

For the cut-stable profile, put $F(S) = b(S) + K'_r(|S| - s)$. On a minimizing cut, $F(S) \leq b_*^{\text{mw}}$. If nonminimizers exist, their cost gap has a positive minimum η , since the shape is fixed. Choose a fixed L bounding $(F(S) - b_*^{\text{mw}})_+$. The sequence sum is at most

$$n^{pr - (p-1)\kappa_*} p^{(p-1)b_*^{\text{mw}}} \left[|\mathcal{S}_*| + (|S| - |\mathcal{S}_*|) n^{-\eta} p^L \right]^{p-1}.$$

Take $p = \lfloor c \log n \rfloor$ for fixed $0 < c < C_0$. The bracket and the root factor $n^{\kappa_*/(2p)}$ are bounded. This proves the expectation bound. Markov at threshold R times the root estimate gives R^{-2p} ; choosing R large enough gives n^{-A} without enlarging the supplied window.

F.5 The factor lower bound

Fix a minimum separator maximizing $a(S) + \Theta(S)$. Condition all occurrences meeting active components and all occurrences contained in S . Their weight is

$$w(x_S) = \prod_{K \text{ active}} T_K(x_S) \prod_{e \subseteq S} \xi_{x_e}^{(e)}.$$

A factor cannot meet two components of the two-section minus S outside S . Assign the remaining occurrences to the boundary side met by their nonseparator roles. Minimality of S observes each separator role on both sides, by the path argument in Lemma 4.2. The weighted coefficient flattening is therefore exactly the same direct sum of weighted all-ones blocks. The general-law stacking inequality reduces the lower bound to one simultaneous maximum of $|w(x_S)|$.

For completeness, the needed active-component estimate follows with full conditioning as follows. Fix an active component K of k roles and a crossing occurrence e_0 . Put $B = e_0 \cap K$, $b = |B| \geq 1$. Let $\mathcal{I}$ contain the full arrays of occurrences wholly inside K , and let O comprise the other crossing slices. For a fixed separator tuple write

$$T_K = \sum_{y \in I_B} z_y(\mathcal{I}, O) \xi_{x_{e_0 \cap S}, y}^{(e_0)}, \quad \sigma^2 = \sum_y z_y^2, \quad V(\mathcal{I}) = \mathbb{E}(T_K^2 \mid \mathcal{I}).$$

Every role of K is covered by a factor; every role outside B occurs in a factor other than e_0 . Distinct complete assignments therefore give orthogonal monomials. The fixed-moment inequality gives

$$\mathbb{E}V \asymp n^k, \quad \mathbb{E}V^2 \leq Cn^{2k}, \quad \mathbb{E}|z_y|^{2M} \leq C_M n^{M(k-b)}.$$

Paley–Zygmund and an upper cutoff give an internal set of positive probability on which $cn^k \leq V \leq Cn^k$. On each of its full realizations, $\mathbb{E}_O \sigma^2 = V$ and $\mathbb{E}_O \sigma^4 \leq CV^2$. Hence $cn^k \leq \sigma^2 \leq Cn^k$ has conditional probability at least a fixed $q_0 > 0$.

The coefficient maximum has unconditional failure probability

$$\mathbb{P}\{\max_y |z_y| > n^{k/2-1/4}\} \leq C_M n^{b-Mb+M/2} = o(1)$$

for sufficiently large fixed M . Its conditional failure probability, as a function of $\mathcal{I}$, has expectation $o(1)$. Markov therefore removes an $o(1)$ set of internal realizations and leaves a set $\mathcal{G}_K$ of positive probability on which

$$\mathbb{P}_O\{cn^k \leq \sigma^2 \leq Cn^k, \max_y |z_y| \leq n^{k/2-1/4} \mid \mathcal{I}\} \geq q_0/2.$$

The crossing slices have identical marginal laws for every separator tuple, so the same full internal set works for all tuples.

To make the conditioning step explicit, let $q_n(\mathcal{I})$ be the conditional probability of the coefficient failure event. Its expectation is $o(1)$ by the preceding union bound. Hence $\mathbb{P}\{q_n(\mathcal{I}) > q_0/2\} \leq 2\mathbb{E}q_n/q_0 = o(1)$. Remove this set from the positive-probability internal variance set. On each remaining full realization, intersecting the variance event with the coefficient event loses at most $q_0/2$ of conditional probability. There is no union bound over separator tuples here: given the same internal arrays, the as-yet unexposed crossing slices have identical product laws for every fixed tuple. Their conditional failure-probability functions are therefore the same function of $\mathcal{I}$. Independence is used later only for tuples with disjoint crossing coordinates.

The tail assumption with $\theta_e \leq 2$ implies $\|\xi_e\|_q \leq Cq$. Its moment generating function is analytic near zero, by the absolute power series and $m! \geq (m/e)^m$. For $k(u) = \log \mathbb{E}e^{u\xi_e}$, symmetry and

variance one give $k(u) = u^2/2 + O(u^4)$, $k'(u) = u + O(u^3)$, and $k''(u) = 1 + O(u^2)$. For $T = \sum_i z_i \xi_i$, tilt with $\lambda = 2t/\sigma^2$. If $t \geq 4\sigma$ and $\lambda \max_i |z_i|$ is sufficiently small, the tilted mean lies in $[1.8t, 2.2t]$ and its variance is at most $2\sigma^2$. Chebyshev gives tilted probability at least $1/2$ to $t \leq T \leq 3t$. Since the logarithmic moment generating function is nonnegative, undoing the tilt gives

$$\mathbb{P}\{T \geq t\} \geq \frac{1}{2}e^{-6t^2/\sigma^2}.$$

At $t = \varepsilon n^{k/2} \sqrt{\log n}$ the hypotheses hold uniformly on the coefficient event above. Thus on every realization in $\mathcal{G}_K$ the component success probability is at least $cn^{-C\varepsilon^2}$.

For a separator-only factor, Paley–Zygmund on $|\xi_e|^q$ and the two-sided moment bounds give

$$\mathbb{P}\{|\xi_e| \geq \frac{1}{2}c_e q^{\theta_e/2}\} \geq \frac{1}{4} \left(\frac{c_e}{C_e 2^{\theta_e/2}} \right)^{2q}.$$

Choose $q = \alpha_e \log n$ with fixed small $\alpha_e > 0$. This gives the required logarithmic threshold with an arbitrarily small positive polynomial success exponent.

Intersect the independent internal good sets. On every full realization in their intersection, use $\Theta(n)$ separator tuples with all role labels distinct across trials. Every crossing or separator-only occurrence has a nonempty separator support, so its coordinates are disjoint across these trials. Choose ε and the finitely many α_e so that their total success exponent is less than one. Conditional independence then gives

$$\mathbb{E} \max_{x_S} |w(x_S)| \geq cn^{|D|/2} (\log n)^{[a(S) + \Theta(S)]/2}.$$

For an empty separator there is no active or separator-only factor and $w = 1$. The flattening proves the core lower bound.

Finally, for a fixed-law detached component, $\mathbb{E} Z_K^2 = \prod_{v \in K} m_v$ and $\mathbb{E} Z_K^4 \leq C(\prod_{v \in K} m_v)^2$. Interpolation gives $\mathbb{E}|Z_K| \asymp n^{|K|/2}$. Restore these independent expectation factors and unused-role sizes in Equation (34). This completes Theorem 5.2, and hence Theorem 1.4.

The full trace restoration is exact:

$$\mathbb{E} \operatorname{tr}((HH^*)^p) = d(n)^{2p} \prod_j \mathbb{E}|Z_j|^{2p} \mathbb{E} \operatorname{tr}((H_c H_c^*)^p).$$

For example, if $H = ZI_n$ and $Z = \sum_{i,j} g_{ij} \sim N(0, n^2)$, its trace moment is $n^{2p+1}(2p-1)!!$. Omitting the detached moment would lose this factorial, which cannot be absorbed into C^{2p} .

F.6 Detached scalars and the moment window

The distinction between a core matrix and detached scalar factors also persists beyond expectation. If d_0 is the number of connected detached random components, the moment statement has the form

$$|||M_\alpha|||_{L^q} \asymp_{\alpha, C_0} n^{(v+h-s)/2} (\log n)^{a_*/2} q^{d_0/2}, \quad 2 \leq q \leq C_0 \log(2n). \quad (75)$$

The logarithmic factor is already present at fixed q . Replacing it by $q^{a_*/2}$ would contradict the diagonal example above. Conversely, a high-probability lower bound for the full matrix cannot be obtained by ignoring the possibility that a detached scalar is small. Expectation, growing moments, and overwhelming-probability statements require separate quantifiers.

G Dimension-free bounded-noise comparison

Proof of Theorem 5.6. First, for deterministic vectors $b_i \in B$ and $|a_i| \leq 1$,

$$\left\| \sum_i a_i \varepsilon_i b_i \right\|_{L^p} \leq \left\| \sum_i \varepsilon_i b_i \right\|_{L^p}. \quad (76)$$

Choose independent signs η_i with $\mathbb{E}\eta_i = a_i$. Conditional Jensen for the convex function $\|\cdot\|^p$ proves the inequality, since the products $\varepsilon_i \eta_i$ are independent uniform signs. If $0 < m \leq a_i \leq M$, applying this inequality both to a_i/M and to m/a_i gives two-sided factors m and M .

For one group, take an independent copy X'_i and put $D_i = X_i - X'_i$. These differences are independent and symmetric, so jointly they have the law $D_i = \varepsilon_i R_i$, with $R_i = |D_i|$ and independent uniform signs independent of all magnitudes. This representation also holds when $R_i = 0$. Since $R_i \leq 2K$ and $\mathbb{E}R_i^2 = 2$,

$$\mu_i := \mathbb{E}R_i \geq K^{-1}.$$

Conditional Jensen using $\mathbb{E}X'_i = 0$, followed by contraction conditional on the magnitudes, yields

$$\left\| \sum_i X_i b_i \right\|_{L^p} \leq \left\| \sum_i D_i b_i \right\|_{L^p} \leq 2K \left\| \sum_i \varepsilon_i b_i \right\|_{L^p}.$$

For the reverse inequality, Jensen over the magnitudes and lower contraction give

$$\left\| \sum_i D_i b_i \right\|_{L^p} \geq \left\| \sum_i \mu_i \varepsilon_i b_i \right\|_{L^p} \geq K^{-1} \left\| \sum_i \varepsilon_i b_i \right\|_{L^p}.$$

Minkowski bounds the left side by $2 \|\sum_i X_i b_i\|_{L^p}$. This proves the comparison for one group. Condition on all other groups, apply these inequalities to conditional p th moments, and integrate. Replacing the groups one at a time gives exactly Q factors of $2K$, proving Equation (38). $\square$

H An independent proof for the gated Khatri–Rao product

Proof of Theorem 5.8. Write $\circ$ for entrywise matrix multiplication. If P, Q are positive semidefinite and $P = \sum_\ell u_\ell u_\ell^*$, then

$$P \circ Q = \sum_\ell \text{diag}(u_\ell) Q \text{diag}(u_\ell)^* \preceq \|Q\| \text{diag}(P).$$

Using $K^* K = (A_1^* A_1) \circ \dots \circ (A_r^* A_r)$ and iterating gives

$$\|K\| \leq \|A_1\| \prod_{t=2}^r \max_j \|A_t[:, j]\|_2. \quad (77)$$

For a standard Gaussian matrix A , $\mathbb{E}\|A\| \leq C\sqrt{n}$. For completeness, a $1/4$ -net of the unit sphere has at most 9^n points by disjoint-ball volume comparison. Approximating both singular vectors bounds $\|A\|$ by twice the largest $|x^* A y|$ on the two nets. Each fixed form is standard normal. The union bound $2 \cdot 9^{2n} \exp(-t^2/2)$, capped at one and integrated, gives the claimed estimate. Since each column norm is at most $\|A\|$, independence in Equation (77) implies $\mathbb{E}\|K\| \leq C_r n^{r/2}$.

Choose J as the smallest maximizer of $|D_{JJ}|$. It depends only on D , and is independent of every A_t . The J th column gives

$$\|KD\| \geq |D_{JJ}| \prod_t \|A_t[:, J]\|_2.$$

For a standard Gaussian vector $g \in \mathbb{R}^n$, the second and fourth moments of its Euclidean norm are n and $n(n+2)$. Paley–Zygmund therefore gives $\mathbb{E} \|g\|_2 \geq c\sqrt{n}$. Conditioning on D and its selected column, and using the upper bound above, yields

$$c_r n^{r/2} \mathbb{E} \max_j |D_{jj}| \leq \mathbb{E} \|KD\| \leq C_r n^{r/2} \mathbb{E} \max_j |D_{jj}|. \quad (78)$$

For $p = \max(2, \log n)$, maximum is bounded by the ℓ_p norm, so

$$\mathbb{E} \max_j \prod_h |g_{jh}| \leq n^{1/p} \prod_h \|g_{1h}\|_{L^p} \leq C_q (\log(2n))^{q/2}.$$

Conversely choose $\alpha > 0$ with $\alpha q/2 < 1$ and set $t = \sqrt{\alpha \log n}$. Integrating the normal density on $[t, t+1/t]$, for $t \geq 1$, gives $\mathbb{P}\{|g| \geq t\} \geq ct^{-1}e^{-t^2/2}$. At each column, all q gates exceed t in absolute value with probability at least $c_q t^{-q} n^{-\alpha q/2}$. Its product with n tends to infinity. Independence across columns thus gives a gate product at least t^q with probability tending to one. This proves the reverse maximum estimate and, by Equation (78), the theorem. $\square$

I Local contraction and the complete aspect reduction

We prove Theorem 6.3 by establishing the reductions in their required order.

I.1 All-moment local contraction

For coefficients in any normed space, write $F_b = \sum B_{i_1, \dots, i_Q} \prod_t b_t(i_t) \varepsilon_{i_t}^{(t)}$. If $0 < c_t \leq |b_t(i)| \leq C_t$, then for every $p \geq 1$,

$$\left(\prod_t c_t \right) \|F_1\|_p \leq \|F_b\|_p \leq \left(\prod_t C_t \right) \|F_1\|_p. \quad (79)$$

For $|d_i| \leq 1$, choose independent auxiliary signs of mean d_i . Conditional convexity bounds $\|\sum_i d_i \varepsilon_i v_i\|^p$ by the auxiliary expectation of $\|\sum_i \eta_i \varepsilon_i v_i\|^p$. The products $\eta_i \varepsilon_i$ are independent uniform signs. Average and iterate through the groups with $d_i = b_t(i)/C_t$. For the reverse inequality, contract the weighted chaos by $c_t/b_t(i)$.

Apply Equation (79) with $b_t = n^{-\gamma_t} a_{t,n}$. This removes all amplitudes with a factor n^Γ and constants independent of both p and n . It is done before selecting separator labels; no assertion of identical conditional weighted laws is needed.

I.2 Positive aspects and the minimum face

Assume first that all remaining roles have positive exponents and all factors have arity at least two. Remove sign factors on a single boundary, deterministic isolated middle roles, and detached components. Let G be the resulting core and $B_G = \sum_{v \in G} \beta_v$.

Factor-legal states are legal in its two-section. Coarea gives the weighted defect

$$\Delta_\beta = \sum_v \beta_v (p - |\pi_v|) - \kappa(p - 1) = \sum_k (\beta(S_k) - \kappa) \geq 0,$$

and hence $\sum_v \beta_v |\pi_v| = p(B_G - \kappa) + \kappa - \Delta_\beta$.

Let $\mathcal{M}$ be the full family of minimum-cost separators and fix $S_0 \in \mathcal{M}$. The subspace

$$L = \{x : x(S) = x(S_0) \text{ for every } S \in \mathcal{M}\}$$

is defined over the rationals. Positive coordinates and the strict inequalities $x(T) > x(S_0)$ for $T \notin \mathcal{M}$ define a relatively open neighborhood of β in L . Rational points are dense in L , so choose one in that neighborhood and clear denominators. This gives a positive integer vector w with precisely the same minimizing family.

Put $\kappa_w = \min_S w(S)$. Finiteness gives a constant $c_0 > 0$ such that

$$\Delta_\beta \geq c_0 t_w, \quad t_w = \sum_k (w(S_k) - \kappa_w) \in \mathbb{N}. \quad (80)$$

Explicitly take the minimum of $(\beta(T) - \kappa)/(w(T) - \kappa_w)$ over nonminimizers; if there are none take $c_0 = 1$.

Replace role v by a clique of w_v clones, retaining all its boundary memberships, and replace each graph edge by a complete join between the two fibers. For a deleted clone set T , let S_T be the original roles whose whole fibers are deleted. Every surviving fiber is connected. Quotient connectivity, boundary touching, and attachment are exactly those of $G - S_T$. Thus T is a separator if and only if S_T is. Undoing a partial fiber deletion preserves separation and decreases size. Every minimum clone separator therefore consists of full fibers, and corresponds precisely to a member of $\mathcal{M}$. Its minimum size is κ_w and its maximum active count is a_* .

Copy each original partition to every clone in its fiber. Within a fiber its meet is the original even partition; between fibers legality follows from the original edge. This is an injection into legal clone states of defect t_w . By Theorem 3.4, their number is at most $C_w^{2p} p^{a_* p + K_w t_w}$. The positive label weights and Equation (80) now give

$$\mathbb{E} \operatorname{tr}((H_G H_G^*)^p) \leq C^{2p} n^{p(B_G - \kappa) + \kappa} p^{a_* p} \sum_{t \geq 0} (p^{K_w} / n^{c_0})^t. \quad (81)$$

For any fixed logarithmic window the ratio tends to zero. Choose $p = \max(2, \lceil \log(2n) \rceil, \lceil q/2 \rceil)$. Taking roots proves the core upper bound at scale $n^{(B_G - \kappa)/2} (\log(2n))^{a_*/2}$, including bounded q .

I.3 The aspect-sensitive lower bound

Positive costs make each minimum-cost separator inclusion-minimal. Its conditional coefficient flattening, proved exactly as in Lemma 4.2, has norm

$$\left(\prod_{v \in G \setminus (D \cup S)} m_v \right)^{1/2} \max_{x_S} \prod_{K \text{ active}} |T_K(x_S)|.$$

The minimality path observes every separator role on both sides. A factor cannot bridge two different components outside S . These facts prove the flattening for the actual unequal dimensions.

We spell out the change in the conditional estimate, since a comparable-size theorem does not imply it. Fix a component K and a crossing occurrence with nonempty inside support B . Put $B_K = \sum_{v \in K} \beta_v$. With the notation of the factor lower proof,

$$\mathbb{E} T_K^2 \asymp n^{B_K}, \quad \mathbb{E} z_y^2 \asymp n^{B_K - \beta(B)}, \quad \mathbb{E} V^2 \leq C n^{2B_K}.$$

Fixed Hilbert moments give the same conditional Paley–Zygmund estimates for σ^2 on a positive-measure set of full internal realizations. For any fixed $M > 2$,

$$\mathbb{P}\{\max_y |z_y| > n^{B_K/2 - \beta(B)/4}\} \leq C_M n^{\beta(B)(1-M/2)} = o(1).$$

Conditional Markov removes an $o(1)$ set of internal realizations. On each remaining realization, the simultaneous variance and coefficient event has probability bounded below, uniformly over separator tuples.

For $t = \varepsilon n^{B_K/2} \sqrt{\log(2n)}$, the sign-tilt hypothesis follows from

$$\frac{t \max_y |z_y|}{\sigma^2} = O\left(n^{-\beta(B)/4} \sqrt{\log(2n)}\right) \rightarrow 0.$$

The one-trial conditional success probability is at least $cn^{-C\varepsilon^2}$. If $S \neq \emptyset$, there are at least cn^{b_S} tuples distinct in every separator role, where $b_S = \min_{v \in S} \beta_v > 0$. Choose ε so that the sum of component success exponents is less than b_S . Full conditional independence yields

$$\mathbb{E} \max_{x_S} \prod_K |T_K(x_S)| \geq cn^{\beta(D)/2} (\log(2n))^{a(S)/2}.$$

When there are no active components the product is one. Combining this with the flattening and a maximizing minimum-cost cut proves the core lower bound.

For a detached higher-arity component, orthogonality and fixed fourth moments give $\mathbb{E}|Z_K| \asymp n^{\beta(K)/2}$. For growing upper moments use Equation (70): a unit ordinary defect loses at least $\min_{v \in K} \beta_v > 0$ in the exponent of n . Thus

$$\mathbb{E}|Z_K|^{2p} \leq C^{2p} n^{p\beta(K)} p^p \sum_{\delta \geq 0} (p^L / n^{\min_{v \in K} \beta_v})^\delta.$$

Its logarithmic-window root is at most $Cn^{\beta(K)/2} \sqrt{q}$ at $p = \lceil q/2 \rceil$, with the exact second moment treating $p = 1$. Independent restoration gives one factor $\sqrt{q}$ per detached component and none for the expectation.

I.4 Bounded roles

Return to nonnegative aspects and write $Z = \{v : \beta_v = 0\}$. Their nonempty label sets have uniformly bounded sizes. Fix consistent zero-boundary labels and let H_b be the corresponding rectangular compression. There are boundedly many such blocks, and $\|H_b\| \leq \|H\| \leq \sum_b \|H_b\|$.

For each occurrence e and tuple u on its zero coordinates, introduce an auxiliary sign multiplying that entire array slice. A fixed zero-middle assignment z acquires the character

$$\chi_z(\eta) = \prod_{e: e \cap Z \neq \emptyset} \eta_{e, (z, b)|_{e \cap Z}}.$$

Two assignments differing on a factor-covered zero-middle role have different characters: the occurrence label is part of the slice identifier. Unused zero-middle roles instead give a deterministic multiplicity L_0 with $1 \leq L_0 \leq C$. If F_z is the fixed-assignment contribution with these unused sums omitted, Fourier projection gives exactly

$$\mathbb{E}_\eta[\chi_z(\eta) H_b(\varepsilon^\eta)] = L_0 F_z(\varepsilon).$$

Every fixed slice flip preserves the independent sign law. Minkowski therefore bounds $L_0 \|F_z\|_p$ by $\|H_b\|_p$. Conversely, each block is a sum of only boundedly many such terms. All consistent assignments have the same reduced incidence law, though they need not be independent. Summing block norms proves two-sided comparison between $\|H\|_p$ and $\|F_z\|_p$, with constants independent of p, n .

The resulting positive scopes are $e \cap P$. Empty scopes give scalar signs of modulus one. Scopes contained in one positive boundary give diagonal sign isometries. Neither changes the norm.

I.5 Unary absorption and completion

Multiply all unary occurrences at a positive middle role v into one independent sign vector $\sigma_v(x_v)$. If v belongs to a higher-arity occurrence, choose one such occurrence $e(v)$. Conditional on all unary arrays, replace its coordinates by

$$\tilde{\varepsilon}_{x_e}^{(e)} = \varepsilon_{x_e}^{(e)} \prod_{v:e(v)=e} \sigma_v(x_v).$$

This deterministic sign flip preserves the entire conditional product law, independently of the conditioned values, while algebraically absorbing the unary factors. It therefore proves an equality in distribution of the whole matrix, not merely a parity upper bound.

A graph-isolated middle role without factors contributes m_v . A graph-isolated middle role with unary factors contributes $\sum_i \sigma_v(i)$, whose expectation is comparable to $\sqrt{m_v}$ and whose L^q norm is at most $C\sqrt{m_v q}$. It is one detached random component regardless of how many unary occurrences were multiplied.

The positive higher-arity argument now applies to the remaining model. Restoring deterministic roles, detached scalars, bounded-role comparisons, and local amplitudes gives the exponent $\Gamma + (B_+ + h_\beta - \kappa)/2$ and the claimed logarithmic and moment factors. If $P = \emptyset$, a fixed-assignment contribution is a scalar product of signs of norm one; the same bounded comparison gives $f = \Gamma$ and $g = g_{\text{tail}} = 0$. Markov at logarithmic order proves the tail claim.

For rational exponents, determine zero roles, perform these finite structural operations, enumerate core separators, compare rational costs exactly, retain all ties, and maximize the active count. This terminates and proves the computability assertion in Theorem 6.3. It does not verify the analytic amplitude bounds of an arbitrary input program. For real exponents, the mathematical formula remains valid but exact evaluation requires a representation that can decide the finite cost comparisons.

J Finite Gaussian lifts and Hermite polarization

J.1 The uniform Gaussian lift: full proof

We prove Theorem 7.1.

For each of Q Gaussian occurrences on a nonempty scope e , adjoin one private middle role of size n and replace it by a sign factor on the enlarged scope. Denote this sign template by $\hat{\mathcal{F}}$, and its sign certificate by $(\hat{f}, \hat{g}, \hat{d}_0)$.

Let $S_m = m^{-1/2} \sum_{j=1}^m \varepsilon_j$ and g be standard Gaussian. For $1 \leq k \leq m$,

$$(2k-1)!! \frac{(m)_k}{m^k} \leq \mathbb{E} S_m^{2k} \leq (2k-1)!! = \mathbb{E} g^{2k}.$$

The lower bound retains exact pair partitions in the expansion. For the upper bound, pair within every even fiber; counting all pairings overcounts each surviving assignment. For $k \leq m/2$, $(m)_k/m^k \geq e^{-k(k-1)/m}$.

Let $T_X(p) = \mathbb{E} \text{tr}((H_X H_X^*)^p)$. In the coefficient-one real model every trace coefficient is nonnegative. Odd primitive multiplicities vanish for both laws. For each occurrence, the half-multiplicities sum to p . Termwise comparison therefore gives

$$T_S(p) \leq T_G(p) \leq e^{Qp(p-1)/m} T_S(p), \quad p \leq m/2. \quad (82)$$

For $m = n$, the private-role sums give the exact identity in law $H_S = n^{-Q/2}H_{\widehat{F}}$. No central limit approximation is involved.

For the reverse norm comparison, set $\mu = \mathbb{E}|g| = \sqrt{2/\pi}$. Independence of Gaussian signs and magnitudes and conditional Jensen show, in any real Banach space and for $q \geq 1$,

$$\left\| \sum_j \varepsilon_j b_j \right\|_{L^q} \leq \mu^{-1} \left\| \sum_j g_j b_j \right\|_{L^q}.$$

Apply this to each entire independent factor group of the lifted polynomial. Replacing all lifted signs by Gaussians makes each normalized private sum standard Gaussian, so

$$\| \| H_G \| \|_q \geq \mu^Q n^{-Q/2} \| \| H_{\widehat{F}} \| \|_q.$$

For the upper bound first split detached components. On a core with Q_c occurrences, choose p proportional to $\log(2n)$ with $2p \geq q$. Its row dimension D is a fixed power of n up to constants. By Equation (82),

$$\| \| H_{c,G} \| \|_q \leq e^{Q_c(p-1)/(2n)} D^{1/(2p)} \| \| H_{c,S} \| \|_{2p} \leq C n^{\widehat{f}_c - Q_c/2} (\log(2n))^{\widehat{g}}.$$

The exponential and dimension factors are bounded. For a detached scalar use $p = \lceil q/2 \rceil$ in the scalar version of the comparison and Equation (71); at expectation use $p = 1$. Its scale is $n^{(\widehat{r}_j - Q_j)/2} \sqrt{q}$, with no $\sqrt{\log n}$ in expectation. Independent restoration proves

$$\mathbb{E} \| H_G \| \asymp n^{\widehat{f} - Q/2} (\log(2n))^{\widehat{g}}, \quad \| \| H_G \| \|_q \asymp n^{\widehat{f} - Q/2} (\log(2n))^{\widehat{g}} q^{\widehat{d}_0/2} \quad (83)$$

for every fixed $C_0 > 0$, uniformly over $2 \leq q \leq C_0 \log(2n)$ and all sufficiently large n . The constants and cutoff may depend on the fixed template, role-size comparison constants, and C_0 .

For circular complex arrays write $Z = (A + iB)/\sqrt{2}$. Expanding Q factors and applying Minkowski gives the real-Gaussian upper comparison with factor $2^{Q/2}$. Conditioning on all A arrays gives $\mathbb{E}(H_Z | A) = 2^{-Q/2}H_A$, and Jensen gives the reverse comparison. No positivity of complex trace coefficients is asserted in this step.

If all original scopes have arity at least two, the lift has the same minimum separator size as the original two-section. A path through a private role can be shortened through its scope clique, and every lifted separator must already separate the old paths. A minimum separator therefore contains no private role. After its deletion, a private role is an active singleton precisely when its whole scope lies in the separator; otherwise it joins the component of the surviving scope. Thus

$$\widehat{f} - Q/2 = (|W| + h - s)/2, \quad \widehat{g} = \frac{1}{2} \max_{|S|=s} (a(S) + |\{e : e \subseteq S\}|).$$

Unary factors are handled by the lift itself, which distinguishes random singleton roles from unused roles.

J.2 Hermite polarization: full proof

We prove Theorem 7.2. We prove the dimension-free comparison. For Banach coefficients b_a , let

$$Y = \sum_a b_a \frac{\text{He}_d(g_a)}{\sqrt{d!}}, \quad D = \sum_a b_a \prod_{j=1}^d g_a^{(j)}, \quad A_d = \frac{d^{d/2}}{\sqrt{d!}}.$$

Then $A_d^{-1}\|D\|_q \leq \|Y\|_q \leq A_d\|D\|_q$ for all $q \geq 1$. Put $h_a = d^{-1/2} \sum_j g_a^{(j)}$. Conditional Gaussian means and covariances, or their generating function, give $\mathbb{E}(\prod_j g_a^{(j)} \mid h_a) = d^{-d/2} \text{He}_d(h_a)$. Jensen proves the upper bound for Y .

For the other direction let $h_a(\varepsilon) = d^{-1/2} \sum_j \varepsilon_j g_a^{(j)}$. The full sign Fourier coefficient is

$$\mathbb{E}_\varepsilon \left[\left(\prod_j \varepsilon_j \right) \text{He}_d(h_a(\varepsilon)) \right] = d! d^{-d/2} \prod_j g_a^{(j)}.$$

All lower-degree terms vanish, and the leading term survives only when every sign is used once. Minkowski and the Gaussian law of each $h_a(\varepsilon)$ prove the reverse bound. Apply the comparison conditionally through the fixed independent occurrence groups, then use Equation (83).

K Complete degree-four construction and positivity verification

We prove Theorem 8.2, including normalization, Boolean and nonedge constraints, the exact size constraint, and positivity on all polynomials of degree at most two. The graph event is chosen first; the size parameter is then tuned pointwise on that same event.

K.1 Exact graph-matrix multiplication

For shapes α, β with compatible inner boundary lengths, matrix multiplication forces equality between their inner boundary labels. Every additional cross-equality is a partial matching of roles, since each original embedding is injective. Let μ range over partial matchings extending these forced identifications. Identify matched roles and keep all other quotient roles distinct. If an edge occurs from both factors, cancel the two copies using $\varepsilon_{ij}^2 = 1$. Denote the resulting shape by $\alpha \star_\mu \beta$. Then, pointwise,

$$M_\alpha M_\beta = \sum_\mu M_{\alpha \star_\mu \beta}, \quad M_\alpha^* = M_{\alpha^*}, \quad (84)$$

where α^* exchanges boundaries.

To prove the product identity, expand pairs of embeddings and group them by their exact cross-collision relation. For a fixed relation, pairs of embeddings are in bijection with injective quotient embeddings. Their sign product is exactly the quotient monomial after cancellation, and the external coordinates agree. The collision classes are disjoint and exhaustive. Thus every fixed finite graph-matrix expression has an exact finite expansion. This algebraic closure is specific to the sign identity used above; Gaussian repeated coordinates cannot be canceled in this way.

We now implement this program using the corrected quartic ansatz of Hopkins et al. [12]. Let W be a symmetric zero-diagonal sign matrix and $A_{ij} = (1 + W_{ij})/2$ for $i \neq j$.

K.2 The common graph event

All expansions below involve a fixed finite list of shapes on at most five roles and at most ten edges. The growing-moment theorem gives a simultaneous bound $Kn^f L$ with $L = (\log(2n))^{10}$, except on an event of probability at most n^{-12} for sufficiently large n . This deliberately loose logarithm suffices for every polynomially subcritical term.

Two families need log-free bounds. For completeness, the following reference argument supplies them independently of an enumeration of logarithmic exponents. Deleting edges from a typed sign shape only removes parity constraints in its positive trace expansion. If the reference is a

disjoint union of boundary edges and two-edge boundary paths, its matrix is a tensor product of independent sign matrices and products of two such matrices. The net proof in Lemma D.2, or its tail integral at arbitrary logarithmic order, bounds these factors at scales $n^{1/2}$ and n , respectively. Independence or Hölder with a fixed enlarged moment window handles their tensor product. Trace domination, taking a moment at least $\log n$ to absorb the coordinate dimension, proves the same scale for a supergraph with the same polynomial exponent. Lemma 2.4 transfers the estimate to global injectivity. We use this only for the explicit references described below.

Unordered pair matrices are compressions of ordered boundary-coordinate matrices, with fixed multiplicative constants for a shared-root orbit. A symmetric sum over two unordered middle roles is replaced by half the *whole* ordered sum before its Fourier expansion. No nonsymmetric individual pattern is restricted by an ordering condition and then treated as a full graph matrix.

Let C_4 be the number of four-cliques and $Z_4 = \sum_T \text{four-clique} \sum_s \prod_{a \in T} W_{sa}$. The graph event also includes

$$|C_4 - \binom{n}{4}/64| \leq Kn^3L, \quad |Z_4| \leq Kn^{5/2}L, \quad \|W\| \leq K\sqrt{n}. \quad (85)$$

For the first bound expand the six clique edges: the 63 nonempty scalar patterns have polynomial exponent at most 3. For the second, the four center-star edges are mandatory and the six clique edges are optional: the 64 patterns are connected with exponent $5/2$. The conversion from unordered quartets has factor $1/24$. We enlarge one fixed K to cover all finite coefficients and all graph bounds. It is independent of the parameters chosen below.

K.3 Quartic moments and the filled matrix

For an internal parameter $\omega \geq 8$ and fixed $\gamma > 0$, put $\tau = \gamma(\omega/n)^5$ and

$$\mu_4(T) = \mathbf{1}_{\{T \text{ clique}\}} \left[\frac{\binom{\omega}{4}}{C_4} + \tau \sum_s \prod_{a \in T} W_{sa} \right].$$

Let ω' solve $\binom{\omega'}{4} = \binom{\omega}{4} + \tau Z_4$ near ω . Extend downwards by $\mu_j(S) = (\omega' - j)^{-1} \sum_{x \notin S} \mu_{j+1}(S \cup \{x\})$ for $j = 3, 2, 1$, and set $\mu_0 = 1$. On Equation (85),

$$2^{-14}n^4 \leq C_4 \leq n^4, \quad d := |\omega' - \omega| \leq K\gamma\omega^2n^{-5/2}L \leq 1 \quad (86)$$

uniformly for $8 \leq \omega \leq c\sqrt{n}$, eventually. This follows from the derivative of $\binom{t}{4}$, which is bounded below by a constant times ω^3 on $[\omega - 1, \omega + 1]$.

Index matrices by unordered pairs. Let D be the signless vertex-pair incidence matrix, put $J_D = D^*D$, and let P project onto $\text{range}(D^*)$, with $P_0 = I - P$. Then $DD^* = (n - 2)I + \mathbf{1}\mathbf{1}^*$ and $\|J_D\| = 2n - 2$. Define $Q_{\{a,b\},s} = W_{sa}W_{sb}$ and $S_W = \text{diag}(W_{ab})$.

The filled uncorrected pair matrix has entries

$$M'(I, J) = \beta_i \sum_{\substack{T \supseteq I \cup J \\ |T|=4}} \prod_{e \in E(T) \setminus (E(I) \cup E(J))} A_e, \quad i = |I \cap J|,$$

where $\beta_0 = \binom{\omega}{4}$, $\beta_1 = \binom{\omega}{3}/4$, and $\beta_2 = \binom{\omega}{2}/6$. Its expectation is

$$E = \alpha_0 \mathbf{1}\mathbf{1}^* + (\alpha_1 - \alpha_0)J_D + (\alpha_2 - 2\alpha_1 + \alpha_0)I, \quad (87)$$

with

$$\alpha_0 = \frac{\binom{\omega}{4}}{16}, \quad \alpha_1 = \frac{\binom{\omega}{3}(n-3)}{64}, \quad \alpha_2 = \frac{\binom{\omega}{2}\binom{n-2}{2}}{192}.$$

Let $b = \alpha_2 - 2\alpha_1 + \alpha_0$. The three eigenvalues on constants, their orthogonal complement in $\text{range}(D^*)$, and $\ker D$ are $b + (2n - 2)(\alpha_1 - \alpha_0) + \binom{n}{2}\alpha_0$, $b + (n - 2)(\alpha_1 - \alpha_0)$, and b . For $n \geq 128$ and $8 \leq \omega \leq n/128$, elementary binomial bounds give respectively the lower bounds $\omega^4 n^2/24576$, $\omega^3 n^2/16384$, and $\omega^2 n^2/8192$. In particular

$$E \succeq aB, \quad a = 2^{-15}, \quad B = \omega^3 n^2 P + \omega^2 n^2 P_0. \quad (88)$$

For example, $\alpha_2 \geq \omega^2 n^2/6144$, $2\alpha_1 \leq \omega^2 n^2/24576$, $\alpha_1 - \alpha_0 \geq \omega^3 n/8192$, and $\binom{n}{2}\alpha_0 \geq \omega^4 n^2/24576$ suffice for these bounds. The two-scale metric is essential.

K.4 Local fluctuations and the exact square

On disjoint pairs $I = \{a, b\}, J = \{c, d\}$, let $\mathcal{X} = \{ac, ad, bc, bd\}$. The local fluctuation is $\alpha_0 \sum_{\emptyset \neq F \subseteq \mathcal{X}} \prod_{e \in F} W_e$. Four of these patterns are single edges, four are two-edge stars, and seven contain a cross perfect matching. Completing the full single-edge and star orbits gives the exact identity

$$L_0 = \alpha_0(D^*WD + D^*Q^* + QD + Z), \quad \|Z\| \leq Kn.$$

The seven remaining patterns have a two-edge matching reference of the same exponent 1, so are log-free. Indeed, among the nonempty subsets of the four edges of $\mathcal{X}$, the four singleton subsets and four two-edge stars are the exceptional orbits already completed. The other subsets are the two perfect matchings, the four three-edge subsets, and the full four-edge set. Each contains one of the two perfect matchings. This accounts for all 15 local patterns without requiring a computer enumeration. The unwanted overlap entries in the three completed matrices have bounded magnitudes and $O(n)$ entries per row and column; their norm is $O(n)$.

On distinct overlapping pairs $\{a, b\}, \{a, c\}$ the local fluctuation is $\alpha_1 W_{bc}$. If $\text{Lift}(T)$ denotes compression of $T \otimes I + I \otimes T$ to symmetric off-diagonal pair coordinates, then $L_1 = \alpha_1 \text{Lift}(W)$ and $\|L_1\| \leq K\omega^3 n^{3/2}$.

Set $\rho = \tau C_4/16$ and define

$$R(I, J) = 16\rho \mathbf{1}_{\{I \cap J = \emptyset\}} \prod_{e \in \mathcal{X}} A_e \sum_s \prod_{u \in I \cup J} W_{su}, \quad \tilde{R}(I, J) = \rho \sum_s \prod_{u \in I \Delta J} W_{su}.$$

The exact all-collision identity is

$$\tilde{R} = \rho(QQ^* + S_W J_D S_W). \quad (89)$$

For disjoint pairs it is immediate. On $\{a, b\}, \{a, c\}$, QQ^* misses exactly the term $W_{ab}W_{ac}$ supplied by $S_W J_D S_W$. On the diagonal the two terms are $n - 2$ and 2 .

On disjoint pairs, $R - \tilde{R}$ has 15 patterns: the center is joined to all four boundary roles and a nonempty subset of $\mathcal{X}$ is present. Choose any edge ac in that subset. The disjoint edge $a - c$ and path $b - s - d$ form a reference with separator size two and exponent $3/2$, equal to that of the parent. Thus these patterns are log-free. On overlapping pairs the retained matrix $\tilde{R}/\rho$ is exactly $\text{Lift}(W^2) + (2 - n)I$, of norm $O(n)$. Using Equation (86),

$$\|R - \tilde{R}\| \leq K\rho(n^{3/2} + n) \leq K\gamma\omega^5\sqrt{n}, \quad \frac{\gamma\omega^5}{2^{18}n} \leq \rho \leq \frac{\gamma\omega^5}{n}. \quad (90)$$

Complete the dangerous square exactly:

$$\rho QQ^* + \alpha_0(D^*Q^* + QD) = \rho \left(Q + \frac{\alpha_0}{\rho} D^* \right) \left(Q + \frac{\alpha_0}{\rho} D^* \right)^* - \frac{\alpha_0^2}{\rho} J_D. \quad (91)$$

Both the displayed square and $\rho S_W J_D S_W$ are positive semidefinite. The negative term is supported on P . Therefore

$$\left\| B^{-1/2} \left(\alpha_0 D^* W D - \frac{\alpha_0^2}{\rho} J_D \right) B^{-1/2} \right\| \leq K \left(\frac{\omega}{\sqrt{n}} + \frac{1}{\gamma} \right).$$

The remaining normalized costs of $\alpha_0 Z$, L_1 , and $R - \tilde{R}$ are at most $K\omega^2/n$, $K\omega/\sqrt{n}$, and $K\gamma\omega^3/n^{3/2}$, respectively. No independence of the random scalar ρ and W has been used.

K.5 Extension-count fluctuations

The shared-root extension fluctuation is exactly

$$\Delta_1(\{a, b\}, \{a, c\}) = \frac{\beta_1}{16} (1 + W_{bc}) \sum_{s \notin \{a, b, c\}} [(1 + W_{as})(1 + W_{bs})(1 + W_{cs}) - 1].$$

Its 14 patterns give $\|\Delta_1\| \leq K\omega^3 n^{3/2} L$. For the kernel block there is the stronger bound $\|P_0 \Delta_1 P_0\| \leq K\omega^3 n L$. Indeed, nine patterns have both exclusive boundary roles incident and a b - c path avoiding a ; their separator size is at least two. The other five have an isolated exclusive boundary role. If c is isolated, write the completed pattern as UD , where

$$U_{I,a} = \mathbf{1}_{\{a \in I\}} \sum_{s \notin I} \prod_{e \in F(s, a, I \setminus \{a\})} W_e.$$

On distinct overlapping pairs the difference is the single excluded term $s = c$; the diagonal has magnitude at most $2(n-2)$. The collision matrix has row and column absolute sums at most $4n$. Since $DP_0 = 0$, only this collision matrix survives the compression. Transpose when b is isolated. This covers all five exceptions.

The nine-five split can be checked directly. Write the chosen nonempty star subset as $F \subseteq \{as, bs, cs\}$ and let the optional edge be bc . If bc is present, all seven choices of F have the required b - c path. If bc is absent, exactly the two choices containing both bs and cs have such a path. The five remaining choices are $\{as\}$, $\{bs\}$, $\{cs\}$, $\{as, bs\}$, $\{as, cs\}$, each with an isolated exclusive boundary role. Thus every one of the 14 patterns belongs to the stated estimate or collision completion.

The diagonal extension fluctuation is

$$\Delta_2(I, I) = \beta_2 \left[\sum_{\substack{s < t \\ s, t \notin I}} A_{as} A_{at} A_{bs} A_{bt} A_{st} - \frac{\binom{n-2}{2}}{32} \right].$$

Replace the symmetric middle sum by half the ordered sum. Its 31 nonempty patterns have equal two-role boundaries and at most one isolated middle role, so $\|\Delta_2\| \leq K\omega^2 n^{3/2} L$. Bounding the four P, P_0 blocks separately yields

$$\|B^{-1/2}(\Delta_1 + \Delta_2)B^{-1/2}\| \leq KL \left[\frac{1 + \sqrt{\omega}}{\sqrt{n}} + \frac{\omega}{n} \right]. \quad (92)$$

K.6 The exact lower-degree cleanup

For $j = 2, 3$ and a j -set S , let $c_4(S)$ be the number of four-cliques containing S , and define the uncorrected moments

$$\mu_j^0(S) = \frac{\binom{\omega}{j} c_4(S)}{C_4 \binom{4}{j}}.$$

These vanish on noncliques. Equivalently, they are the downward recursion of the uncorrected fourth moments with the internal parameter ω , not the corrected parameter ω' .

Let G_{pair} project onto graph-edge pair coordinates and let $N'_{I,J} = \mu_{|I \cup J|}(I \cup J)$. For a clique S of size two or three, put $h_S = \sum_{T \supseteq S, T \text{ four-clique}} \sum_s \prod_{v \in T} W_{sv}$. The downward recursion gives exactly

$$\mu_3 = r_3 \mu_3^0 + \frac{\tau h_S}{\omega' - 3}, \quad \mu_2 = r_2 \mu_2^0 + \frac{2\tau h_S}{(\omega' - 2)(\omega' - 3)},$$

where $r_3 = (\omega - 3)/(\omega' - 3)$ and $r_2 = (\omega - 2)(\omega - 3)/[(\omega' - 2)(\omega' - 3)]$. The factor two counts the two orders of the added vertices. Also $|r_i - 1| \leq Kd/\omega$.

Define filled matrices supported on shared-root off-diagonal and diagonal pairs, respectively:

$$\begin{aligned} K_3(\{a, b\}, \{a, c\}) &= A_{bc} \sum_{d \notin \{a, b, c\}} A_{ad} A_{bd} A_{cd} \sum_s W_{sa} W_{sb} W_{sc} W_{sd}, \\ K_2(\{a, b\}, \{a, b\}) &= \sum_{\substack{c < d \\ c, d \notin \{a, b\}}} A_{ac} A_{ad} A_{bc} A_{bd} A_{cd} \sum_s W_{sa} W_{sb} W_{sc} W_{sd}. \end{aligned}$$

The 16 and 32 Fourier patterns have five roles, no isolated middle, and separator size two. Thus $\|K_3\| + \|K_2\| \leq Kn^{3/2}L$. Let $M_1^{\text{cl}}, M_2^{\text{cl}}$ denote $C_4\mu_3^0$ on overlapping clique pairs and $C_4\mu_2^0$ on clique-pair diagonals. Nonnegative count bounds give norms at most $K\omega^3n^2$ and $K\omega^2n^2$. Set

$$\begin{aligned} \text{Err} &= (r_3 - 1)M_1^{\text{cl}} + (r_2 - 1)M_2^{\text{cl}} \\ &+ \frac{C_4\tau}{\omega' - 3} G_{\text{pair}} K_3 G_{\text{pair}} + \frac{2C_4\tau}{(\omega' - 2)(\omega' - 3)} G_{\text{pair}} K_2 G_{\text{pair}}. \end{aligned}$$

For $N = M' + R$, direct comparison in each overlap case gives the exact identity

$$G_{\text{pair}}(N + \text{Err})G_{\text{pair}} = C_4N'. \quad (93)$$

Moreover

$$\frac{\|\text{Err}\|}{\omega^2 n^2} \leq K\gamma L \left[\frac{\omega^2 + \omega}{n^{3/2}} + \frac{\omega^2}{n^{5/2}} \right].$$

This is an orthogonal coordinate compression, not a general entrywise masking inequality.

K.7 Budget, simultaneous sizes, and full positivity

Every error above is included in

$$\begin{aligned} \varepsilon_n &= K \left\{ \frac{1}{\gamma} + \frac{\omega}{\sqrt{n}} + \frac{\omega^2}{n} + \frac{\gamma\omega^3}{n^{3/2}} \right. \\ &\quad \left. + L \left[\frac{1 + \sqrt{\omega}}{\sqrt{n}} + \frac{\omega}{n} + \frac{\gamma(\omega^2 + \omega)}{n^{3/2}} + \frac{\gamma\omega^2}{n^{5/2}} \right] \right\}. \end{aligned}$$

After discarding only positive squares, $N + \text{Err} \succeq (a - \varepsilon_n)B$. Choose $\gamma = 32K/a$ and $c = a/(128K\gamma)$. The four nonvanishing terms are a fixed fraction of a for $\omega \leq c\sqrt{n}$; all L terms tend to zero uniformly on this interval. For sufficiently large n , $\varepsilon_n < a/4$ throughout $8 \leq \omega \leq c\sqrt{n}$. By Equation (93), $N' \succeq 0$ with zero nonedge rows.

The event was defined without any parameter choice. For every real $9 \leq k \leq (c/2)\sqrt{n}$, solve

$$\binom{t}{4} + \gamma(t/n)^5 Z_4 = \binom{k}{4} \quad (94)$$

on $[k-1, k+1]$. The derivative of the first term is bounded below by a constant times k^3 . Both the perturbation divided by k^3 and its derivative divided by k^3 tend to zero uniformly over this interval of k , by Equation (85). Thus the endpoints bracket the target and the derivative is positive. The unique root satisfies $|t-k| \leq C\gamma K k^2 n^{-5/2} L = o(1)$. Use t as the internal parameter and k in the downward recursion. All previous identities are pointwise and all budgets hold at this t . No union over real k or independence from the chosen parameter is required.

Define the linear functional $\mathcal{L}_k$ by these squarefree moments and Boolean reduction. Downward counting yields $\sum_{|S|=j} \mu_j(S) = \binom{k}{j}$ for $1 \leq j \leq 4$. The recursion gives $\mathcal{L}_k((\sum_i x_i - k)x_S) = 0$ for $|S| \leq 3$. Boolean constraints hold by reduction. A nonclique set has zero moment since every superseding quartet is a nonclique, proving the nonedge constraints.

To include the constant and linear rows in positivity, let T have pair rows and columns indexed by subsets of size at most two:

$$T_{\{i,j\},\emptyset} = \frac{2}{k(k-1)}, \quad T_{\{i,j\},\{a\}} = \frac{\mathbf{1}_{\{a \in \{i,j\}\}}}{k-1}, \quad T_{\{i,j\},\{a,b\}} = \mathbf{1}_{\{\{i,j\}=\{a,b\}\}}.$$

The full moment matrix is exactly $T^* N' T$. Indeed these columns replace constants and linear monomials by homogeneous quadratics. The required identities are

$$\begin{aligned} \sum_{j \neq i} x_i x_j - (k-1)x_i &= x_i \left(\sum_j x_j - k \right) - (x_i^2 - x_i), \\ 2 \sum_{i < j} x_i x_j - k(k-1) &= \left(\sum_i x_i - k \right) \left(\sum_i x_i + k - 1 \right) - \sum_i (x_i^2 - x_i). \end{aligned}$$

For a degree-two polynomial p and its quadratic replacement h , the product $(h-p)(h+p)$ uses these constraints only through degree four. Thus $\mathcal{L}_k(p^2) = \mathcal{L}_k(h^2) \geq 0$. This proves all the asserted constraints and full degree-four positivity simultaneously.

The finite template families used above have sizes 7, 15, 14, 31, 16, 32, 64, 63, totaling 242. Their explicit descriptions, collision completions, and norm references were given in the proof; a finite computational check is not used as a substitute for any of these arguments.

L RTN deviations, probability lower bounds, and entropy

We prove Theorems 9.3 and 9.4. Finite Schatten thresholds, the expected operator endpoint, and the sample-state conclusion require different moment inputs; the proof does not infer growing-order control from a fixed-order expansion.

L.1 A fixed-order trace lemma

For a coefficient-one incidence matrix J with R roles, K independent real or circular complex Gaussian occurrences on nonempty scopes, and role sizes N , let h be the number of unused middle roles and s the two-section separator size. For every fixed integer $p \geq 1$,

$$\mathbb{E} \operatorname{tr}((JJ^*)^p) \leq \operatorname{Bell}(2p)^{R-h} ((2p)!)^K N^{p(R+h-s)+s}. \quad (95)$$

Remove unused middle roles, giving the matrix multiplier N^h . In an exact equality state a real Gaussian factor requires even multiplicities; a circular complex factor requires balanced conjugate and unconjugate multiplicities, which also implies evenness. Each nonzero occurrence moment is nonnegative and at most $(2p)!$. Factor legality implies the interval compatibility of Lemma 3.6.

The separator layers yield $\sum_v |\pi_v| \leq p(R - h - s) + s$; for $p = 1$ the layer sum is empty and the same bound holds. There are at most $\text{Bell}(2p)^{R-h}$ partition assignments. Bound falling factorials by powers of N and restore N^{2ph} to prove Equation (95). Detached components are allowed. The constant need not be controlled as p grows.

L.2 Centering, decoupling, and doubled cuts

At an occurrence write $\bar{g}_i g_j = \delta_{ij} + W_{ij}$. Expand the product over occurrences and remove the all-covariance term. The remaining terms K_T are indexed by nonempty active tensor sets T .

For arbitrary Banach coefficients B_{ij} and $q \geq 1$, introduce an independent copy g' and rotate to $u = (g + g')/\sqrt{2}$, $v = (g - g')/\sqrt{2}$. Conditional Jensen, followed by the triangle inequality, gives

$$\left\| \sum B_{ij} (\bar{g}_i g_j - \delta_{ij}) \right\|_{L^q} \leq \left\| \sum B_{ij} (\bar{u}_i v_j + \bar{v}_i u_j) \right\|_{L^q} \leq 2 \left\| \sum B_{ij} \bar{u}_i v_j \right\|_{L^q}.$$

The last two summands have the same law by swapping u, v . Iterating across active groups and using the Schatten-2p norm yields

$$(\mathbb{E} \|F_N\|_{S_{2p}}^{2p})^{1/(2p)} \leq \sum_{T \neq \emptyset} 2^{|T|} N^{-(r-b)} (\mathbb{E} \|D_T\|_{S_{2p}}^{2p})^{1/(2p)}. \quad (96)$$

Here D_T has two independent copies at active tensors and covariance identifications at inactive tensors. This is only an upper decoupling inequality.

Write e_T, e_C for internal edges within T and its complement, and a_T, a_C, b_T, b_C for their terminal counts. The roles of D_T are as follows:

Original edge type	Roles after covariance contraction
Internal within T	$2e_T$ copied internal roles
Internal outside T	e_C unused middle roles
Internal crossing T	$c(T)$ bridge roles
Output on T	a_T bridge roles
Output outside T	a_C unused middle roles
Input on T	$2b_T$ separate boundary roles
Input outside T	b_C common boundary roles

In particular $R_T = r + e_T + b_T$ and $h_T = e_C + a_C$.

Assume $\Delta \geq 0$. In one copy of the induced T -network, regard crossing and output bridges as output terminals. The excess of a cut indexed by $\emptyset \neq S \subseteq T$ over its b_T input cut is $\delta(S) \geq 0$. Integral max-flow/min-cut gives b_T edge-disjoint paths from its inputs to distinct bridges. Reflect and concatenate them in the other copy. They give b_T disjoint boundary paths in the edge-role two-section. The input edges give a matching upper cut. Adding the mandatory b_C identity roles shows that the separator size of D_T is exactly b . This also covers $b_T = 0$.

Apply Equation (95). The exponent after normalization is determined by

$$R_T + h_T - b - 2(r - b) = -c(T) - a_T + b_T = -\delta(T).$$

The finite sum in Equation (96) therefore proves

$$\mathbb{E}[d^{-1} \text{tr} |F_N|^{2p}] \leq C_p N^{-p\Delta}, \quad \Delta \geq 0, \quad (97)$$

for each fixed p .

L.3 Second moments and nonrare spectral mass

At two Gram replicas a circular complex tensor has two Wick pairings, identity or swap, including when coordinates coincide. Let T be the swapped set. Each crossing internal edge, output leg on T , or input leg outside T loses one free label. After normalization its contribution is $N^{-\delta(T)}$. The empty set contributes one and is exactly removed by centering. Hence

$$\mathbb{E}[d^{-1} \operatorname{tr} F_N^2] = \sum_{T \neq \emptyset} N^{-\delta(T)}. \quad (98)$$

Put $\epsilon = N^{-\Delta/2}$ and $L = 2^Q - 1$. On the Gaussian sample space enlarged by an independent uniform eigenvalue index j , let $Y = |\lambda_j(F_N)|$. Equations (97) and (98) give $\epsilon^2 \leq \mathbb{E}Y^2 \leq L\epsilon^2$ and $\mathbb{E}Y^4 \leq C_2\epsilon^4$. Interpolation gives $\mathbb{E}Y \geq C_2^{-1/2}\epsilon$. For $Z = d^{-1}\|F_N\|_1$, $\mathbb{E}Z = \mathbb{E}Y$ and $Z^2 \leq d^{-1} \operatorname{tr} F_N^2$. Paley–Zygmund yields

$$\mathbb{E}Z \geq C_2^{-1/2}\epsilon, \quad \mathbb{P}\{Z \geq \epsilon/(2\sqrt{C_2})\} \geq (4C_2L)^{-1}. \quad (99)$$

Thus the lower scale is not inferred merely from a nonzero second moment.

For fixed finite $t \geq 1$, choose a fixed integer p with $2p \geq t$. Power means and Equation (97) give

$$\mathbb{E}\|F_N\|_{S_t} \leq d^{1/t-1/(2p)}(\mathbb{E} \operatorname{tr} |F_N|^{2p})^{1/(2p)} \leq C_t d^{1/t} \epsilon.$$

Conversely $\|F_N\|_{S_t} \geq d^{1/t-1}\|F_N\|_1$. Equation (99) gives matching expectation and positive-probability lower bounds. This proves the finite Schatten assertion in Theorem 9.3.

If $\Delta > 2b/t$, Markov proves convergence to zero. For $0 \leq \Delta \leq 2b/t$, the probability lower bound excludes convergence, including equality. If $\Delta < 0$, a cut of size $b + \Delta$ factors the map through at most $N^{b+\Delta}$ coordinates. At least $d - N^{b+\Delta}$ eigenvalues of F_N are then -1 . This excludes convergence in any of the stated norms.

For the operator norm, if $\Delta > 0$, choose one fixed integer $p > b/\Delta$. Then for each fixed $\eta > 0$,

$$\mathbb{P}\{\|F_N\|_{\text{op}} > \eta\} \leq C_p \eta^{-2p} N^{b-p\Delta} \longrightarrow 0.$$

For $\Delta = 0$ use Equation (99); for $\Delta < 0$ use rank. This proves the qualitative criterion using fixed moments alone. Fixed moments give an expected upper rate $N^{-\Delta/2+\zeta}$ for each fixed $\zeta > 0$, not the exact endpoint $\zeta = 0$.

L.4 The uniform operator endpoint

We now use the additional uniform Gaussian reduction in Appendix J. For a standard network the role two-section is the line graph, and role separators correspond to terminal-separating edge cuts. Adding private roles does not change the minimum separator size: every detour through a private role can be shortened in its incident clique.

For an inclusion-minimal terminal-separating cut, adding back any one cut edge creates an input-output path. Its endpoints therefore lie in components meeting opposite terminal sets. A terminal-free component of the network minus the cut cannot touch a cut edge; it would consequently have been terminal-free already, contrary to the model. Every tensor retains a path to a terminal and an uncut incident edge. Its private role attaches to a boundary-touching component. Thus every minimum lifted cut has active count zero.

For D_T , remove its unused middle roles and common identity roles. The other components are standard doubled networks. Boundary-touching ones have zero active count by the preceding

argument. A component of the induced T -network with no input produces a detached scalar after doubling through bridges. When $\Delta > 0$ such a component has at least one bridge. Let k_T count these components. The uniform Gaussian bound and the same decoupling give

$$\| \|F_N\|_{\text{op}} \|_{L^q} \leq C \sum_{T \neq \emptyset} N^{-\delta(T)/2} q^{k_T/2} \quad (100)$$

in every fixed logarithmic window. For expectation, the detached q powers are absent. Together with Equation (99), this proves $\mathbb{E}\|F_N\|_{\text{op}} \asymp N^{-\Delta/2}$ for $\Delta > 0$.

The cut function δ is additive on connected components of an induced tensor set. Since each nonempty component has gap at least $\Delta > 0$, a minimizing set is connected. Its k_T is zero or one according as it has an input or not. Every nonminimizer has integer gap at least $\Delta + 1$, whose additional $N^{-1/2}$ absorbs any fixed logarithmic power. Consequently, if $\kappa_0 = 1$ when a minimizing set has no input and $\kappa_0 = 0$ otherwise,

$$\mathbb{P}\{\|F_N\|_{\text{op}} > C_A N^{-\Delta/2} (\log(2N))^{\kappa_0/2}\} \leq N^{-A}$$

for every fixed $A > 0$. This is only an upper tail.

L.5 The sample-state endpoint

Let s be the ordinary minimum input-output edge cut. The original network has no unused middle role or detached component, and its minimum lifted cuts have zero active count. Thus the Gaussian uniform bound gives $\|H_N\|_{\text{op}} \leq C_A N^{(r-s)/2}$ with failure probability at most N^{-A} .

Put $X = N^{-r} \|H_N\|_F^2$. Covariance gives $\mathbb{E}X = 1$. In its second moment a swapped tensor set T loses one free label at every crossing internal edge and every incident open leg. Hence

$$\mathbb{E}X^2 = \sum_{T \subseteq [Q]} N^{-\partial(T)}, \quad \partial(T) = c(T) + a(T) + b(T).$$

Every nonempty T has $\partial(T) \geq 1$, since every network component meets a terminal. Therefore $\text{Var } X \leq (2^Q - 1)/N$ and $\mathbb{P}\{|X - 1| > 1/2\} \leq 4(2^Q - 1)/N$.

The map is nonzero almost surely: some entry is a nonzero polynomial in Gaussian coordinates, as seen by evaluating all coordinates at one. Cut factorization gives $\text{rank } H_N \leq N^s$. On the preceding event and the operator upper event with $A = 2$,

$$N^{-s} \leq \|\rho_A\|_{\text{op}} = \frac{\|H_N\|_{\text{op}}^2}{\|H_N\|_F^2} \leq C N^{-s}.$$

Since $H_\infty \leq H_\alpha \leq H_0$ for all nonnegative Rényi orders, including endpoints, this proves $s \log N - \log C \leq H_\alpha(\rho_A) \leq s \log N$ simultaneously for $0 \leq \alpha \leq \infty$ with probability $1 - O(N^{-1})$. Normalizing each Gaussian tensor by its own nonzero norm leaves ρ_A unchanged, but does not preserve mean-Gram normalization.

M Complete proof of the exact RTN spectral edge

This appendix proves Theorems 10.3 and 10.5. The quotient and comparison are established before the external strong-convergence theorem is used. Concentration and exceptional-event moment control are then proved separately, so that the moment order may grow with the original dimension.

M.1 Exact Wick moments and the flow quotient

Lemma M.1 (Positive moment polynomial). *For every integer $p, N \geq 1$,*

$$F_G(p, N) := \mathbb{E}[N^{-s} \operatorname{tr} Y_N^p] = \sum_{\delta \geq 0} A_G(p, \delta) N^{-\delta}. \quad (101)$$

All coefficients are nonnegative integers.

Proof. Wick's rule at vertex u pairs its p unconjugated occurrences with its p conjugated occurrences by π_u . An edge $\{u, w\}$ then has $p - d(\pi_u, \pi_w)$ free index loops. Hence

$$\mathbb{E} \operatorname{tr}(H_N H_N^*)^p = \sum_{\pi} N^{pR - \mathcal{H}_G^{(p)}(\pi)}.$$

Coincident coordinate values retain their Wick multiplicities; there is no injective-label restriction here. Choose s edge-disjoint simple terminal paths by integral max flow. Triangle inequality on each path gives $\mathcal{H}_G^{(p)} \geq s(p-1)$. Assigning id and γ_p on the two sides of a minimum cut attains equality. The normalization in Equation (41) proves the identity. The opposite cyclic-trace convention gives γ_p^{-1} ; inverting all labels preserves the distances and counts. $\square$

Fix such a family of s paths and orient their union F forward. Form a directed graph D with one forward arc for each used edge and both arcs for each unused edge. Contract its complete strongly connected components (SCCs), discard internal edges, and retain cross-component edges with multiplicity. Denote this quotient by Q and its terminal components by B_0, A_0 .

Lemma M.2 (Balanced acyclic quotient). *Q is an acyclic directed multigraph, $B_0 \neq A_0$, and every quotient vertex lies on a projected terminal flow path. All cross-component edges are used edges. The projected paths remain edge-disjoint, and*

$$\deg^-(B_0) = 0, \quad \deg^+(B_0) = s, \quad \deg^-(A_0) = s, \quad \deg^+(A_0) = 0.$$

Each other quotient vertex C has $\deg^-(C) = \deg^+(C) = k_C \geq 1$. The minimum terminal cut of Q is s .

Proof. An unused edge is bidirectional, so its endpoints belong to the same SCC. A condensation is acyclic. Across a cut of size s , each of the s edge-disjoint paths crosses exactly once: all cut edges are used and directed toward the sink side. No arc crosses back. Thus the terminal components are distinct.

A vertex off the flow can be joined to a first flow vertex by a path consisting only of unused edges, using connectedness. Its SCC therefore meets the flow. A directed path cannot leave an SCC and return, since this would create a cycle in the condensation. Thus projected paths are simple and cover every quotient vertex. Summing flow conservation inside each component cancels internal used edges. This gives balanced external degrees away from the terminal components. Since every component lies on a source-sink path in a DAG, the source has no incoming edge and the sink no outgoing edge. Their degrees are consequently s . The projected flow and the cut immediately after B_0 give the matching lower and upper cut bounds. $\square$

Contracting only the unused-edge connected components would not suffice: used edges can create directed cycles between those components. The complete SCC construction is used throughout. Choose one anchor $r(C)$ per SCC, using the appropriate terminal in B_0, A_0 , and an actual vertex otherwise. Define the single global map

$$\omega_C = \pi_{r(C)}, \quad \bar{\pi}_u = \omega_{C(u)}. \quad (102)$$

M.2 All-defect comparison at an auxiliary dimension

Lemma M.3 (Variation inside an SCC). *If $\Delta_G(\pi) = \delta$ and $C(u) = C(w)$, then $d(\pi_u, \pi_w) \leq 2\delta$.*

Proof. Write $r_u = |\pi_u|$ and assign to each directed arc $u \rightarrow w$ the nonnegative reduced cost $c_{uw} = d(\pi_u, \pi_w) - (r_w - r_u)$. Flow conservation yields

$$\delta = \sum_{e \in F} c_e + \sum_{e \notin F} d_e, \quad \sum_{\text{arcs of } D} c_e = \delta + \sum_{e \notin F} d_e \leq 2\delta.$$

For unused edges the two arc costs sum to $2d_e$. Choose simple directed paths $P : u \rightarrow w$ and $P' : w \rightarrow u$ inside the SCC. Triangle inequality and telescoping give

$$d(\pi_u, \pi_w) \leq r_w - r_u + c(P), \quad d(\pi_u, \pi_w) \leq r_u - r_w + c(P').$$

Each path cost is at most 2δ ; their edges need not be disjoint. Adding proves the assertion. $\square$

Define $A_Q, \mathcal{H}_Q, \Delta_Q$ using the same terminal labels and minimum cut s , and let $C_0 = 1 + 4R$.

Lemma M.4 (Defect and fiber bounds). *The map Equation (102) satisfies $0 \leq \Delta_Q(\omega) \leq C_0\delta$. For $p \geq 2$, each fiber at fixed ω, δ has at most $p^{4v\delta}$ original labelings. Consequently*

$$A_G(p, \delta) \leq p^{4v\delta} \sum_{j=0}^{C_0\delta} A_Q(p, j), \quad A_G(p, 0) = A_Q(p, 0). \quad (103)$$

Proof. Deleted edges have equal labels after applying the map, so $\mathcal{H}_Q(\omega) = \mathcal{H}_G(\bar{\pi})$. The terminal labels are unchanged, and

$$\mathcal{H}_G(\bar{\pi}) \leq \mathcal{H}_G(\pi) + \sum_{u \in V_G} \deg_G(u) d(\pi_u, \bar{\pi}_u) \leq \mathcal{H}_G(\pi) + 4R\delta.$$

Here we used Lemma M.3 and $\sum_{u \in V_G} \deg_G(u) \leq 2R$. The quotient cut bound gives nonnegative defect.

A radius- 2δ ball in S_p has size at most

$$\sum_{h=0}^{2\delta} \binom{p}{2}^h \leq \left(1 + \binom{p}{2}\right)^{2\delta} \leq p^{4\delta}.$$

For $\delta = 0$ the size is one. Every original label lies in this ball about its anchor. Multiplying over v vertices is a bound on possible choices, not an assertion of independence. At defect zero, all labels inside an SCC agree. Conversely, every zero-defect quotient labeling lifts uniquely to a component-constant labeling with the same energy. $\square$

We now prove the auxiliary-dimension theorem stated in Section 10. Its notation is the positive moment polynomial of Lemma M.1 and the quotient just constructed.

Proof. By positivity of the quotient coefficients,

$$\sum_{j \leq C_0\delta} A_Q(p, j) \leq M^{C_0\delta} \sum_{j \geq 0} A_Q(p, j) M^{-j} = M^{C_0\delta} F_Q(p, M).$$

Insert this in Equation (103), multiply by $N^{-\delta}$, and extend the finite sum to an infinite geometric series with ratio $p^{4v} M^{C_0}/N$. Taking $M = p^2$ gives the second bound. $\square$

The quotient comparison retains its positive-defect states instead of projecting onto zero-defect states alone. It evaluates their entire positive moment polynomial at a new dimension. This is the point at which a separately sampled Gaussian quotient can be used without changing the distributional assumptions on the original RTN.

M.3 The overlapping Ginibre circuit and its strong limit

Label the flow paths $1, \dots, s$. For each internal quotient vertex C , let $K_C \subseteq [s]$ be the paths through it; $|K_C| = k_C$. Order these vertices topologically as $C_1, \dots, C_t$. Independently sample normalized square Ginibre matrices $G_C^{(M)}$ of size $d_C = M^{k_C}$, with entry variance d_C^{-1} . Embed each gate on tensor factors K_C of $(\mathbb{C}^M)^{\otimes s}$, using identity on the other factors, and set

$$B_M = \hat{G}_{C_t}^{(M)} \cdots \hat{G}_{C_1}^{(M)}. \quad (104)$$

Lemma M.5 (Exact circuit realization). *Let $H_{Q,M}$ be the raw independent-Gaussian contraction of Q , with direct terminal edges interpreted as identity wires. In the wire-labeled coordinates,*

$$B_M = M^{-\frac{1}{2}} \sum_C k_C H_{Q,M}, \quad Y_{Q,M} = B_M B_M^*, \quad F_Q(p, M) = \mathbb{E}[M^{-s} \text{tr}(B_M B_M^*)^p].$$

Proof. An internal quotient tensor has k_C inputs and k_C outputs. Its matricization is a square iid Gaussian matrix. Path labels give consistent input and output tensor factors, and topological contraction is exactly multiplication of the embedded gates. Different boundary ordering contributes only permutation unitaries. Counting edges at their tails gives $R_Q = s + \sum_C k_C$, so M^{s-R_Q} is the product of the squared gate normalizations. The Wick formula also holds for direct identity wires, whose terminal permutations give their usual index-loop factors. $\square$

If $t = 0$, $B_M = I$, $F_Q(p, M) = 1$, and $E_G = 1$. Suppose henceforth that $t > 0$. A normalized complex Ginibre gate has the distribution

$$G_C^{(M)} = (S_{C,1}^{(M)} + iS_{C,2}^{(M)})/\sqrt{2},$$

where the $2t$ matrices S are independent normalized GUE matrices of the indicated base dimensions. The covariance of this array is $\mathbb{E}G_{ij}\overline{G_{kl}} = d_C^{-1}\mathbf{1}_{i=k,j=l}$, and its pseudocovariance is zero, proving the asserted iid circular Gaussian law.

In Chen et al. [5, Section 9.4, Theorem 9.8], take the number of tensor factors to be s , the number of GUE variables to be $2t$, their supports to be $K_{C_1}, K_{C_1}, \dots, K_{C_t}, K_{C_t}$, and the coefficient dimension to be one. These supports are nonempty; the theorem does not require them to be distinct or disjoint. The polynomial is the ordered product in Equation (104), of fixed degree t , independent of p and M . We obtain a limit b in a C^* -probability space with faithful trace τ such that

$$\|B_M\| \longrightarrow \|b\| =: L \quad \text{almost surely.} \quad (105)$$

The weak moment limit of Charlesworth and Collins [4, Theorem 4], also recalled directly before that strong-convergence theorem, gives, for fixed p ,

$$\lim_{M \rightarrow \infty} F_Q(p, M) = \tau((bb^*)^p) = A_Q(p, 0) = A_G(p, 0). \quad (106)$$

Expected moments are justified directly by that weak-limit interface, or by uniform integrability: the gate moment bound proved below implies

$$\sup_{M \geq 1} \mathbb{E} \left[(M^{-s} \text{tr}(B_M B_M^*)^p)^2 \right] \leq [C(1 + \sqrt{4p})]^{4pt} < \infty$$

for fixed p . This bound does not require the growing-moment conclusion.

The scalar spectral measure of bb^* has compact support and therefore is uniquely determined by these moments. It is μ_G . Faithfulness implies that its support is the spectrum of bb^* : a nonzero nonnegative continuous function of bb^* has positive trace. It follows that

$$E_G = \|bb^*\| = L^2. \quad (107)$$

Also $m_1 = 1$, so $E_G \geq 1$.

M.4 From strong convergence to growing moments

Qualitative norm convergence does not by itself control growing moments. We supply both concentration and control of the size of the exceptional contribution.

Lemma M.6 (Gate tails and moments). *For a normalized d by d complex Ginibre matrix G , universal constants $c, C > 0$ satisfy*

$$\mathbb{P}\{\|G\| > 16\} \leq e^{-cd}, \quad (\mathbb{E} \|G\|^r)^{1/r} \leq C(1 + \sqrt{r/d}) \quad (r \geq 1).$$

Proof. A $1/4$ -net of the complex unit sphere has at most 9^{2d} points. Two such nets give $\|G\| \leq 2 \max_{u,w} |w^* G u|$. Each scalar in this maximum is circular Gaussian of variance $1/d$. Thus

$$\mathbb{P}\{\|G\| > x\} \leq \min\{1, \exp(4d \log 9 - dx^2/4)\}.$$

For $x \geq 16$ the second term is at most $\exp(-dx^2/8)$. Integrating the tail of $(\|G\| - 16)_+$ gives $\mathbb{E}[(\|G\| - 16)_+^r] \leq (8/d)^{r/2} \Gamma(r/2 + 1)$. The bound $\Gamma(r/2 + 1)^{1/r} \leq C\sqrt{r}$ and Minkowski's inequality prove the moment estimate. $\square$

Lemma M.7 (Concentration about the limiting norm). *For each $\varepsilon > 0$ there are positive constants $c_\varepsilon, C_\varepsilon$ and M_0 , depending on Q, ε , such that*

$$\mathbb{P}\{\|B_M\| > L + \varepsilon\} \leq C_\varepsilon e^{-c_\varepsilon M} \quad (M \geq M_0).$$

Proof. Write gate entries as $(\xi + i\eta)/\sqrt{2d_C}$ in independent standard real Gaussian coordinates. Let Ω_M be the set on which every gate norm is at most 16. Since $d_C \geq M$, Lemma M.6 gives $\mathbb{P}(\Omega_M^c) \leq te^{-c_0 M}$. In particular $\mathbb{P}(\Omega_M) \geq 3/4$ once $M \geq \max\{1, \lceil c_0^{-1} \log(4t) \rceil\}$.

For any two coordinate vectors in Ω_M , telescope their gate products. A gate difference is bounded in operator norm by its base-matrix Frobenius norm; embedding with identity factors preserves operator norm. Cauchy-Schwarz over gates gives the pairwise Lipschitz constant

$$\ell_M = 16^{t-1} \left(\sum_C (2d_C)^{-1} \right)^{1/2} \leq 16^{t-1} \sqrt{t/(2M)}$$

for $f_M = \|B_M\|$ on Ω_M . No convexity of Ω_M is required, and no ambient M^s Frobenius factor is introduced.

The restricted Gaussian concentration inequality [5, Lemma 5.6] yields

$$\mathbb{P}\{|f_M - \text{med } f_M| > u\} \leq \mathbb{P}(\Omega_M^c) + Ce^{-cu^2/\ell_M^2}.$$

By Equation (105), every choice of median tends to L . For all sufficiently large M , take $u = \varepsilon/2$. Combining the two exponential bounds proves the assertion. $\square$

M.5 The analytic bridge at an auxiliary dimension

The preceding concentration estimate is one input to the following general criterion. The other is high-moment control of the exceptional contribution. We state the criterion before verifying both inputs for the comparison circuit.

Lemma M.8 (From reference concentration to growing moments). *Let $Z_M \geq 0$ and $L \geq 0$. Suppose that for every $\varepsilon > 0$ there are $C_\varepsilon, c_\varepsilon > 0$ and M_ε such that, for a fixed $\beta > 0$,*

$$\mathbb{P}\{Z_M > L + \varepsilon\} \leq C_\varepsilon e^{-c_\varepsilon M^\beta} \quad (M \geq M_\varepsilon).$$

Choose integers M_p with $M_p^\beta/p \rightarrow \infty$ and suppose $\sup_{p \geq p_0} \|Z_{M_p}\|_{L^{4p}} \leq D < \infty$. Then for every $\eta > 0$ and all sufficiently large p ,

$$\mathbb{E} Z_{M_p}^{2p} \leq 2(L^2 + \eta)^p. \quad (108)$$

Proof. Choose $\varepsilon > 0$ with $(L + \varepsilon)^2 < L^2 + \eta$. Below the threshold $L + \varepsilon$, the contribution is at most $(L + \varepsilon)^{2p}$. Cauchy–Schwarz bounds the remaining contribution by $D^{2p} C_\varepsilon^{1/2} \exp(-c_\varepsilon M_p^\beta/2)$. Its p -th root tends to zero, so each contribution is eventually at most $(L^2 + \eta)^p$. $\square$

Corollary M.9 (Logarithmic-order transfer with the right base). *Under Theorem 10.4, suppose $r_p \leq p^b$ for fixed $b \geq 0$. Choose $M_p = \lceil p^a \rceil$ with fixed $a > 0$ and $a\beta > 1$. If $F_\Lambda(p, M_p) \leq \mathbb{E} Z_{M_p}^{2p}$ for a family satisfying Lemma M.8, then for every $\eta > 0$ and all sufficiently large p ,*

$$F_\Omega(p, N) \leq \frac{2(L^2 + \eta)^p}{1 - p^b \lceil p^a \rceil^\kappa / N} \quad \text{whenever the denominator is positive.}$$

The denominator tends to one when $p \asymp \log N$. The coefficient $a_{p,\delta}$ is bounded by the same numerator times $p^{b\delta} \lceil p^a \rceil^{\kappa\delta}$.

Proof. Insert Equation (108) into Equation (48) and its coefficient bound. A fixed power of $\log N$ is $o(N)$. $\square$

The high-moment assumption is not a consequence of qualitative strong convergence. Let $Z_M = e^{\sqrt{M}}$ with probability e^{-M} and $Z_M = 1$ otherwise. The event probabilities are summable, so in any common coupling $Z_M \rightarrow 1$ almost surely. For fixed k ,

$$\mathbb{E} Z_M^k = 1 - e^{-M} + e^{-M+k\sqrt{M}} \rightarrow 1, \quad \mathbb{E} Z_{p^2}^{2p} \geq e^{p^2}.$$

This family even has exponentially rare upper deviations. It shows that fixed moments and small exceptional probability alone do not control growing moments. The uniform $4p$ -moment bound is a sufficient way to control that contribution here, not a claim that it is the only possible condition for such a transfer.

For the RTN, take $Z_M = \|B_M\|$, $\beta = 1$, $a = 2$, $b = 4v$, and $\kappa = 1 + 4R$. The normalized trace is bounded by the operator norm moment, and $L^2 = E_G$. The following proof supplies the uniform $4p$ bound and gives the original quantitative statement with all model constants retained.

Lemma M.10 (Auxiliary growing moments). *For every $\eta > 0$, there is $p_0(Q, \eta)$ such that*

$$F_Q(p, p^2) \leq 2(E_G + \eta)^p \quad (p \geq p_0). \quad (109)$$

Consequently, for every $\delta \geq 0$ and the same threshold,

$$A_G(p, \delta) \leq 2(E_G + \eta)^p p^{K_G \delta}. \quad (110)$$

Proof. For $p \leq M$, independence and Lemma M.6 give

$$\mathbb{E} \|B_M\|^{4p} \leq \prod_C \mathbb{E} \|G_C^{(M)}\|^{4p} \leq D_Q^{4p}$$

with D_Q independent of M, p . Fix $\varepsilon > 0$. On the exceptional event of Lemma M.7, apply Cauchy–Schwarz to obtain

$$F_Q(p, M) \leq \mathbb{E} \|B_M\|^{2p} \leq (L + \varepsilon)^{2p} + D_Q^{2p} C_\varepsilon^{1/2} e^{-c_\varepsilon M/2}. \quad (111)$$

Fix η first and choose ε so that $(L + \varepsilon)^2 < E_G + \eta$. At $M = p^2$, the logarithm of the second term divided by p is $2 \log D_Q + (\log C_\varepsilon)/(2p) - c_\varepsilon p/2$, which tends to $-\infty$. This proves Equation (109). Combining it with Equation (103) and positivity at $M = p^2$ gives Equation (110), uniformly in δ . When $t = 0$ the same assertions follow directly from $F_Q = 1$. $\square$

This estimate preserves the exact exponential base up to any fixed $\eta > 0$. It does not assert the stronger relative bound $A_G(p, \delta) \leq A_G(p, 0) p^{C_G \delta}$.

M.6 The upper edge, fixed moments, and normalization

Fix $\varepsilon > 0$. By Theorem 10.5 and Lemma M.10,

$$\mathbb{E}[N^{-s} \operatorname{tr} Y_N^p] \leq \frac{2(E_G + \varepsilon/2)^p}{1 - p^{K_G}/N}$$

for all sufficiently large p satisfying $p^{K_G} < N$. Choose

$$p_N = \lceil c_\varepsilon \log N \rceil, \quad c_\varepsilon > \frac{s}{\log((E_G + \varepsilon)/(E_G + \varepsilon/2))}.$$

Then $p_N^{K_G}/N \rightarrow 0$ and all fixed large-parameter thresholds hold eventually. Positivity and Markov's inequality give

$$\mathbb{P}\{\|Y_N\| > E_G + \varepsilon\} \leq \frac{2N^s}{1 - p_N^{K_G}/N} \left(\frac{E_G + \varepsilon/2}{E_G + \varepsilon} \right)^{p_N} \rightarrow 0. \quad (112)$$

The trace dimension N^s is essential in this choice of p_N .

For the lower bound it suffices to use fixed moments. We give a direct Wick proof, including a useful covariance identity. Write $T_{N,p} = N^{-s} \operatorname{tr} Y_N^p$.

Lemma M.11 (Nonnegative covariance polynomial). *For fixed positive integers p, q ,*

$$\operatorname{Cov}(T_{N,p}, T_{N,q}) = \sum_{\delta \geq 1} B_G(p, q; \delta) N^{-\delta}, \quad B_G(p, q; \delta) \in \mathbb{Z}_{\geq 0},$$

and

$$\mathbb{E}[(T_{N,p} - m_p)^2] \leq \frac{((2p)!)^v}{N} + \frac{(p!)^{2v}}{N^2}. \quad (113)$$

Proof. Use labels in S_{p+q} , terminal identity at $\mathbf{b}$, and $\Gamma = (1 \cdots p)(p+1 \cdots p+q)$ at $\mathbf{a}$. The mixed Wick sum is

$$\mathbb{E}[T_{N,p} T_{N,q}] = \sum_{\pi \in S_{p+q}^{VG}} N^{s(p+q-2) - \mathcal{H}_G^\Gamma(\pi)}.$$

All labelings preserving the two trace blocks split into an S_p labeling and an S_q labeling. Their energies add, so their entire contribution, including positive defects, is $F_G(p, N) F_G(q, N)$.

No block-mixing labeling has zero defect. Indeed, at minimum energy every unused edge has equal labels and each flow path is geodesic from $\mathbf{id}$ to Γ . Every vertex connects to a flow vertex through unused edges. Thus each label lies in $[\mathbf{id}, \Gamma]$ in absolute order. If σ lies in that interval,

concatenate reduced transposition words for σ and $\sigma^{-1}\Gamma$. The resulting word has length $|\Gamma|$. Starting at identity, every step must merge two cycles; a splitting step would prevent the total cycle-count decrease from equaling the word length. A merge across the two final cycle supports could never be undone. Hence σ preserves both supports.

Subtracting the block-preserving contribution leaves positive terms with integer defects at least one. There are at most $((p+q)!)^v$ terms, giving the asserted covariance polynomial and the variance bound $((2p)!)^v/N$. The ordinary Wick sum also gives $0 \leq \mathbb{E}T_{N,p} - m_p \leq (p!)^v/N$. Adding squared bias proves Equation (113). $\square$

Because $\text{rank } Y_N \leq N^s$, $T_{N,p} \leq \|Y_N\|^p$. Compactness and positivity of μ_G imply $m_p^{1/p} \rightarrow E_G$. For $0 < \varepsilon < E_G$, choose a fixed p with $m_p > (E_G - \varepsilon)^p$. Then Equation (113) gives

$$\mathbb{P}\{\|Y_N\| \leq E_G - \varepsilon\} \leq \mathbb{P}\{T_{N,p} \leq (E_G - \varepsilon)^p\} \rightarrow 0.$$

For $\varepsilon \geq E_G$ the lower estimate is trivial. Together with Equation (112) this proves the first convergence in Theorem 10.3.

Finally, $\mathbb{E}Z_N = 1$. In the second moment of Z_N , each vertex has either the identity or the transposition label in S_2 , and both terminal labels are identity. If T is the transposed vertex set, its term is $N^{-|\partial T|}$. Thus

$$\text{Var } Z_N = \sum_{\emptyset \neq T \subseteq V_G} N^{-|\partial T|} \leq (2^v - 1)/N. \quad (114)$$

Every nonempty T has a boundary by connectedness, including $T = V_G$. Some entry of H_N is a nonzero polynomial in the Gaussian coordinates, so $Z_N > 0$ almost surely. Slutsky's theorem applied to $X_N = Y_N/Z_N$ proves the second convergence. Since $E_G \geq 1$, continuity of the logarithm gives Equation (44), completing the proof.

N An independent Wishart calibration

Let X be P by d with independent standard circular complex coordinates, $P \geq d$, and $F = X^*X/P - I_d$. For $\gamma = (1 \cdots k)$, direct Wick expansion gives

$$\mathbb{E} \text{tr}(X^*X/P)^k = P^{-k} \sum_{\pi \in \mathfrak{S}_k} P^{\#\pi} d^{\#(\gamma\pi)}.$$

Pairing conjugate position s with unconjugated position $\pi(s)$ identifies row labels along cycles of π and column labels along cycles of $\gamma\pi$. Coinciding labels do not erase pairing multiplicities.

For $k = 1, 2, 3, 4$, this yields

$$\begin{aligned} \mathbb{E} \text{tr } W &= d, \\ \mathbb{E} \text{tr } W^2 &= d + d^2/P, \\ \mathbb{E} \text{tr } W^3 &= d + 3d^2/P + (d^3 + d)/P^2, \\ \mathbb{E} \text{tr } W^4 &= d + 6d^2/P + (6d^3 + 5d)/P^2 + (d^4 + 5d^2)/P^3, \end{aligned}$$

where $W = X^*X/P$. Subtracting the identity gives

$$\mathbb{E} \text{tr } F^2 = d^2/P, \quad \mathbb{E} \text{tr } F^4 = (2d^3 + d)/P^2 + (d^4 + 5d^2)/P^3.$$

For a uniform eigenvalue index let $Y = |\lambda_j(F)|$ and $u = d/P$. Then $\mathbb{E}Y^2 = u$ and $\mathbb{E}Y^4 \leq 9u^2$. Interpolation and Cauchy-Schwarz give

$$\frac{1}{3}d\sqrt{d/P} \leq \mathbb{E}\|F\|_1 \leq d\sqrt{d/P}, \quad \mathbb{P}\{\|F\|_1 \geq d\sqrt{d/P}/6\} \geq 1/36. \quad (115)$$

For the probability bound use $(d^{-1}\|F\|_1)^2 \leq d^{-1} \text{tr } F^2$ and Paley–Zygmund. These finite-dimensional bounds hold uniformly over every $P \geq d$, without a fixed-aspect limit. At $P = N^a, d = N^b$, they confirm the trace threshold $a > 3b$. For $a < b$, rank already excludes convergence.

O Proof of the sparse phase diagram

This appendix proves Proposition 5.7 for its specified iid family. Let $D_i = \sum_j B_{ij}$, $D_{\max} = \max_i D_i$, and $\lambda = n\rho$. The $h = 1$ case of Segner [19, Theorem 1.1] and the deterministic maximum-row-norm lower bound imply

$$\mathbb{E} \|X\| \asymp \rho^{-1/2} \mathbb{E} \sqrt{D_{\max}}. \quad (116)$$

Indeed, each row norm is $\rho^{-1/2} \sqrt{D_i}$, and the maximum column norm has the same distribution as the maximum row norm. The comparison constant is independent of dimension and distribution. This is the only external norm theorem used in the proof. We now estimate the expectation on the right, rather than only a typical value.

For every integer $k \geq 1$, a binomial union bound gives

$$\mathbb{P}\{D_{\max} \geq k\} \leq n(e\lambda/k)^k. \quad (117)$$

Also, the exponential-moment estimate $\mathbb{E} e^{D_i} = (1 + \rho(e - 1))^n \leq \exp(\lambda(e - 1))$ gives

$$\mathbb{P}\{D_{\max} \geq k\} \leq n \exp(\lambda(e - 1) - k).$$

Polynomially growing row occupancy: $0 \leq \beta < 1$. Here $\lambda = n^{1-\beta}$ grows polynomially. For $k_0 = \lceil C\lambda \rceil$ with a sufficiently large fixed C , the last bound has a negligible geometric tail after k_0 . Consequently $\mathbb{E} D_{\max} = O(\lambda)$ and Jensen gives $\mathbb{E} \sqrt{D_{\max}} = O(\sqrt{\lambda})$. One row has mean λ and variance at most λ , so Chebyshev’s inequality gives the matching lower bound. At $\beta = 0$ each row count is deterministically n , with the same conclusion.

Critical occupancy: $\beta = 1$. Now $\lambda = 1$. Put $k_0 = \lceil 3 \log n / \log \log n \rceil$. The bound $n(e/k)^k$ is $o(1)$ at k_0 , and its ratio at successive integers is at most $e/(k + 1)$. Summing the tail yields $\mathbb{E} D_{\max} = O(\log n / \log \log n)$. For the lower bound, fix $0 < \delta < 1$ and set $k = \lfloor (1 - \delta) \log n / \log \log n \rfloor$. Since $k = o(\sqrt{n})$,

$$\mathbb{P}\{D_i = k\} = \binom{n}{k} n^{-k} (1 - 1/n)^{n-k} = \exp(-k \log k + O(k) + o(1)).$$

Thus $n\mathbb{P}\{D_i = k\} = n^{\delta+o(1)} \rightarrow \infty$. The rows are independent, so $D_{\max} \geq k$ with probability tending to one. Jensen supplies the matching upper bound for $\mathbb{E} \sqrt{D_{\max}}$.

Sparse but nonempty: $1 < \beta < 2$. Choose a fixed integer $k_0 > 2/(\beta - 1)$. The first term of Equation (117) at k_0 tends to zero, and successive terms have ratio at most $e\lambda/(k + 1) = o(1)$. Thus $\mathbb{E} D_{\max} = O(1)$. There is at least one occupied entry with probability $1 - (1 - \rho)^{n^2} \rightarrow 1$, proving $\mathbb{E} \sqrt{D_{\max}} \asymp 1$.

Bounded total occupancy: $\beta = 2$. Let $N = \sum_{i,j} B_{ij}$. Then $\mathbb{E} N = 1$ and $\mathbb{E} \sqrt{D_{\max}} \leq \mathbb{E} \sqrt{N} \leq 1$. On the other hand $\mathbb{P}\{N = 1\} \rightarrow e^{-1}$, giving a constant lower bound.

Rare occupancy: $\beta > 2$. Here $\mu = \mathbb{E}N = n^2\rho \rightarrow 0$. Pointwise $\sqrt{D_{\max}} \leq N$, so its expectation is at most μ . Exactly one entry is occupied with probability $\mu(1 - \rho)^{n^2-1} \sim \mu$, yielding the reverse bound. Multiplication by $\rho^{-1/2}$ in Equation (116) gives all regimes of Equation (39).

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